

SCHEM, MBP 15" MLB

08/18/2008

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ZONE	ECN	DESCRIPTION OF CHANGE	CK APPD DATE	ENG APPD DATE
?		?	?	?	?

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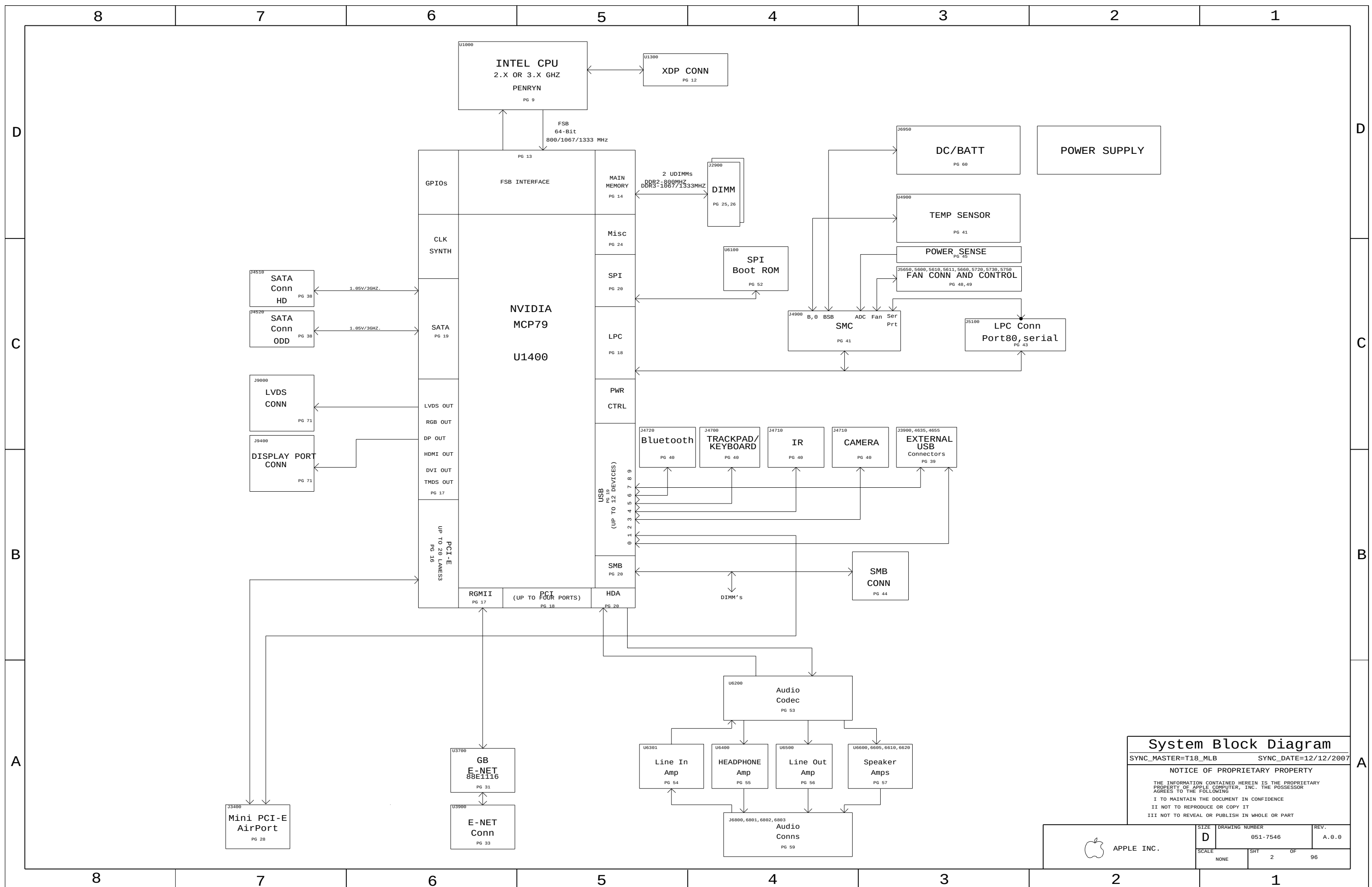
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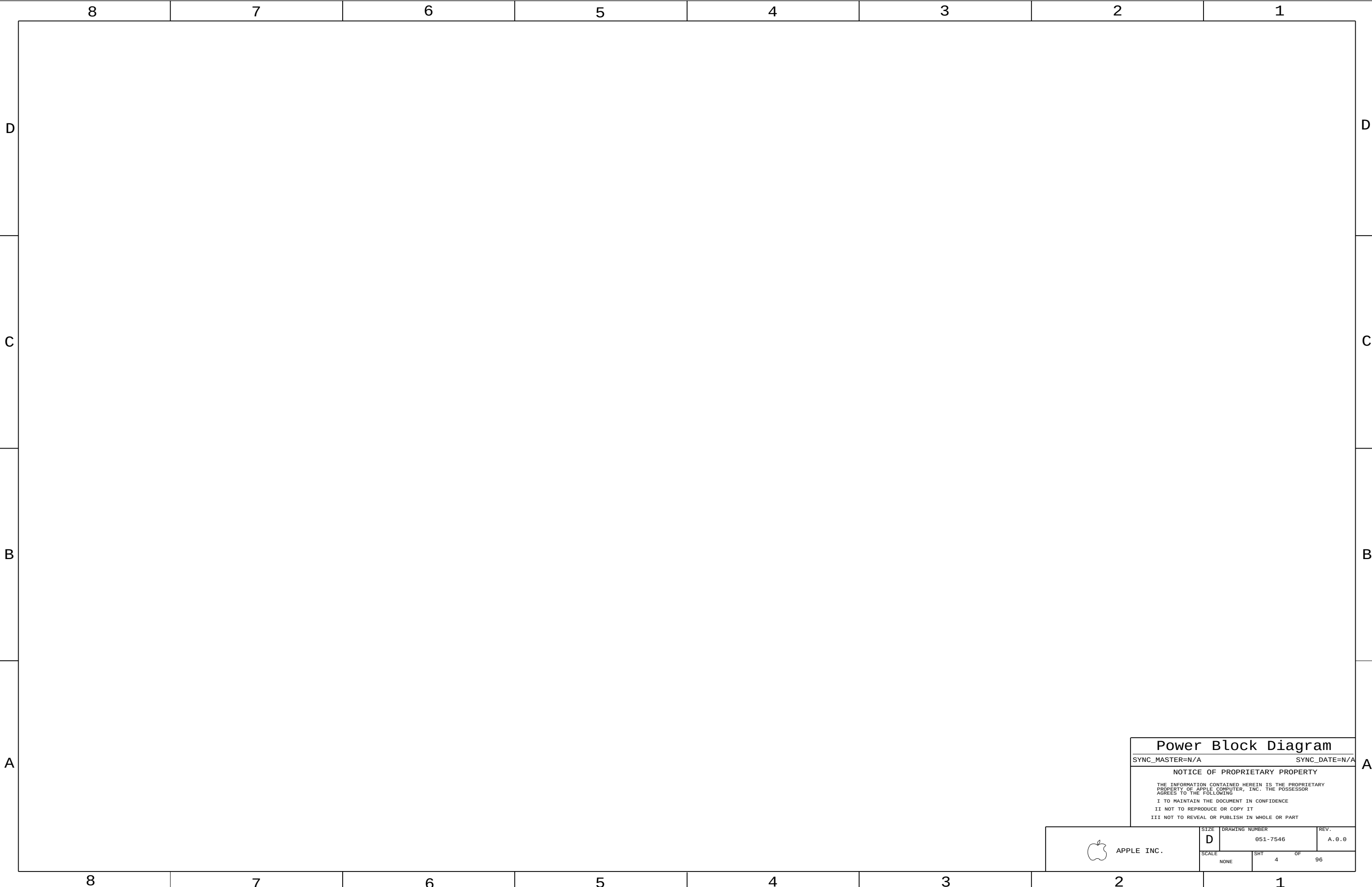
Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-7546	1	SCHEM, FIBBO, M98	SCH	CRITICAL	
820-2330	1	PCBF, FIBBO, M98	PCB	CRITICAL	

DRAWING
TITLE=MLB
ABBREV=DRAWING
LAST_MODIFIED=Mon Aug 18 01:48:34 2008

DIMENSIONS ARE IN MILLIMETERS XX ± _____ X.XX ± _____ X.XXX ± _____ ANGLES ± _____ DO NOT SCALE DRAWING  THIRD ANGLE PROJECTION	METRIC		 APPLE INC.	
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	DRAFTER	DESIGN CK	TITLE	
	ENG APPD	MFG APPD	SCHEM, MBP 15MLB	
	QA APPD	DESIGNER	DRAWING NUMBER 051-7546 REV: A.0.0	
RELEASE		SCALE NONE	SHT 1 OF 96	
MATERIAL/FINISH NOTED AS APPLICABLE		SIZE D		





Power Block Diagram		
SYNC_MASTER=N/A		SYNC_DATE=N/A
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SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	4	96

	APPLE INC.
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
630-9334	PCBA, 2.4GHZ, 256SAM_VRAM, M98	M98_COMMON, EEE_0ZA, CPU_2_4GHZ, FB_256_SAMSUNG
630-9335	PCBA, 2.4GHZ, 256HYN_VRAM, M98	M98_COMMON, EEE_0ZB, CPU_2_4GHZ, FB_256_HYNIX
630-9336	PCBA, 2.5GHZ, 512SAM_VRAM, M98	M98_COMMON, EEE_0ZC, CPU_2_5GHZ, FB_512_SAMSUNG
630-9337	PCBA, 2.5GHZ, 512QIM_VRAM, M98	M98_COMMON, EEE_0ZD, CPU_2_5GHZ, FB_512_QIMONDA
630-9585	PCBA, 2.8GHZ, 512SAM_VRAM, M98	M98_COMMON, EEE_2NH, CPU_2_8GHZ, FB_512_SAMSUNG
630-9586	PCBA, 2.8GHZ, 512QIM_VRAM, M98	M98_COMMON, EEE_2NJ, CPU_2_8GHZ, FB_512_QIMONDA

M98 BOM Groups

BOM GROUP	BOM OPTIONS
M98_COMMON	ALTERNATE, COMMON, M98_COMMON1, M98_COMMON2, M98_COMMON3, M98_DEBUG, M98_PROGPARTS
M98_COMMON1	ONEWIRE_PU, ISL6258A, MEMRESET_HW, MEMRESET_MCP, MCP_B02, MCP_PROD, MCPSEQ_SMC
M98_COMMON2	BKLT_PLL_NOT, BMON_ENG, MIKEY, BOOT_MODE_USER, GPUVID_1P00V, MUXGFX
M98_COMMON3	DPMUX_EN_S0, DP_ESD, EG_PWRSEQ_HW, DP_CA_DET_EG_PLD, MCP_CS1_N0
M98_DEBUG	SMC_DEBUG_YES, XDP, LPCPLUS, VREFMRGN
M98_PROGPARTS	GMUX_PROG, BOOTROM_PROG, SMC_PROG, TPAD_PROG

BOM GROUP	BOM OPTIONS
FB_256_SAMSUNG	VRAM4, VRAM_256_SAMSUNG
FB_256_HYNIX	VRAM4, VRAM_256_HYNIX
FB_512_SAMSUNG	VRAM4, VRAM_512_SAMSUNG
FB_512_QIMONDA	VRAM4, VRAM_512_QIMONDA

Bar Code Labels / EEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:0ZA]	CRITICAL	EEE_0ZA
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:0ZB]	CRITICAL	EEE_0ZB
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:0ZC]	CRITICAL	EEE_0ZC
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:0ZD]	CRITICAL	EEE_0ZD
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:2NH]	CRITICAL	EEE_2NH
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:2NJ]	CRITICAL	EEE_2NJ

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3639	1	IC, PDC, SLB4N, PRQ, 2.4G, 25W, 1066, M0, 3M, BGA	U1000	CRITICAL	CPU_2_4GHZ
337S3640	1	IC, PDC, SL3BX, PRQ, 2.53G, 35W, 1066, C0, 6M, BGA	U1000	CRITICAL	CPU_2_5GHZ
338S0554	1	IC, GPU, 55nm, NV G96-GS, BGA969, LF	U8000	CRITICAL	
338S0570	1	IC, RTL8211CL, GIGE TRANSCEIVER, 48P TQFP	U3700	CRITICAL	
338S0523	1	IC, FW643-06, 1394B PHY/OHCI LINK/PCI-E, 12	U4100	CRITICAL	
338S0600	1	IC, GMCP, MCP79-B01, 35x35MM, BGA1437	U1400	CRITICAL	MCP_B01
338S0563	1	IC, SMC, HS8/2117, 9MMX9MM, TLP	U4900	CRITICAL	SMC_BLANK
341S2289	1	IC, SMC, DEVELOPMENT, M98	U4900	CRITICAL	SMC_PROG
335S0384	1	IC, 32MBIT 8-PIN SPI SERIAL FLASH, SOIC8	U6100	CRITICAL	BOOTROM_BLANK
341S2366	1	IC, EFI ROM, DEVELOPMENT, M98	U6100	CRITICAL	BOOTROM_PROG
341S2272	1	IC, HDCP ROM, NVG96, 8 PIN SOIC, LF, HF	U8770	CRITICAL	HDCP_YES
341S2384	1	IR, ENCORE II, CY7C63803-LQXC	U4800	CRITICAL	
338S0635	1	IC, GMCP, MCP79-B02, 35x35MM, BGA1437	U1400	CRITICAL	MCP_B02
341S2383	1	IC, PSOC +W/USB, 56PIN, MLF, M98	U5701	CRITICAL	TPAD_PROG
337S3641	1	IC, PDC, SLB43, PRQ, 2.8G, 35W, 1066, C0, 6M, BGA	U1000	CRITICAL	CPU_2_8GHZ
333S0482	4	IC, SGRAM, GDDR3, 16Mx32, 800MHZ, 136 FBGA	U8400, U8450, U8500, U8550	CRITICAL	VRAM_256_SAMSUNG
333S0483	4	IC, SGRAM, GDDR3, 16Mx32, 900MHZ, 136 FBGA	U8400, U8450, U8500, U8550	CRITICAL	VRAM_256_HYNIX
333S0481	4	IC, SGRAM, GDDR3, 32Mx32, 900MHZ, 136 FBGA	U8400, U8450, U8500, U8550	CRITICAL	VRAM_512_SAMSUNG
333S0472	4	IC, SGRAM, GDDR3, 32Mx32, 900MHZ, 136 FBGA	U8400, U8450, U8500, U8550	CRITICAL	VRAM_512_QIMONDA

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0603	138S0602		ALL	Murata alt to Samsung
353S1081	353S1294		ALL	LMV2011_OPAMP- GSW
152S0276	152S0683		ALL	Maglayers alt to Gale/Vishay
341S2367	341S2366		ALL	Macronix alt to SST
152S0876	152S0867		ALL	Maglayer alt to Delta
157S0058	157S0055		ALL	Delta alt to TOK Magnetics
353S2312	353S1466		ALL	INTERSil ALT TO INTERSiL
514-0612	514-0607		ALL	FOXLINK RCVR ALT TO FOXCONN
514-0613	514-0608		ALL	FOXLINK RCVR ALT TO FOXCONN
152S0915	152S0796		ALL	Maglayers alt to Cyntec IND

BOM Configuration

SYNC_MASTER=N/A

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SIZE

DRAWING NUMBER

REV.

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051-7546

A.0.0

SCALE

NONE

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OF

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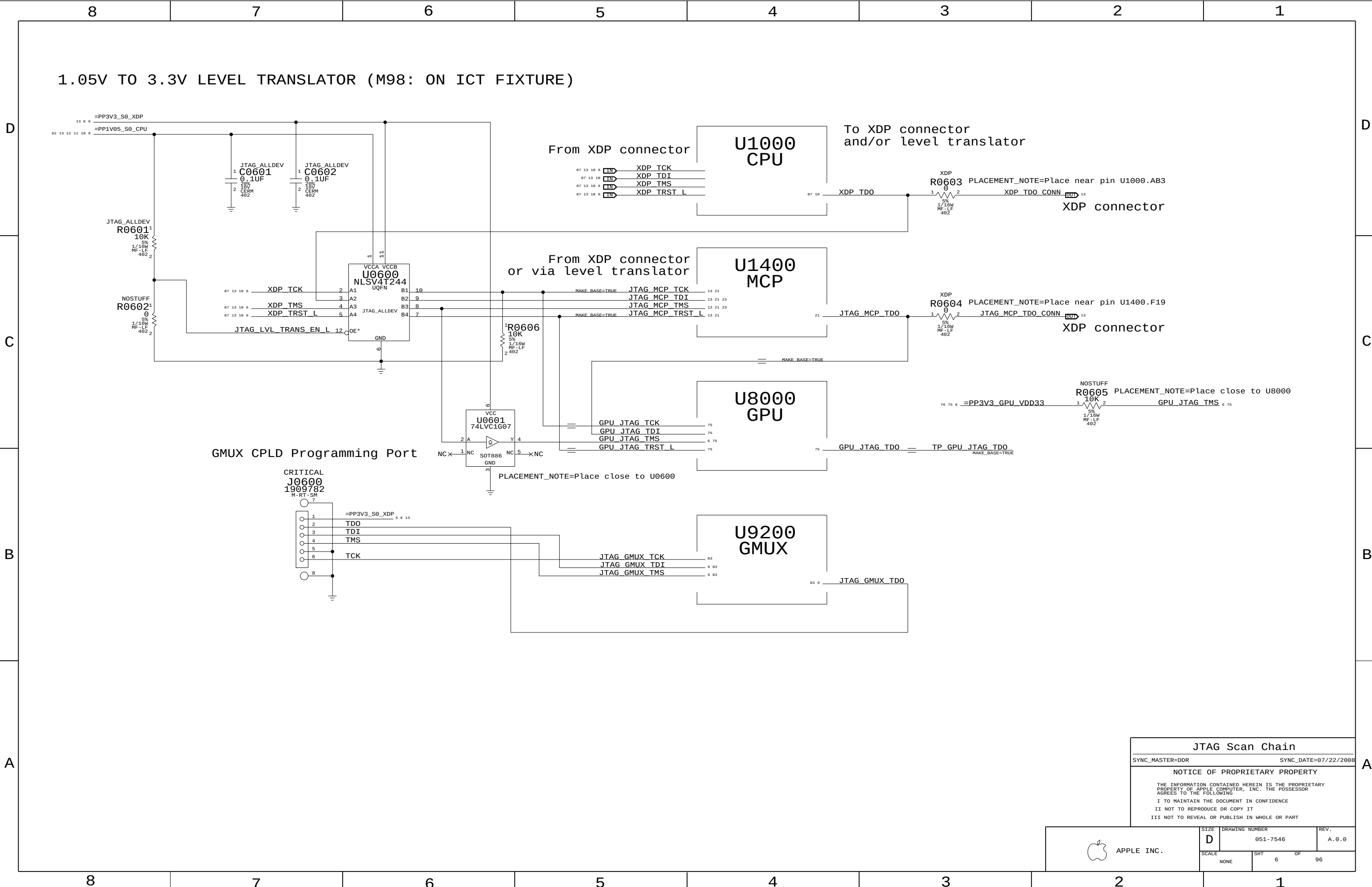
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JTAG Scan Chain

SYNC_MASTER=DDR

SYNC_DATE=07/22/2008

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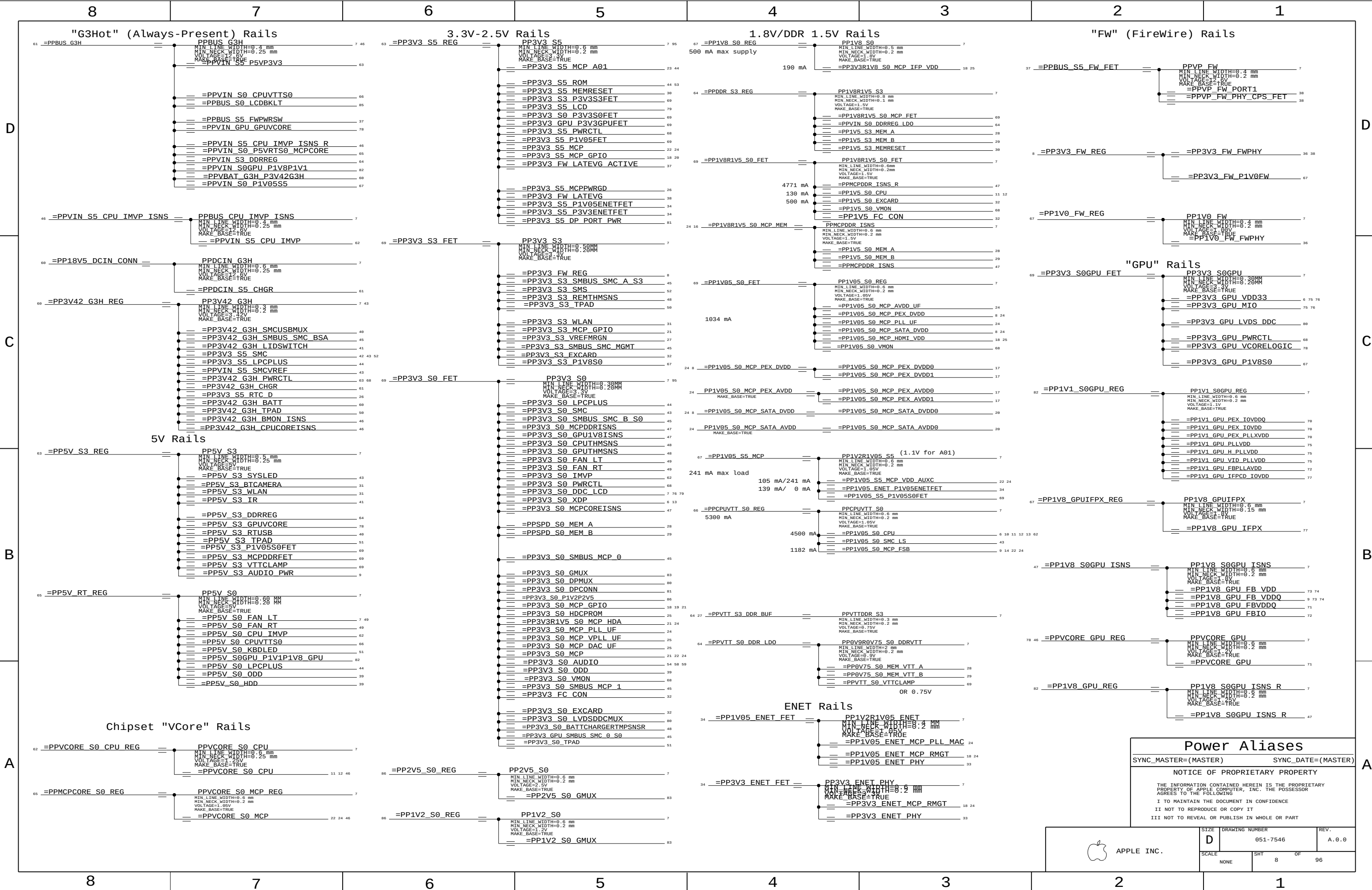
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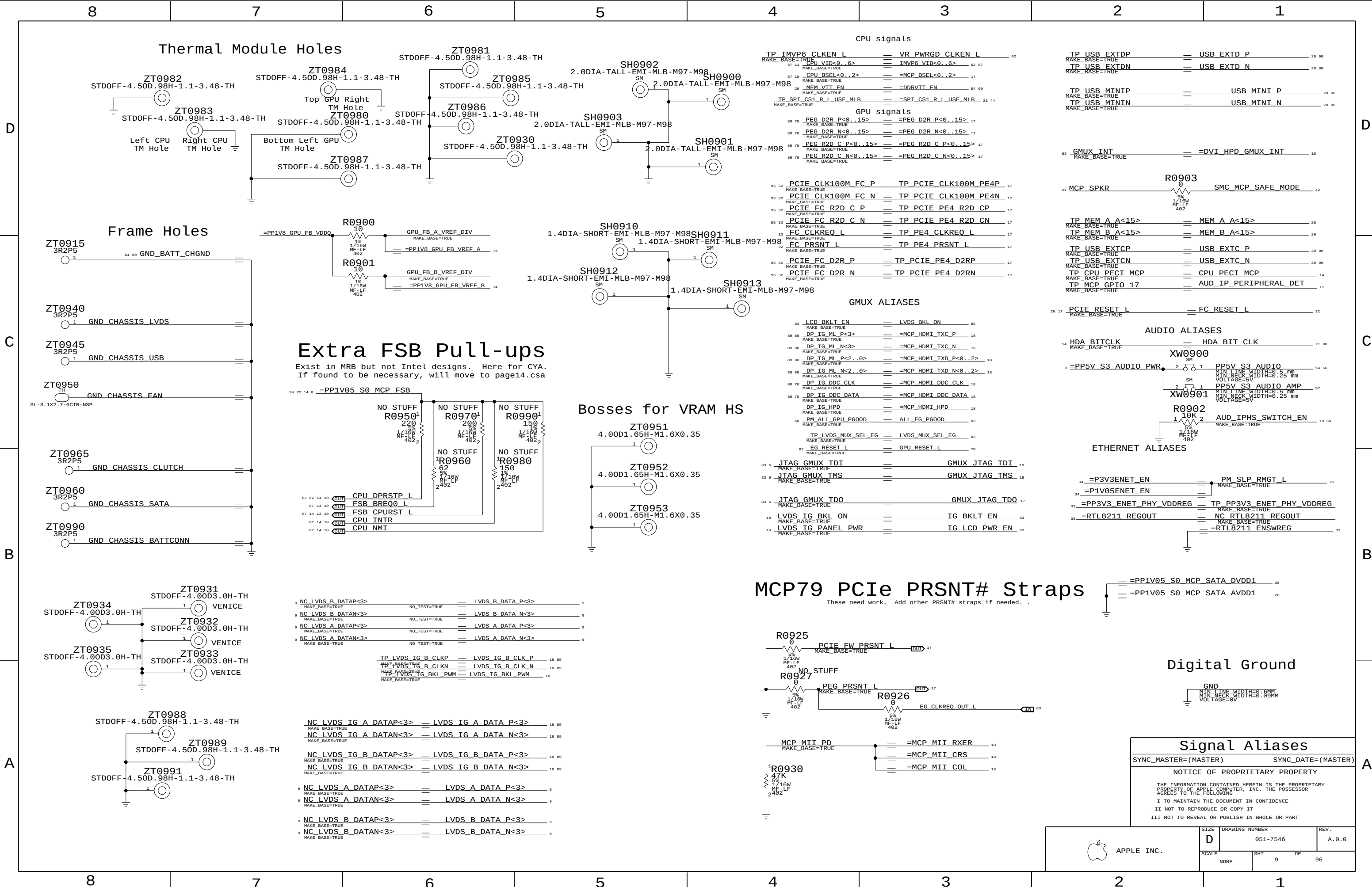
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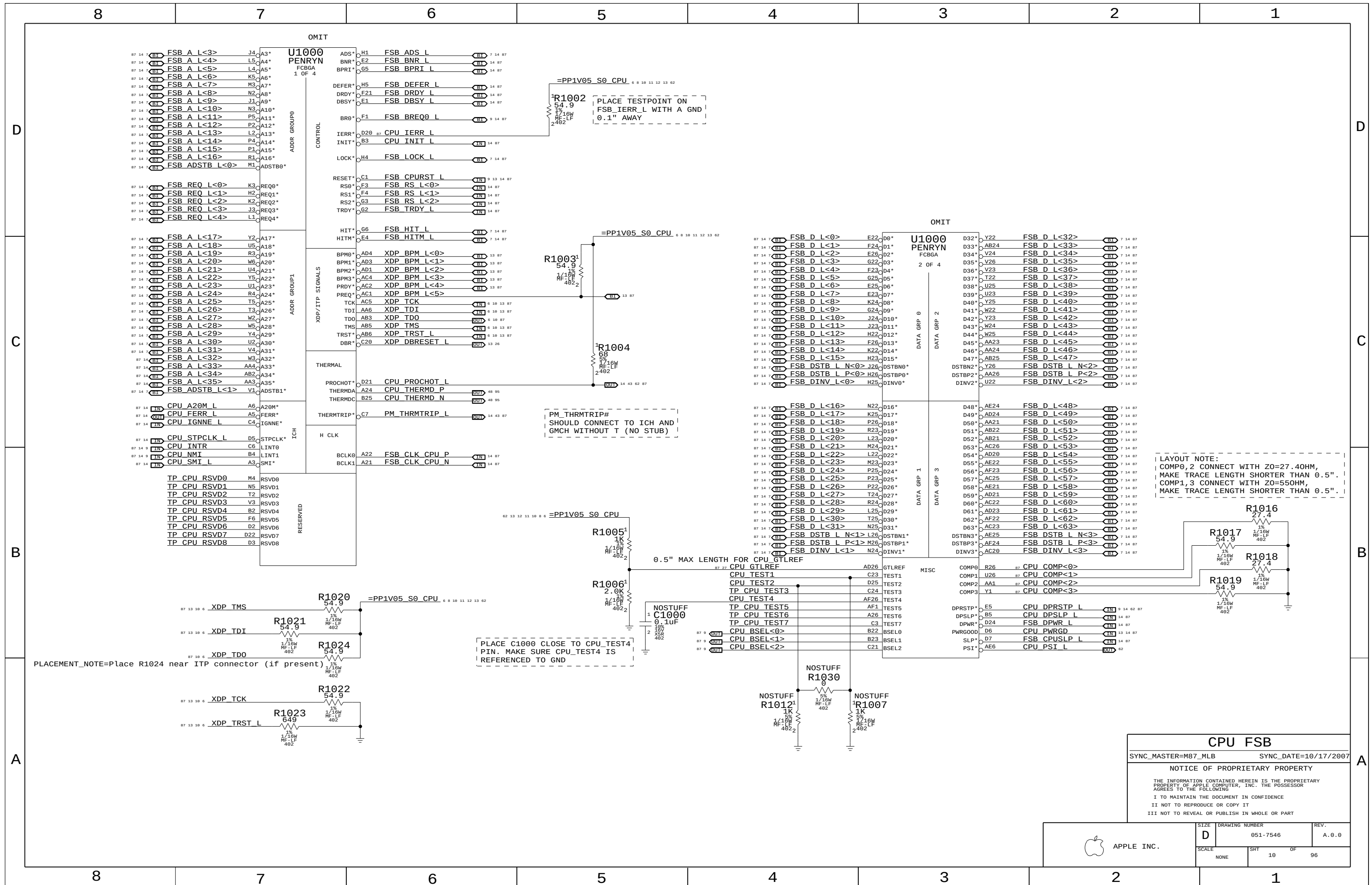
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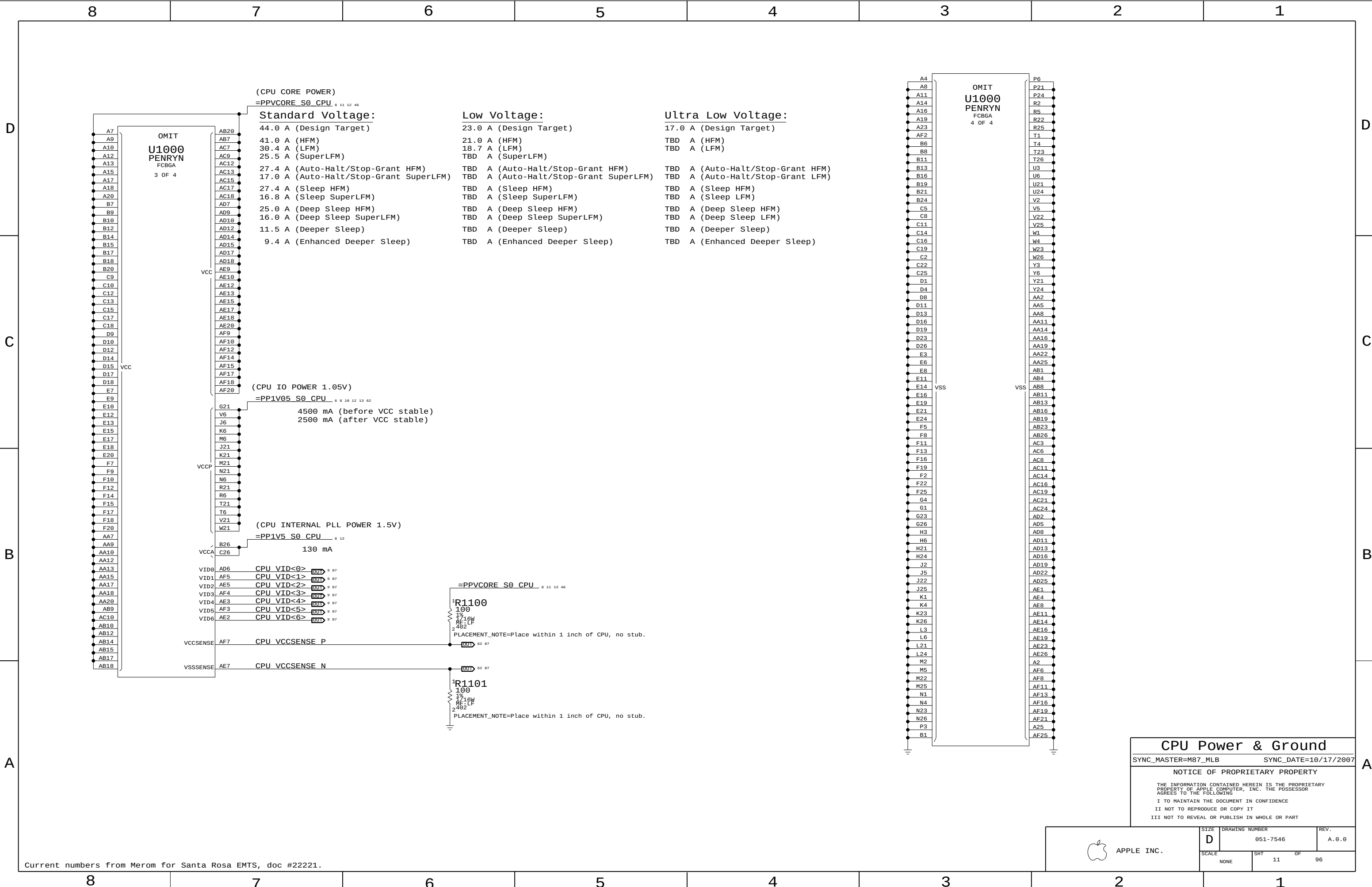
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8		7		6		5		4		3		2		1																	
Functional Test Points																ICT Test Points															
Fan Connectors																CPU FSB NO_TESTS															
FUNC_TEST																NO_TEST															
D	TRUE		=PP5V_S0_FAN_LT		8 49		3 TPS per Fan	TRUE		FSB_A_L<31..3>		10 14 87		D																	
	TRUE		FAN_LT_PWM		49			TRUE		FSB_ADS_L		10 14 87																			
	TRUE		FAN_LT_TACH		49			TRUE		FSB_ADSTB_L<1..0>		10 14 87																			
	TRUE		FAN_RT_PWM		49			TRUE		FSB_D_L<63..0>		10 14 87																			
	TRUE		FAN_RT_TACH		49			TRUE		FSB_DINV_L<3..0>		10 14 87																			
TRUE		GND					5 TPS per Fan	TRUE		FSB_DSTB_L_N<3..0>		10 14 87		C																	
								TRUE		FSB_DSTB_L_P<3..0>		10 14 87																			
								TRUE		FSB_HIT_L		10 14 87																			
								TRUE		FSB_HITM_L		10 14 87																			
								TRUE		FSB_LOCK_L		10 14 87																			
								TRUE		FSB_REQ_L<4..0>		10 14 87																			
LVDS Connectors								IPD_FLEX_CONN																							
FUNC_TEST								TRUE																							
TRUE		=PP3V3_S0_DDC_LCD		8 76 79			6 TPS	TRUE		PP3V3_S3_LDO		51		B																	
TRUE		PP3V3_SW_LCD		79				TRUE		PP18V5_S3		51																			
TRUE		BKL_SYNC		79 84				TRUE		TPAD_GND_F		7 51																			
TRUE		LVDS_DDC_CLK		79 88				TRUE		Z2_CS_L		50 51																			
TRUE		LVDS_DDC_DATA		79 88				TRUE		Z2_DEBUG3		50 51																			
TRUE		LVDS_CONN_A_DATA_N<0>		79 88 94			4 TPS	TRUE		Z2_MOSI		50 51		C																	
TRUE		LVDS_CONN_A_DATA_P<0>		79 88 94				TRUE		Z2_MISO		50 51																			
TRUE		LVDS_CONN_A_DATA_N<1>		79 88 94				TRUE		Z2_SCLK		50 51																			
TRUE		LVDS_CONN_A_DATA_P<1>		79 88 94				TRUE		Z2_BOOT0_EN		51																			
TRUE		LVDS_CONN_A_DATA_N<2>		79 88 94				TRUE		Z2_HOST_INTN		50 51																			
TRUE		LVDS_CONN_A_DATA_P<2>		79 88 94			5 TPS	TRUE		Z2_BOOT_CFG1		50 51		B																	
TRUE		LVDS_CONN_A_DATA_N<3>		79 88 94				TRUE		Z2_CLKIN		50 51																			
TRUE		LVDS_CONN_A_DATA_P<3>		79 88 94				TRUE		Z2_KEY_ACT_L		50 51																			
TRUE		LVDS_CONN_B_DATA_N<0>		79 88 94				TRUE		PSOC_MISO		50 51																			
TRUE		LVDS_CONN_B_DATA_P<0>		79 88 94				TRUE		PSOC_MOSI		50 51																			
TRUE		LVDS_CONN_B_DATA_N<1>		79 88 94			4 TPS	TRUE		PSOC_SCLK		50 51		C																	
TRUE		LVDS_CONN_B_DATA_P<1>		79 88 94				TRUE		SMBUS_SMC_A_S3_SDA		45 93																			
TRUE		LVDS_CONN_B_DATA_N<2>		79 88 94				TRUE		SMBUS_SMC_A_S3_SCL		45 93																			
TRUE		LVDS_CONN_B_DATA_P<2>		79 88 94				TRUE		PSOC_F_CS_L		50 51																			
TRUE		LVDS_CONN_B_CLK_F_N		79 94				TRUE		PICKB_L		50 51																			
TRUE		LVDS_CONN_B_CLK_F_P		79 94			5 TPS	TRUE		KEYBOARD_CONN				B																	
TRUE		LED_RETURN_1		79 84				TRUE		PP3V42_G3H		7 8 43																			
TRUE		LED_RETURN_2		79 84				TRUE		WS_KBD1		50																			
TRUE		LED_RETURN_3		79 84				TRUE		WS_KBD2		50																			
TRUE		LED_RETURN_4		79 84				TRUE		WS_KBD3		50																			
TRUE		LED_RETURN_5		79 84			4 TPS	TRUE		WS_KBD4		50		C																	
TRUE		LED_RETURN_6		79 84				TRUE		WS_KBD5		50																			
TRUE		LED_RETURN_7		79 84				TRUE		WS_KBD6		50																			
TRUE		LED_RETURN_8		79 84				TRUE		WS_KBD7		50																			
TRUE		LED_RETURN_9		79 84				TRUE		WS_KBD8		50																			
EXCARD Connector								KEYBOARD_CONN																							
FUNC_TEST								TRUE																							
TRUE		USB2_EXCARD_CONN_N		32 95			4 TPS	TRUE		WS_KBD9		50		C																	
TRUE		USB2_EXCARD_CONN_P		32 95				TRUE		WS_KBD10		50																			
TRUE		PCIE_CLK100M_EXCARD_CONN_N		32 95				TRUE		WS_KBD11		50																			
TRUE		PCIE_CLK100M_EXCARD_CONN_P		32 95				TRUE		WS_KBD12		50																			
TRUE		PCIE_EXCARD_R2D_N		32 89 95				TRUE		WS_KBD13		50																			
TRUE		PCIE_EXCARD_R2D_P		32 89 95			5 TPS	TRUE		WS_KBD14		50		B																	
TRUE		PCIE_EXCARD_D2R_P		17 22 89				TRUE		WS_KBD15_CAP		50																			
TRUE		PCIE_EXCARD_D2R_N		17 32 89				TRUE		WS_KBD16_NUM		50																			
TRUE		PP3V3_S3_EXCARD_SWITCH		32				TRUE		WS_KBD17		50																			
TRUE		PP3V3_S0_EXCARD_SWITCH		32				TRUE		WS_KBD18		50																			
TRUE		PP1V5_S0_EXCARD_SWITCH		32			4 TPS	TRUE		WS_KBD19		50		C																	
TRUE		PLT_RESET_SWITCH_L		32				TRUE		WS_KBD20		50																			
TRUE		EXCARD_CPPE_L		32																											









CPU Power & Ground

SYNC_MASTER=M87_MLB

SYNC_DATE=10/17/2007

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APPLE INC.

SIZE D

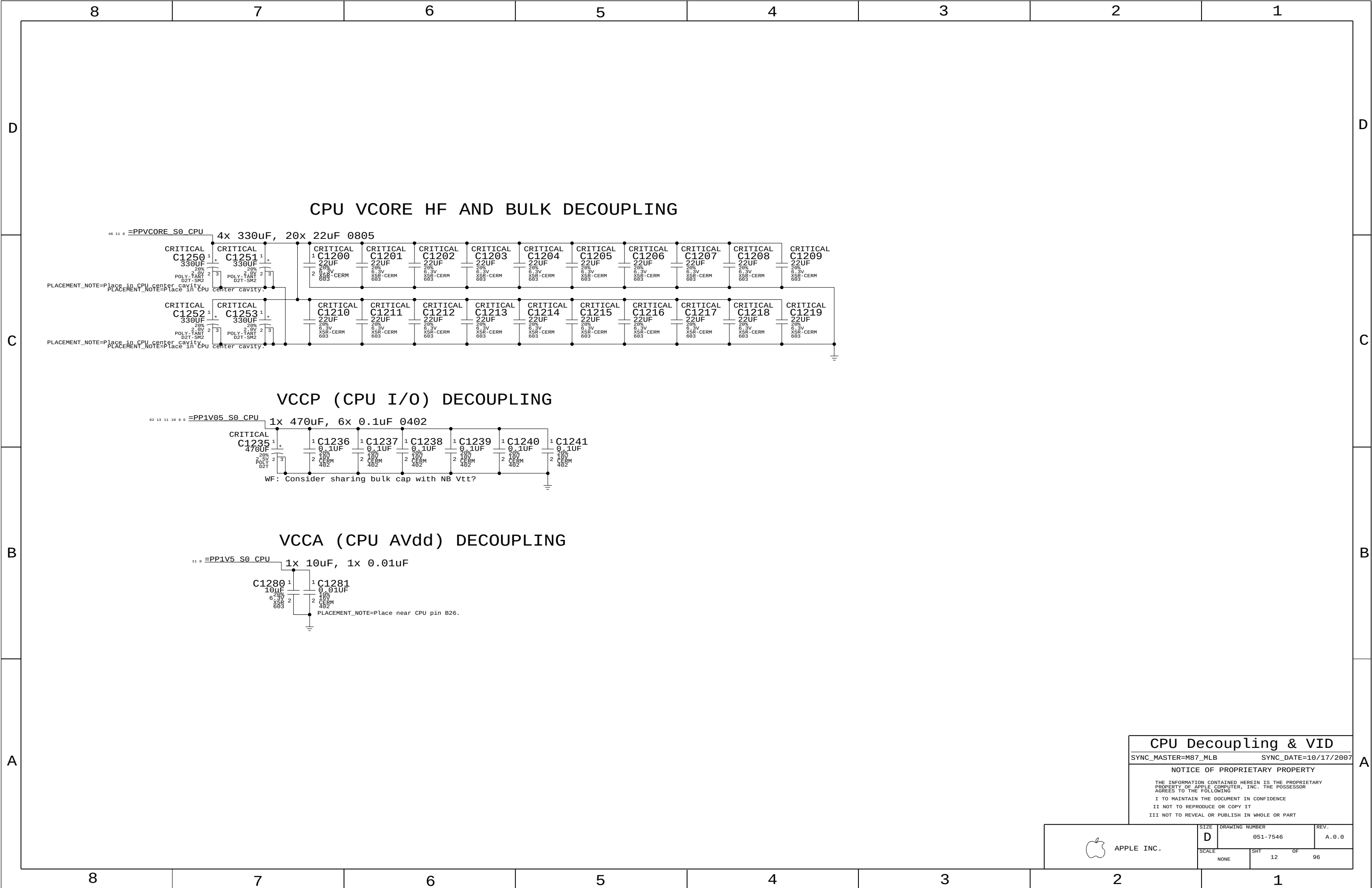
DRAWING NUMBER 051-7546

REV. A.0.0

SCALE NONE

SHT 11

OF 96



CPU Decoupling & VID

SYNC_MASTER=M87_MLB

SYNC_DATE=10/17/2007

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 APPLE INC.	SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 12	OF 96

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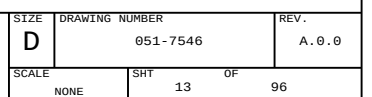
A

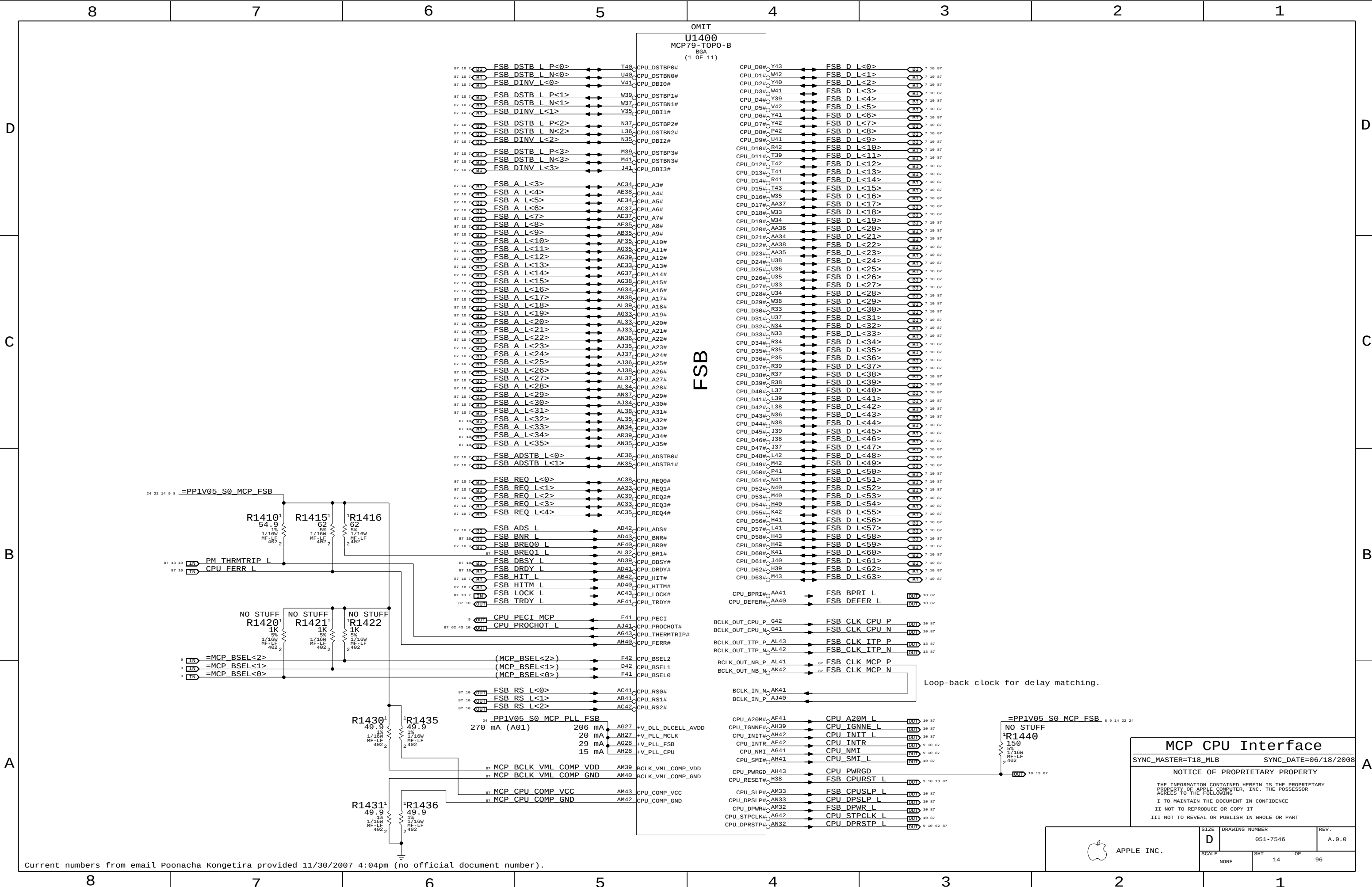
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SCALE		SHT	OF
NONE		14	96



MCP Memory Interface

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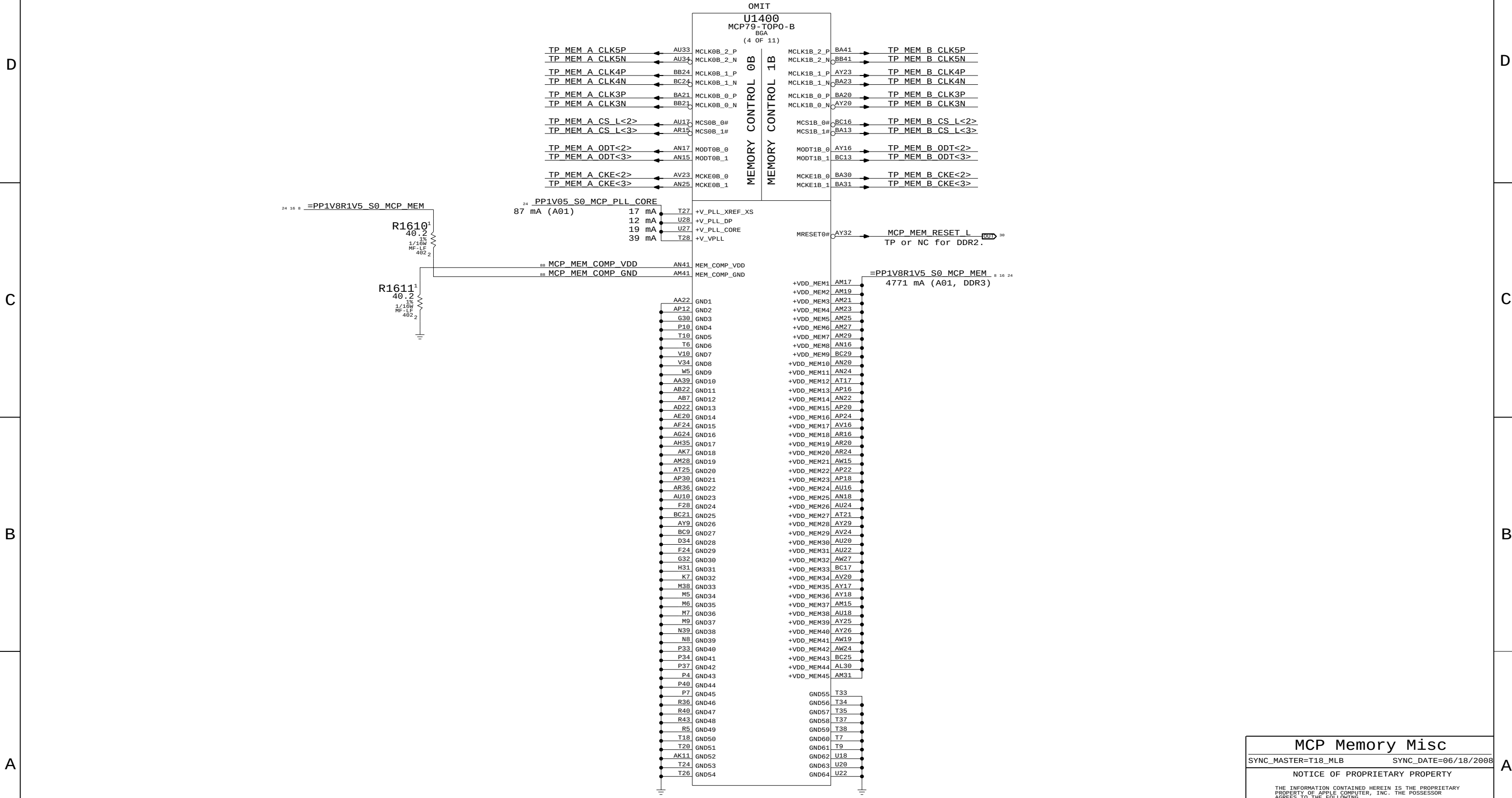
A

D

C

B

A



Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

MCP Memory Misc

SYNC_MASTER=T18_MLB SYNC_DATE=06/18/2008

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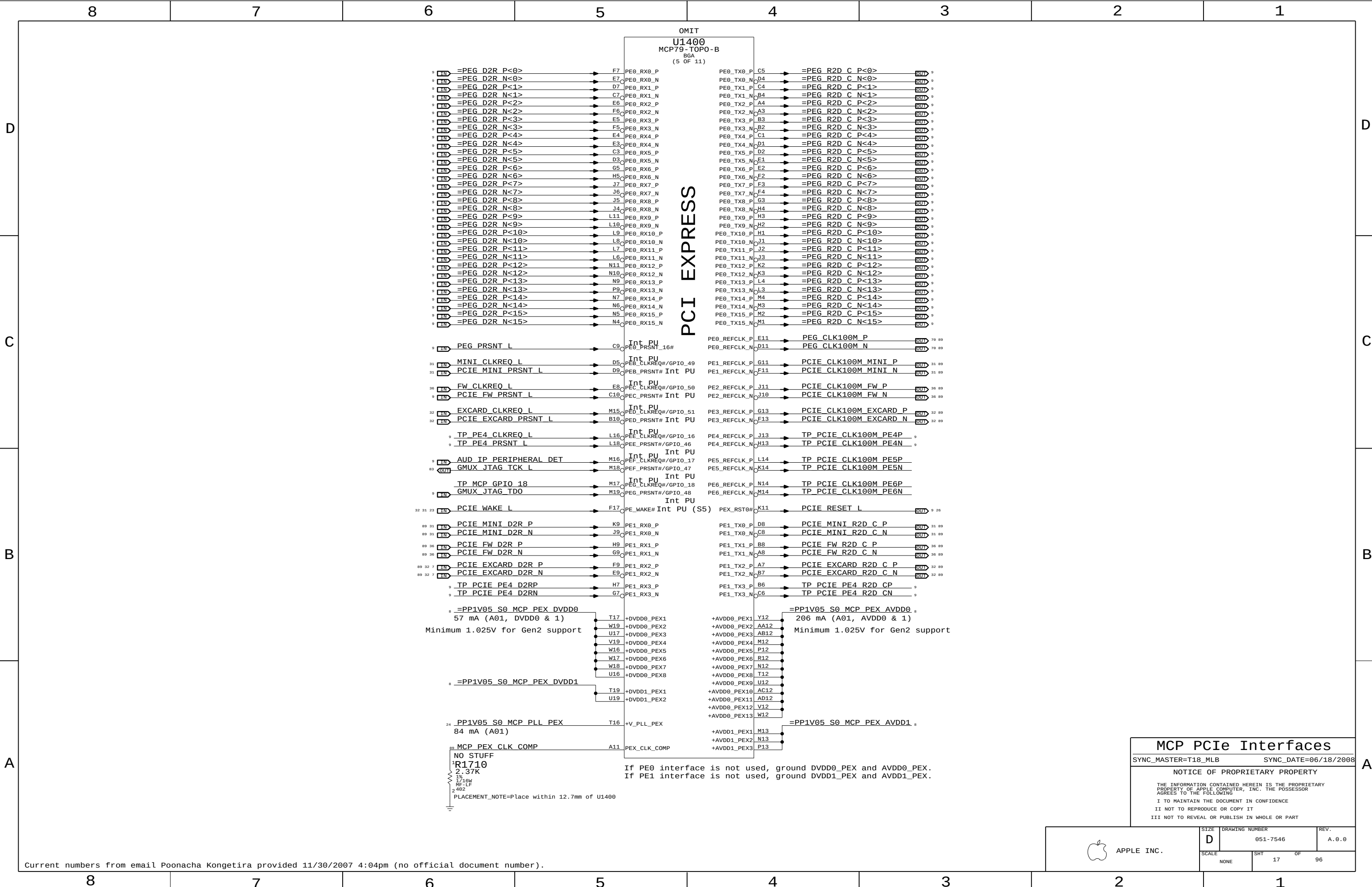
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 APPLE INC.	SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 16	OF 96



OMIT
U1400
MCP79-TOPO-B
BGA
(5 OF 11)

PCI EXPRESS

PE0_RX0_P F7
PE0_RX0_N E7
PE0_RX1_P D7
PE0_RX1_N C7
PE0_RX2_P E6
PE0_RX2_N F6
PE0_RX3_P E5
PE0_RX3_N F5
PE0_RX4_P E4
PE0_RX4_N E3
PE0_RX5_P C3
PE0_RX5_N D3
PE0_RX6_P G5
PE0_RX6_N H5
PE0_RX7_P J7
PE0_RX7_N J6
PE0_RX8_P J5
PE0_RX8_N J4
PE0_RX9_P L11
PE0_RX9_N L10
PE0_RX10_P L9
PE0_RX10_N L8
PE0_RX11_P L7
PE0_RX11_N L6
PE0_RX12_P N11
PE0_RX12_N N10
PE0_RX13_P N9
PE0_RX13_N P9
PE0_RX14_P N7
PE0_RX14_N N6
PE0_RX15_P N5
PE0_RX15_N N4

Int PU
PE0_PRSTNT# Int PU

Int PU
PEB_CLKREQ#/GPIO_49 Int PU

Int PU
PEC_CLKREQ#/GPIO_50 Int PU

Int PU
PEO_CLKREQ#/GPIO_51 Int PU

Int PU
PEE_CLKREQ#/GPIO_16 Int PU

Int PU
PEF_CLKREQ#/GPIO_17 Int PU

Int PU
PEG_CLKREQ#/GPIO_18 Int PU

Int PU
PE_WAKE# Int PU (S5)

PE1_RX0_P K9
PE1_RX0_N J9
PE1_RX1_P H9
PE1_RX1_N G9
PE1_RX2_P F9
PE1_RX2_N E9
PE1_RX3_P H7
PE1_RX3_N G7

+DVDD0_PEX1 T17
+DVDD0_PEX2 W19
+DVDD0_PEX3 U17
+DVDD0_PEX4 V19
+DVDD0_PEX5 W16
+DVDD0_PEX6 W17
+DVDD0_PEX7 W18
+DVDD0_PEX8 U16
+DVDD1_PEX1 T19
+DVDD1_PEX2 U19
+V_PLL_PEX T16
PEX_CLK_COMP A11

PE0_TX0_P C5
PE0_TX0_N D4
PE0_TX1_P C4
PE0_TX1_N B4
PE0_TX2_P A4
PE0_TX2_N A3
PE0_TX3_P B3
PE0_TX3_N B2
PE0_TX4_P C1
PE0_TX4_N D1
PE0_TX5_P D2
PE0_TX5_N E1
PE0_TX6_P E2
PE0_TX6_N F2
PE0_TX7_P F3
PE0_TX7_N F4
PE0_TX8_P G3
PE0_TX8_N H4
PE0_TX9_P H3
PE0_TX9_N H2
PE0_TX10_P H1
PE0_TX10_N J1
PE0_TX11_P J2
PE0_TX11_N J3
PE0_TX12_P K2
PE0_TX12_N K3
PE0_TX13_P L4
PE0_TX13_N L3
PE0_TX14_P M4
PE0_TX14_N M3
PE0_TX15_P M2
PE0_TX15_N M1

=PEG R2D C P<0>
=PEG R2D C N<0>
=PEG R2D C P<1>
=PEG R2D C N<1>
=PEG R2D C P<2>
=PEG R2D C N<2>
=PEG R2D C P<3>
=PEG R2D C N<3>
=PEG R2D C P<4>
=PEG R2D C N<4>
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=PEG R2D C P<14>
=PEG R2D C N<14>
=PEG R2D C P<15>
=PEG R2D C N<15>

PE0_REFCLK_P E11
PE0_REFCLK_N D11

PCIE CLK100M MINI P G11
PCIE CLK100M MINI N F11

PCIE CLK100M FW P J11
PCIE CLK100M FW N J10

PCIE CLK100M EXCARD P G13
PCIE CLK100M EXCARD N F13

TP PCIE CLK100M PE4P J13
TP PCIE CLK100M PE4N H13

TP PCIE CLK100M PE5P L14
TP PCIE CLK100M PE5N K14

TP PCIE CLK100M PE6P N14
TP PCIE CLK100M PE6N M14

PCIE RESET L K11

PCIE MINI R2D C P D8
PCIE MINI R2D C N C8
PCIE FW R2D C P B8
PCIE FW R2D C N A8
PCIE EXCARD R2D C P A7
PCIE EXCARD R2D C N B7
TP PCIE PE4 R2D CP B6
TP PCIE PE4 R2D CN C6

=PP1V05 S0 MCP_PEX_DVDD0
57 mA (A01, DVDD0 & 1)
Minimum 1.025V for Gen2 support

=PP1V05 S0 MCP_PEX_DVDD1

PP1V05 S0 MCP_PLL_PEX
84 mA (A01)

MCP_PEX_CLK_COMP
NO STUFF
R1710
2.37K
1710W
MF-LF
2 462
PLACEMENT_NOTE=Place within 12.7mm of U1400

=PP1V05 S0 MCP_PEX_AVDD0
206 mA (A01, AVDD0 & 1)
Minimum 1.025V for Gen2 support

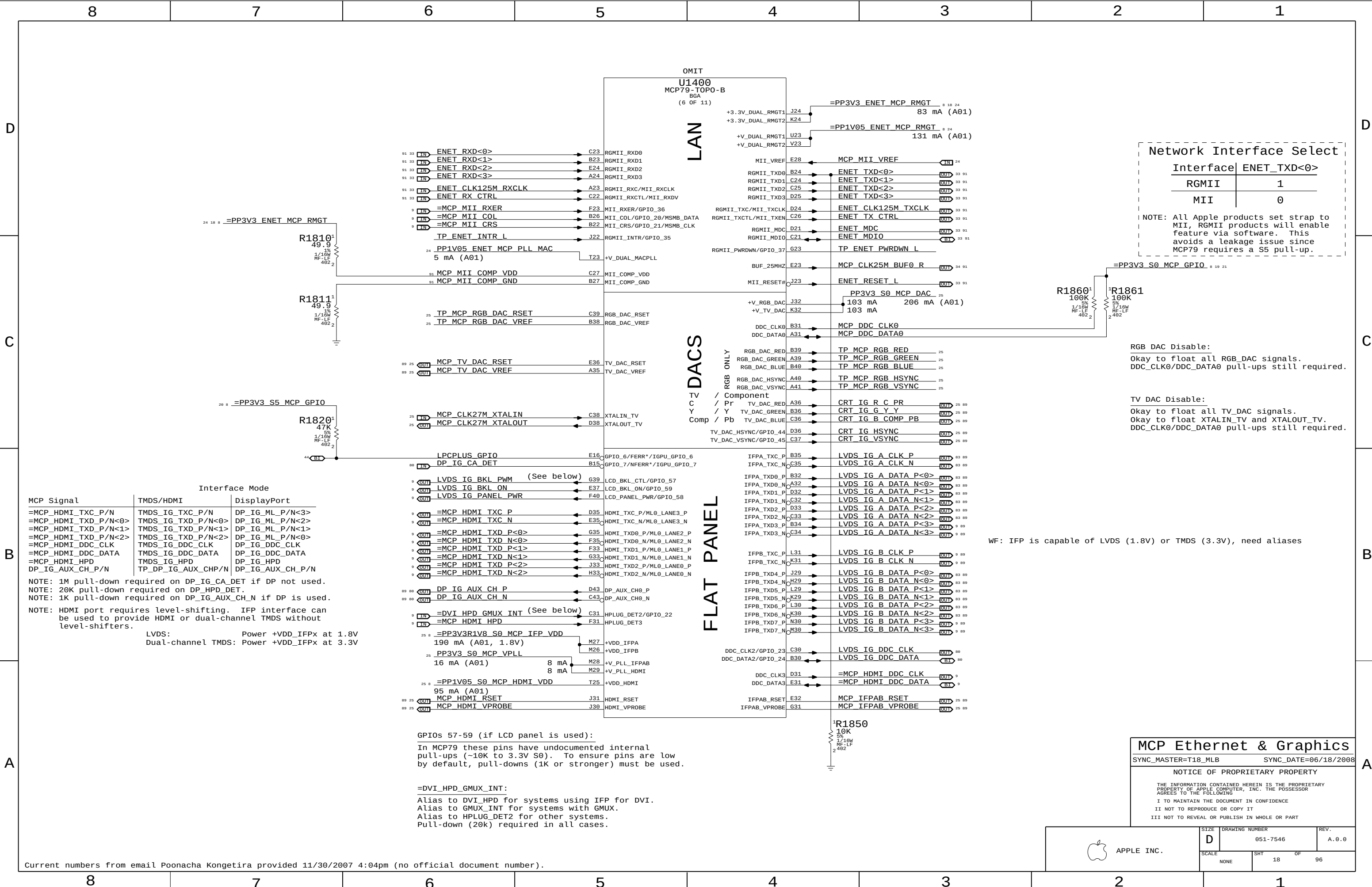
=PP1V05 S0 MCP_PEX_AVDD1

+AVDD0_PEX1 Y12
+AVDD0_PEX2 AA12
+AVDD0_PEX3 AB12
+AVDD0_PEX4 M12
+AVDD0_PEX5 P12
+AVDD0_PEX6 R12
+AVDD0_PEX7 N12
+AVDD0_PEX8 T12
+AVDD0_PEX9 U12
+AVDD0_PEX10 AC12
+AVDD0_PEX11 AD12
+AVDD0_PEX12 V12
+AVDD0_PEX13 W12
+AVDD1_PEX1 M13
+AVDD1_PEX2 N13
+AVDD1_PEX3 P13

If PE0 interface is not used, ground DVDD0_PEX and AVDD0_PEX.
If PE1 interface is not used, ground DVDD1_PEX and AVDD1_PEX.

MCP PCIe Interfaces		
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	SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 17	OF 96



Interface Mode		
MCP Signal	TMDS/HDMI	DisplayPort
=MCP_HDMI_TXC_P/N	TMDS_IG_TXC_P/N	DP_IG_ML_P/N<3>
=MCP_HDMI_TXD_P/N<0>	TMDS_IG_TXD_P/N<0>	DP_IG_ML_P/N<2>
=MCP_HDMI_TXD_P/N<1>	TMDS_IG_TXD_P/N<1>	DP_IG_ML_P/N<1>
=MCP_HDMI_TXD_P/N<2>	TMDS_IG_TXD_P/N<2>	DP_IG_ML_P/N<0>
=MCP_HDMI_DDC_CLK	TMDS_IG_DDC_CLK	DP_IG_DDC_CLK
=MCP_HDMI_DDC_DATA	TMDS_IG_DDC_DATA	DP_IG_DDC_DATA
=MCP_HDMI_HPD	TMDS_IG_HPD	DP_IG_HPD
DP_IG_AUX_CH_P/N	TP_DP_IG_AUX_CH_P/N	DP_IG_AUX_CH_P/N

NOTE: 1M pull-down required on DP_IG_CA_DET if DP not used.
NOTE: 20K pull-down required on DP_HPD_DET.
NOTE: 1K pull-down required on DP_IG_AUX_CH_N if DP is used.
NOTE: HDMI port requires level-shifting. IFP interface can be used to provide HDMI or dual-channel TMDS without level-shifters.

LVDS:	Power +VDD_IFPx at 1.8V
Dual-channel TMDS:	Power +VDD_IFPx at 3.3V

GPIOs 57-59 (if LCD panel is used):
In MCP79 these pins have undocumented internal pull-ups (~10K to 3.3V S0). To ensure pins are low by default, pull-downs (1K or stronger) must be used.

=DVI_HPD_GMUX_INT:
Alias to DVI_HPD for systems using IFP for DVI.
Alias to GMUX_INT for systems with GMUX.
Alias to HPLUG_DET2 for other systems.
Pull-down (20k) required in all cases.

Network Interface Select	
Interface	ENET_TXD<0>
RGMII	1
MII	0

NOTE: All Apple products set strap to MII, RGMII products will enable feature via software. This avoids a leakage issue since MCP79 requires a S5 pull-up.

RGB DAC Disable:
Okay to float all RGB_DAC signals.
DDC_CLK0/DDC_DATA0 pull-ups still required.

TV DAC Disable:
Okay to float all TV_DAC signals.
Okay to float XTALIN_TV and XTALOUT_TV.
DDC_CLK0/DDC_DATA0 pull-ups still required.

WF: IFP is capable of LVDS (1.8V) or TMDS (3.3V), need aliases

MCP Ethernet & Graphics

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SYNC_DATE=06/18/2008

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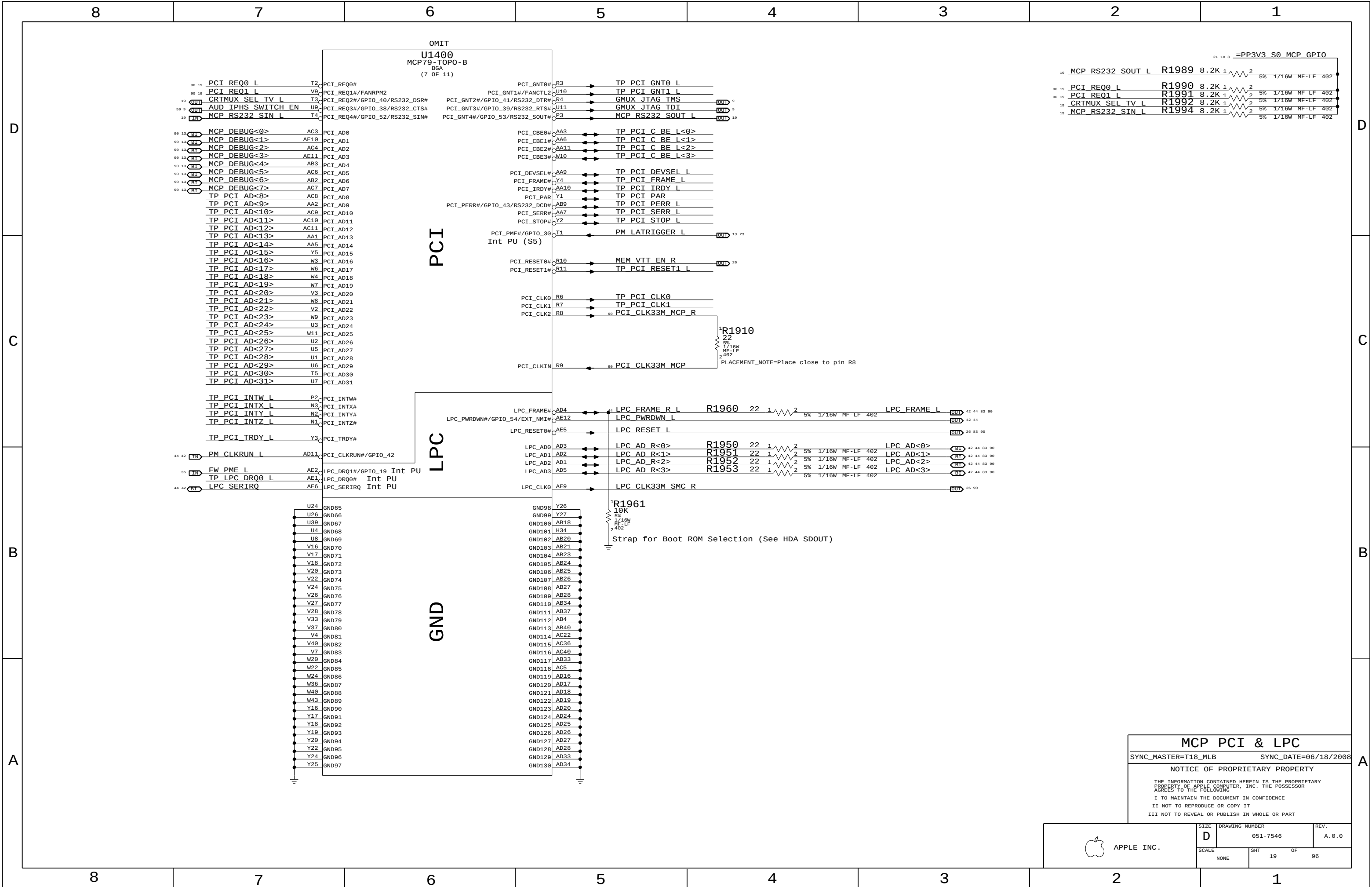
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	SCALE NONE	SHT 18	OF 96



MCP PCI & LPC

SYNC_MASTER=T18_MLB SYNC_DATE=06/18/2008

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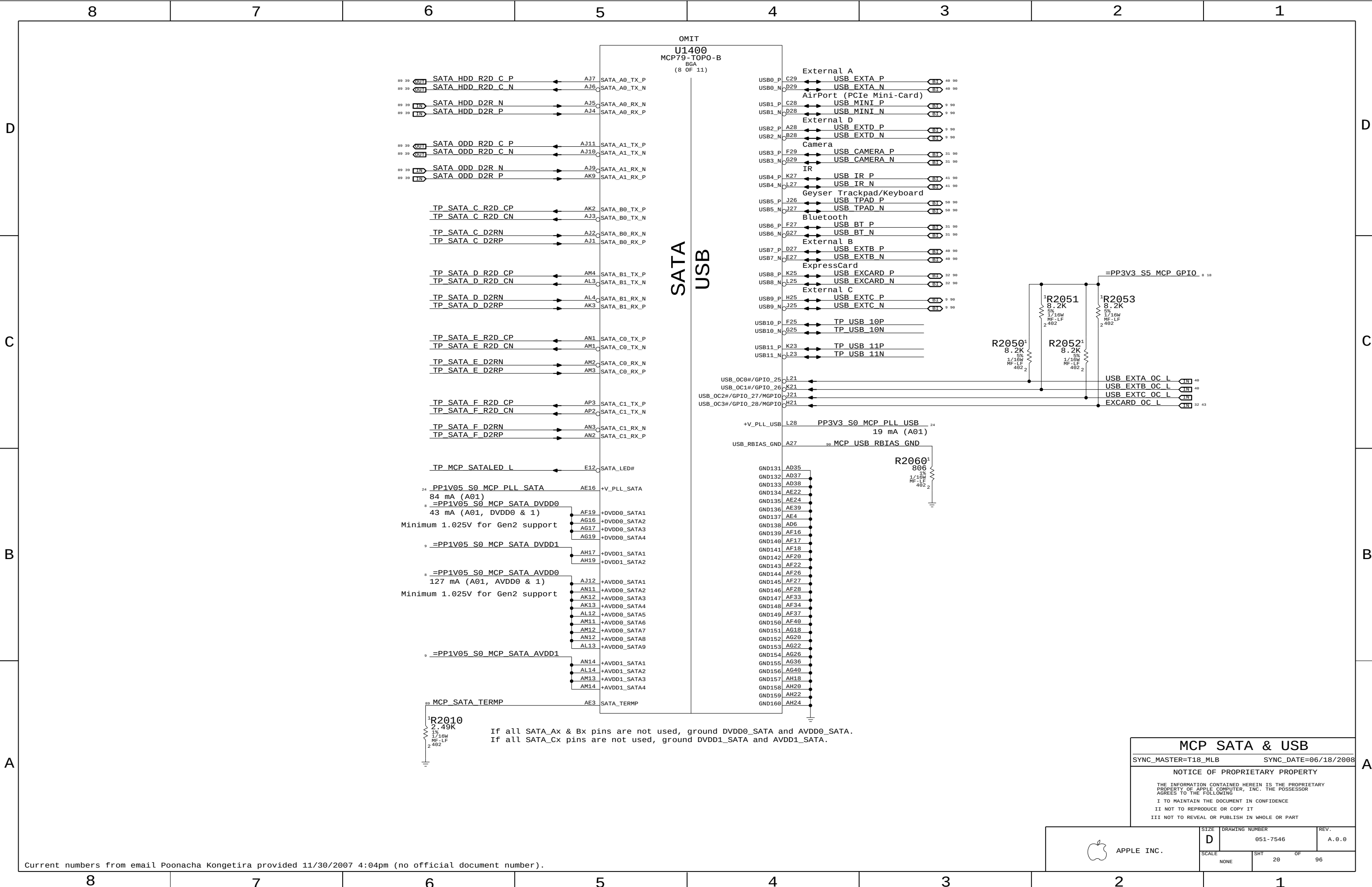
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SCALE		SHT	OF
NONE		19	96



MCP SATA & USB

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SYNC_DATE=06/18/2008

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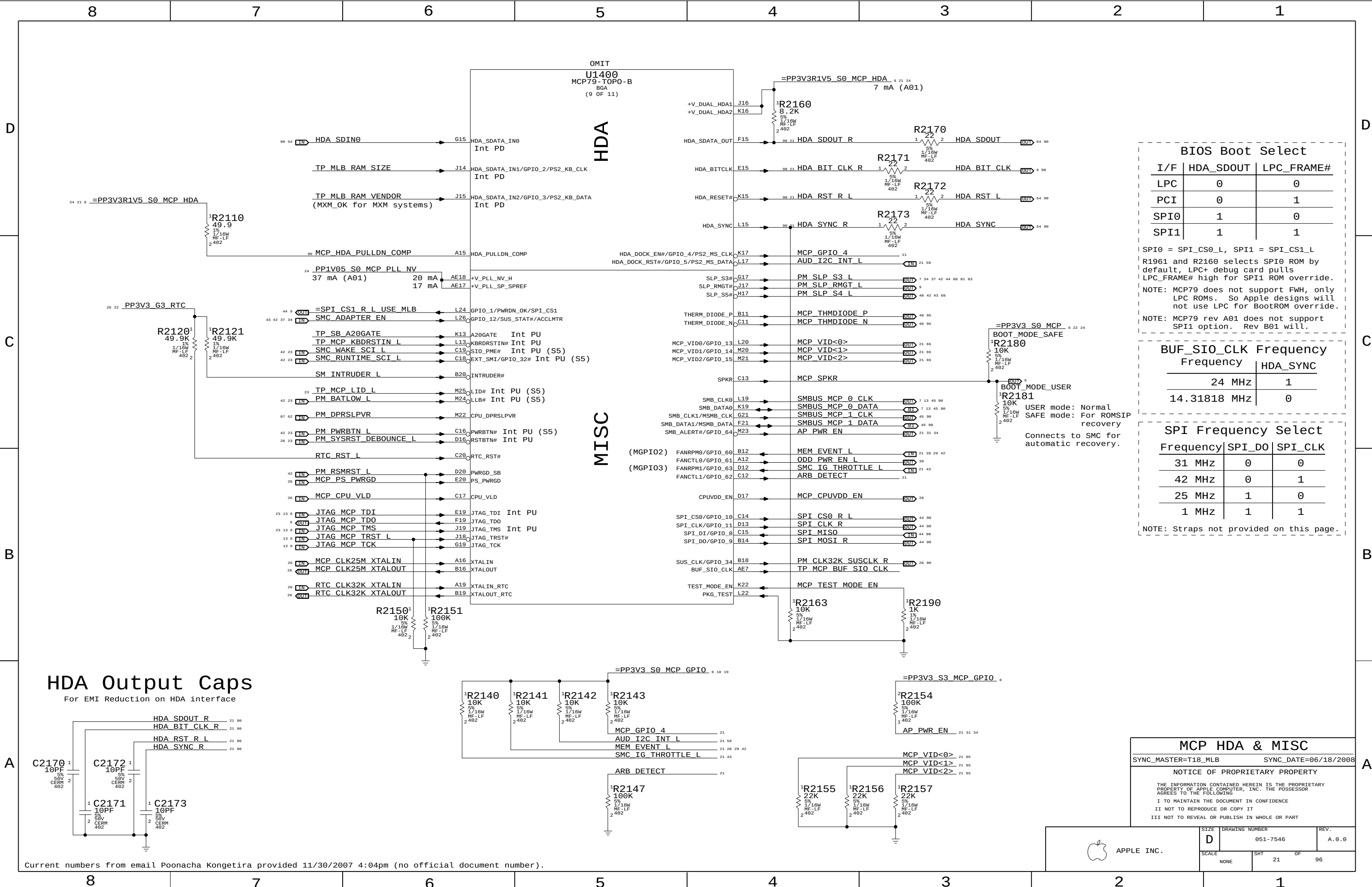
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051-7546		A.0.0
SCALE	SHT	OF
NONE	20	96



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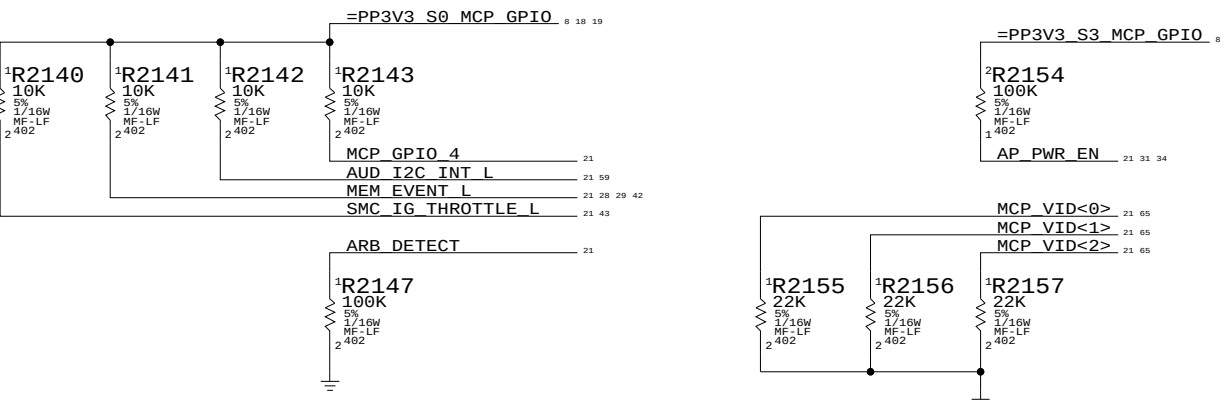
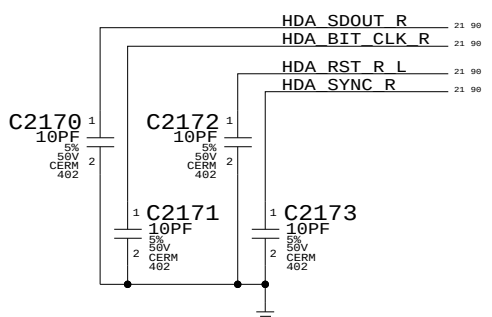
C

B

A

HDA Output Caps

For EMI Reduction on HDA interface



BIOS Boot Select		
I/F	HDA_SDOUT	LPC_FRAME#
LPC	0	0
PCI	0	1
SPI0	1	0
SPI1	1	1

SPI0 = SPI_CS0_L, SPI1 = SPI_CS1_L
R1961 and R2160 selects SPI0 ROM by default, LPC+ debug card pulls LPC_FRAME# high for SPI1 ROM override.
NOTE: MCP79 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.
NOTE: MCP79 rev A01 does not support SPI1 option. Rev B01 will.

BUF_SIO_CLK Frequency		HDA_SYNC
	Frequency	
	24 MHZ	1
	14.31818 MHZ	0

SPI Frequency Select		
Frequency	SPI_D0	SPI_CLK
31 MHZ	0	0
42 MHZ	0	1
25 MHZ	1	0
1 MHZ	1	1

NOTE: Straps not provided on this page.

MCP HDA & MISC

SYNC_MASTER=T18_MLB SYNC_DATE=06/18/2008

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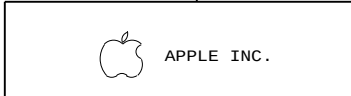
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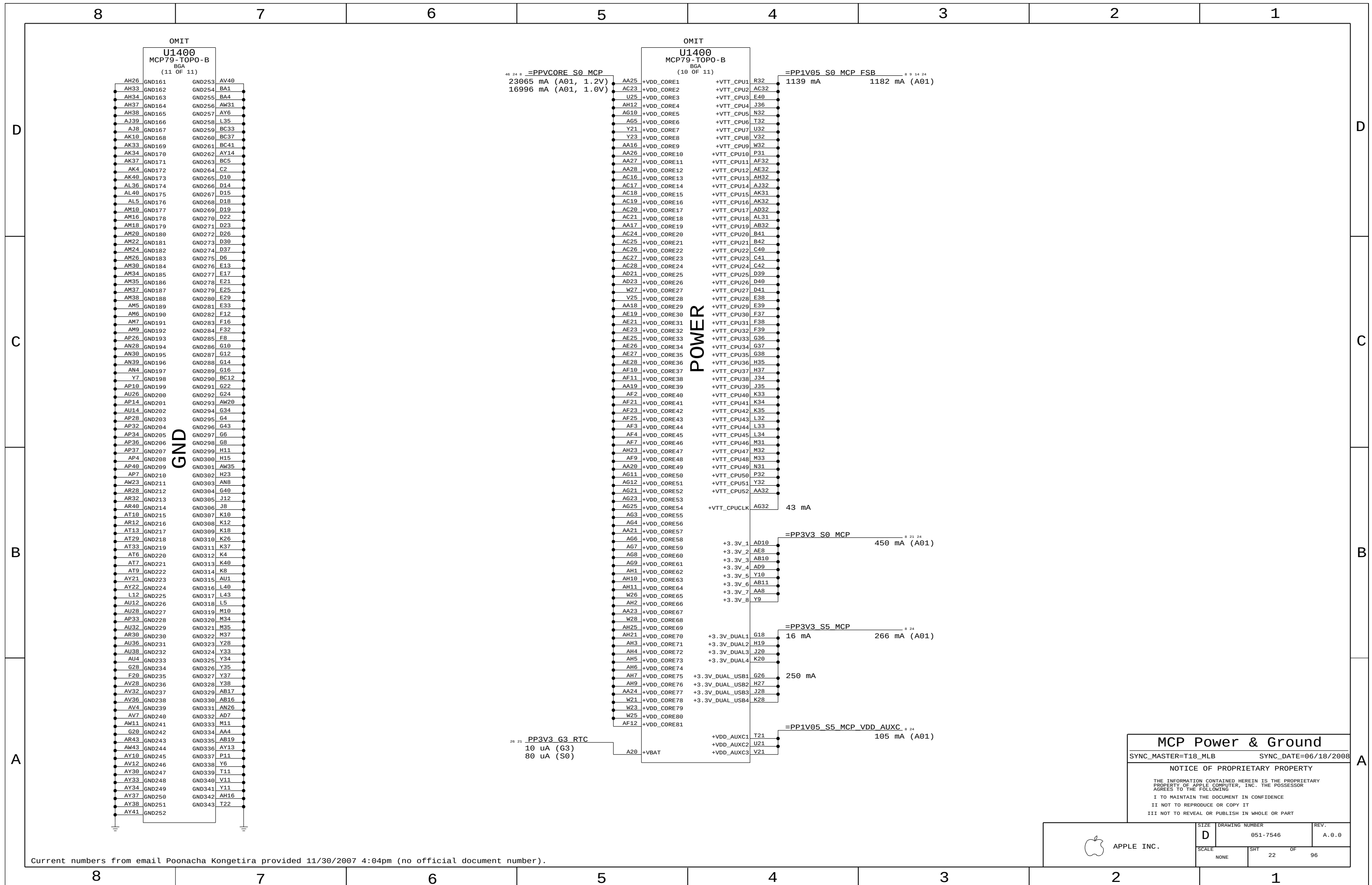
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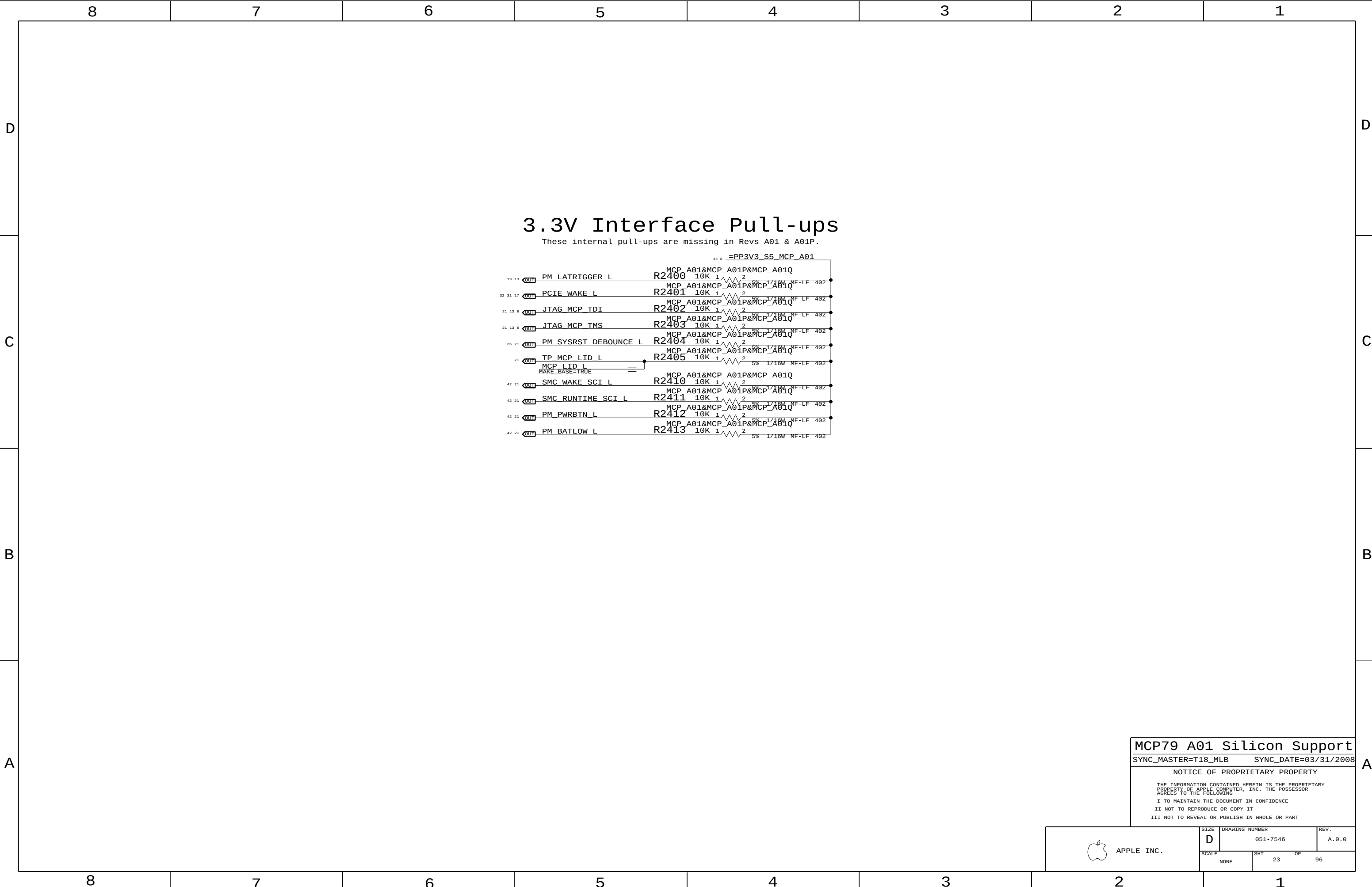
II NOT TO REPRODUCE OR COPY IT

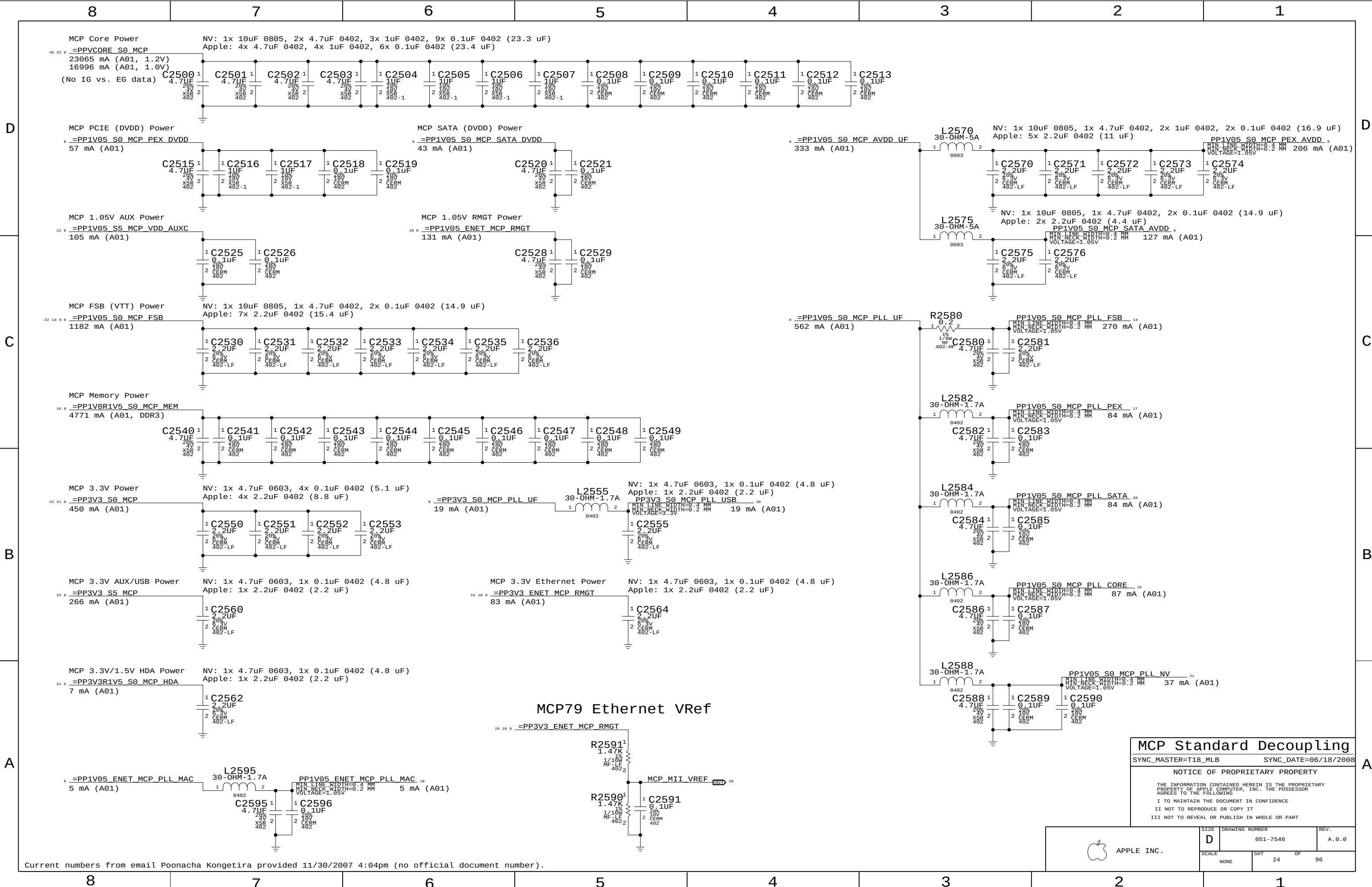
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SCALE	SHT	OF
NONE	21	96









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MCP Standard Decoupling

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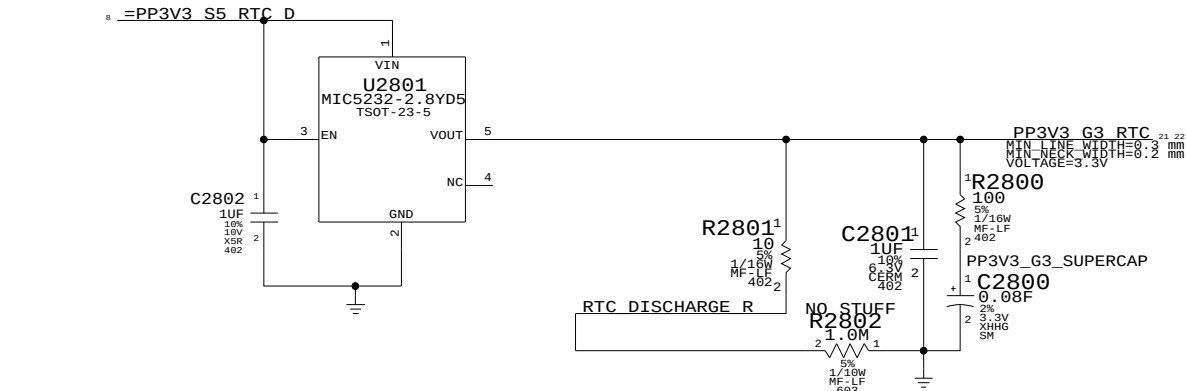
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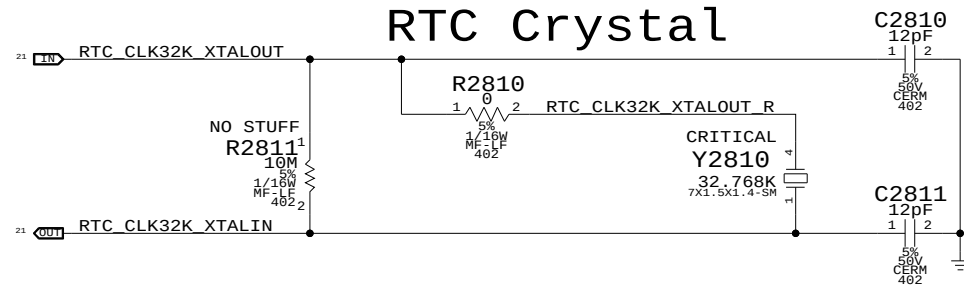
APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	24	96

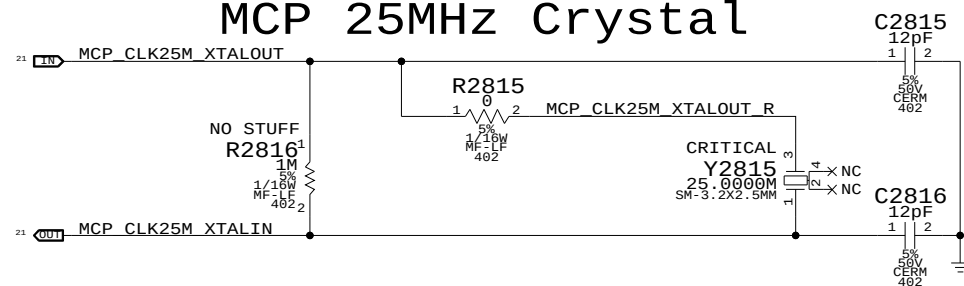
RTC Power Sources



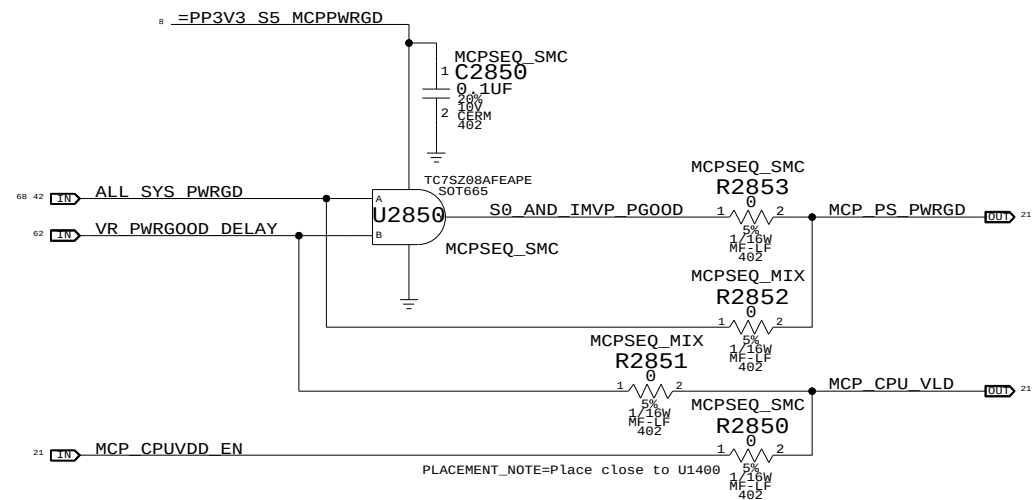
RTC Crystal



MCP 25MHz Crystal



MCP S0 PWRGD & CPU_VLD

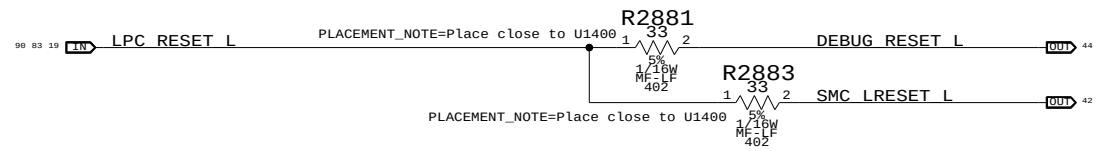


MCPSEQ_SMC represents MCP79 'MLB' power sequencing connections, but results in MCP79 ROMSIP sequence happening after CPU powers up. MCPSEQ_MIX is cross between MLB and internal power sequencing, which results in earlier ROMSIP and MCP FSB I/O interface initialization. SMC 99ms delay from ALL_SYS_PWRGD to INVP_VR_ON plus INVP6 delay for VR_PWRGOOD_DELAY should guarantee CPU_VLD does not go high before CPUVDD_EN (which is 40-100ms after PS_PWRGD assertion).

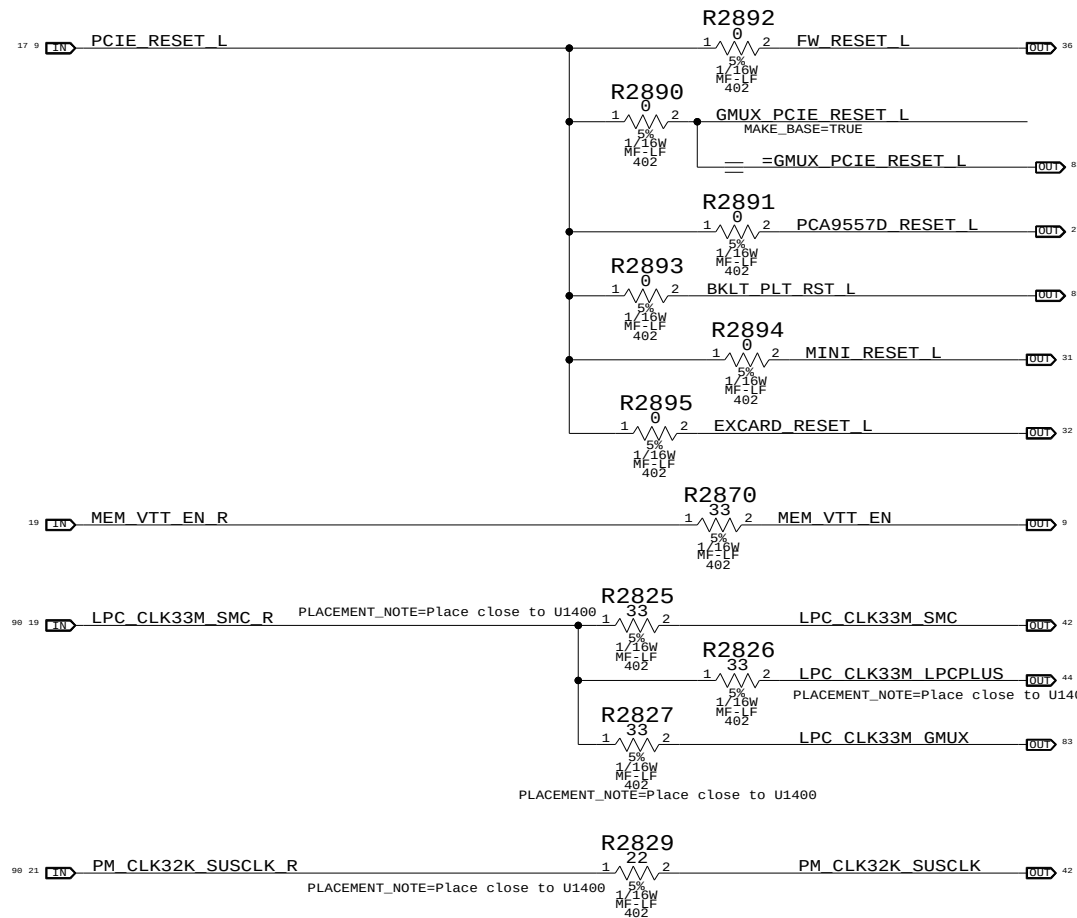
NOTE: If CPU_VLD deasserts during S0 MCP79 will take system to S5 immediately.

Platform Reset Connections

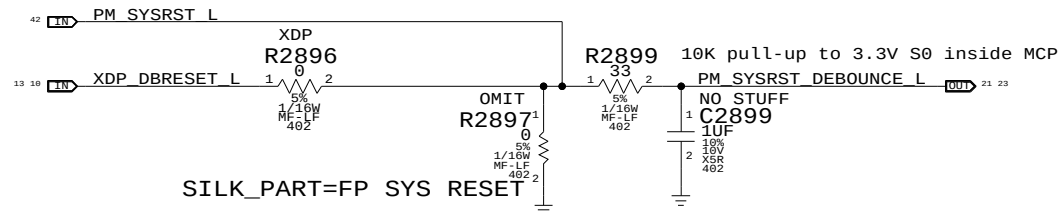
LPC Reset (Unbuffered)



PCIE Reset (Unbuffered)



Reset Button



SB Misc

SYNC_MASTER=T18_MLB	SYNC_DATE=12/17/2007
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SIZE	DRAWING NUMBER	REV.
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D	051-7546	A
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APPLE INC.

SIZE	D
------	---

DRAWING NUMBER

051-7546

SCALE

SMT

REV.

A

Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PP3V3_S5_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- VREFMRGN
- NO_VREFMRGN

DAC channel
Min DAC code
Max DAC code
Max sink I
Max source I
Nominal Vref
Min Vref
Max Vref
Vref Stepping
(per DAC LSB)

MEM A VREF DQ
A
0x00
0x87
-3.75 mA
5 mA
0.75 V
0.375 V
1.250 V
6.5 mV

MEM A VREF CA
B
0x00
0x87
-3.75 mA
5 mA
0.75 V
0.375 V
1.250 V
6.5 mV

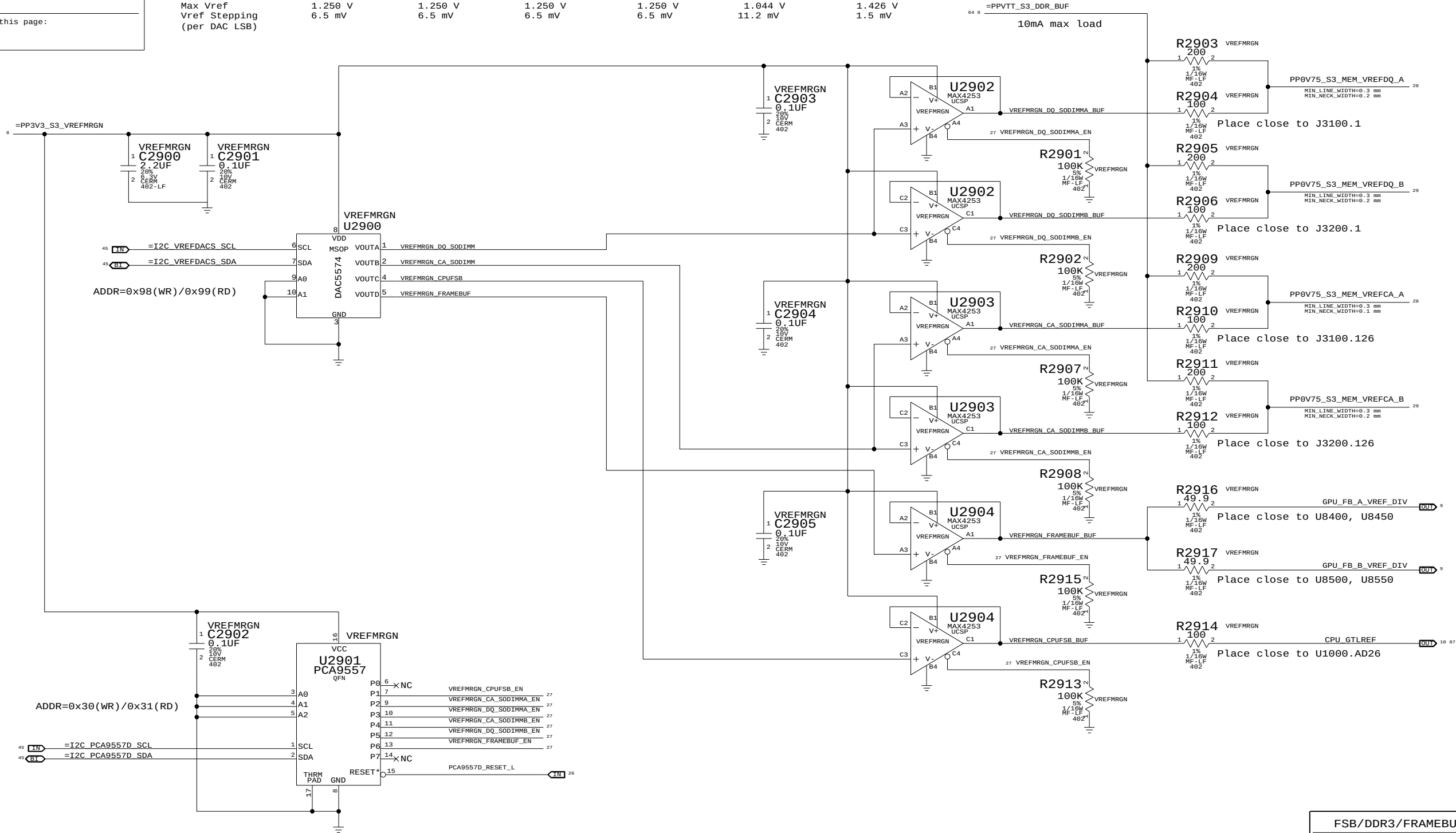
MEM B VREF DQ
A
0x00
0x87
-3.75 mA
5 mA
0.75 V
0.375 V
1.250 V
6.5 mV

MEM B VREF CA
B
0x00
0x87
-3.75 mA
5 mA
0.75 V
0.375 V
1.250 V
6.5 mV

CPU FSB VREF
C
0x00
0x55
-0.91 mA
0.52 mA
0.70 V
0.091 V
1.044 V
11.2 mV

FRAME BUFFER VREF
D
0x00
0xFF
-59.04 mA
51.15 mA
1.248 V
1.042 V
1.426 V
1.5 mV

S0-DIMM A and S0-DIMM B Vref settings should be margined separately (i.e. not simultaneously) due to current limitation of TPS51116 regulator.



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2903	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2905	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2909	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2911	CRITICAL	NO_VREFMRGN

FSB/DDR3/FRAMEBUF Vref Margining

SYNC_MASTER=DDR SYNC_DATE=07/22/2008

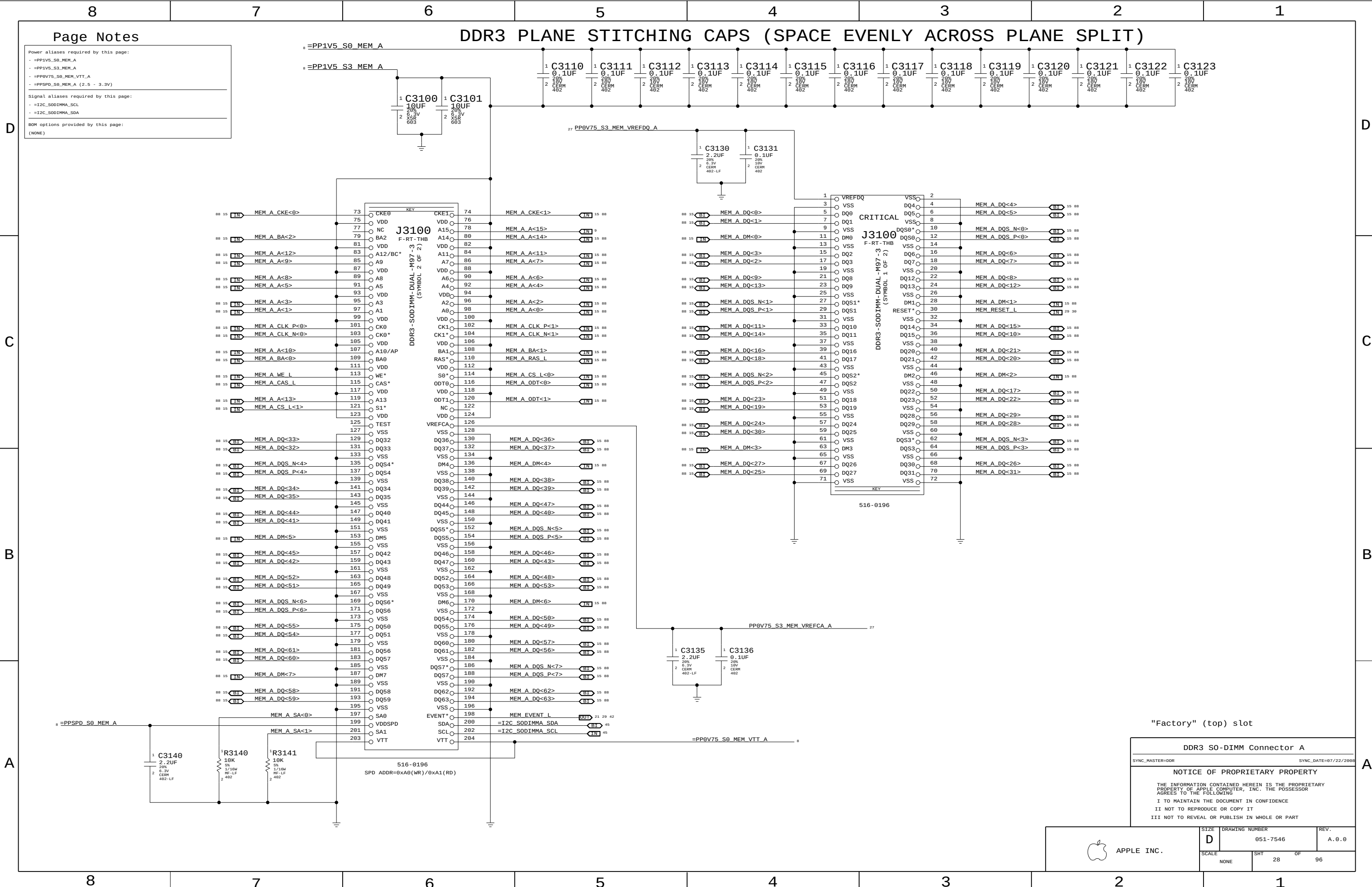
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APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	27	96



Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

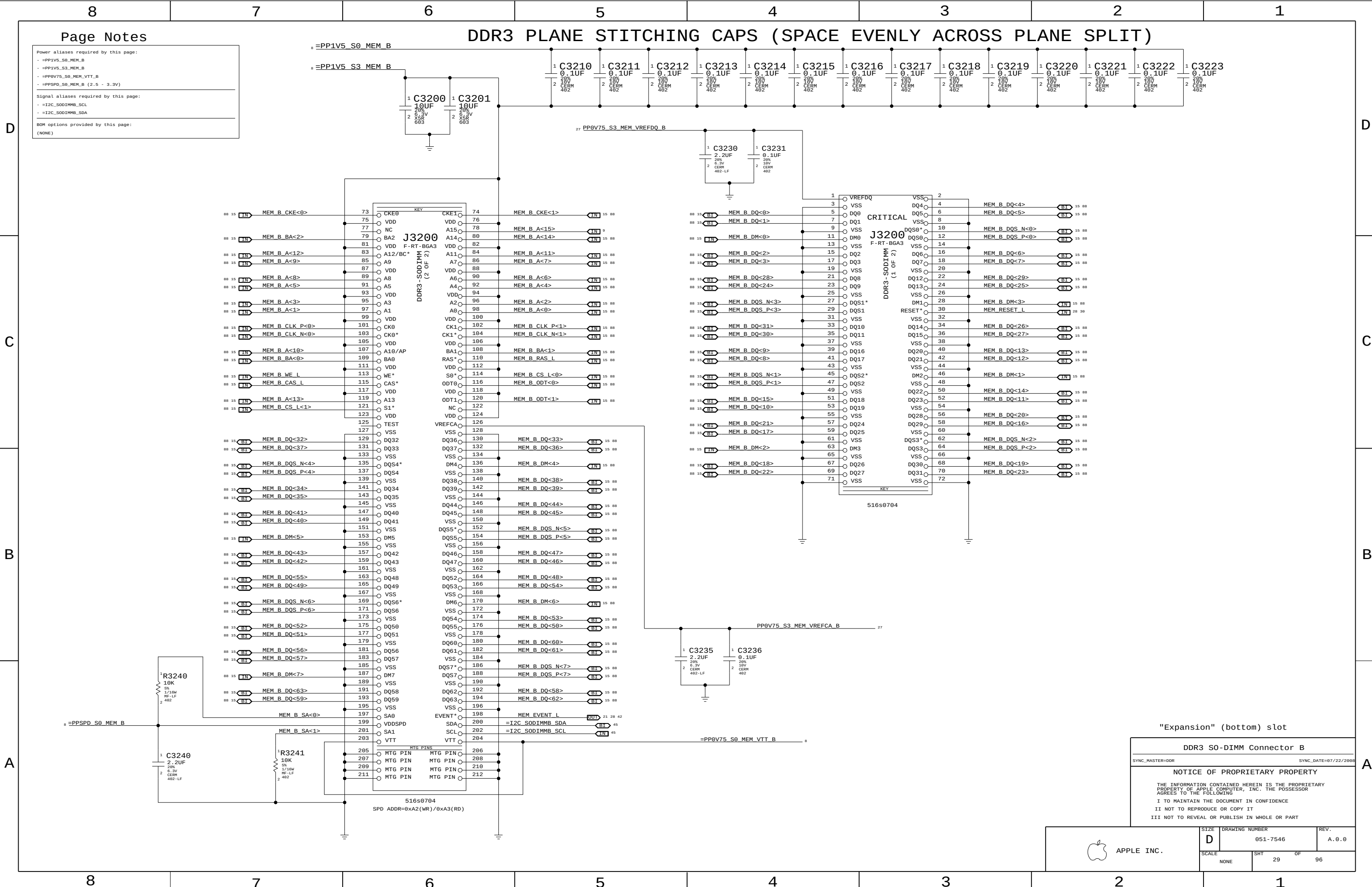
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

"Factory" (top) slot

DDR3 S0-DIMM Connector A		
SYNC_MASTER=DDR		SYNC_DATE=07/22/2008
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Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMB_SCL
- =I2C_SODIMMB_SDA

BOM options provided by this page:
(NONE)

DDR3 PLANE STITCHING CAPS (SPACE EVENLY ACROSS PLANE SPLIT)

"Expansion" (bottom) slot

DDR3 S0-DIMM Connector B

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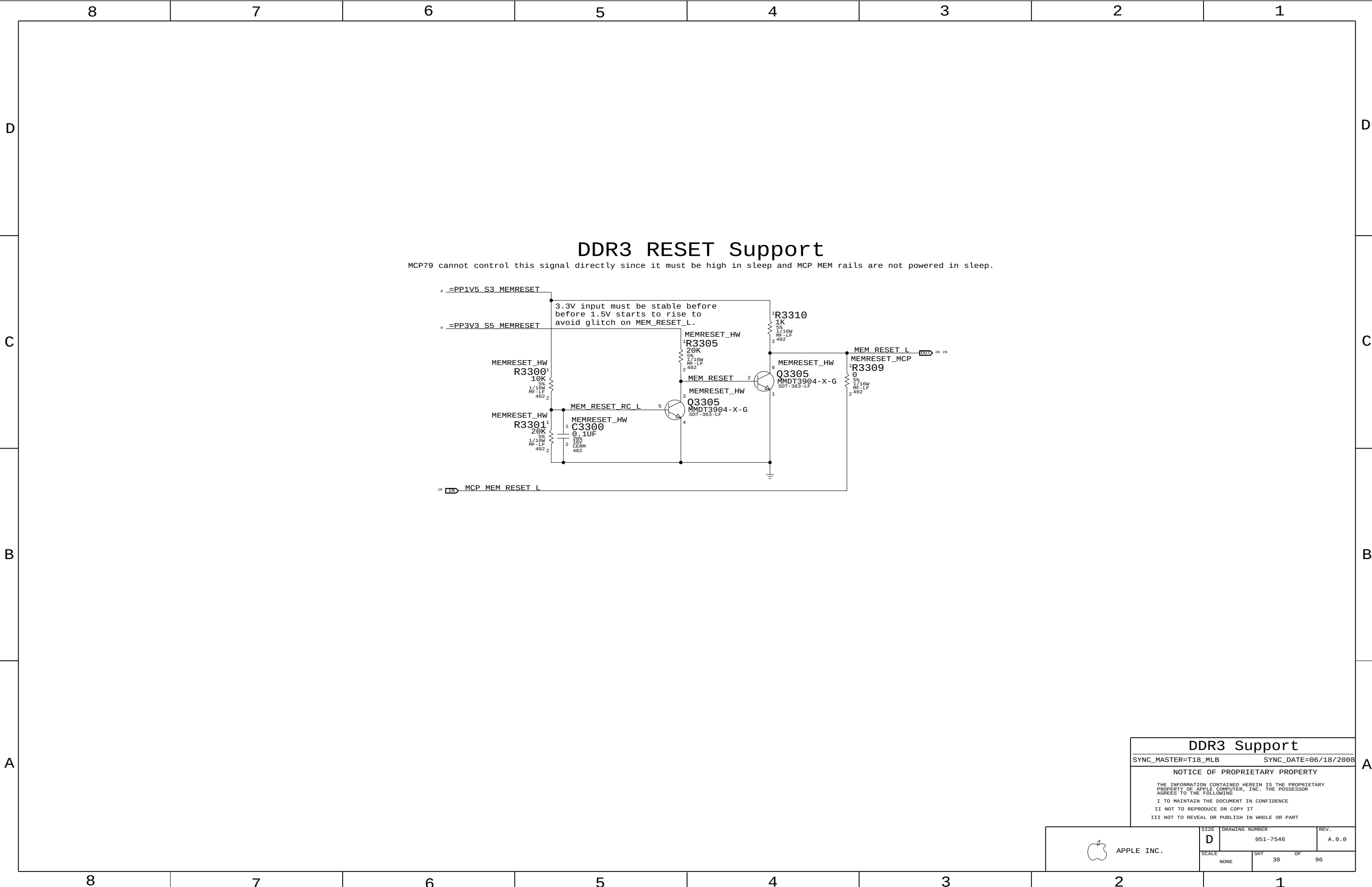
II NOT TO REPRODUCE OR COPY IT

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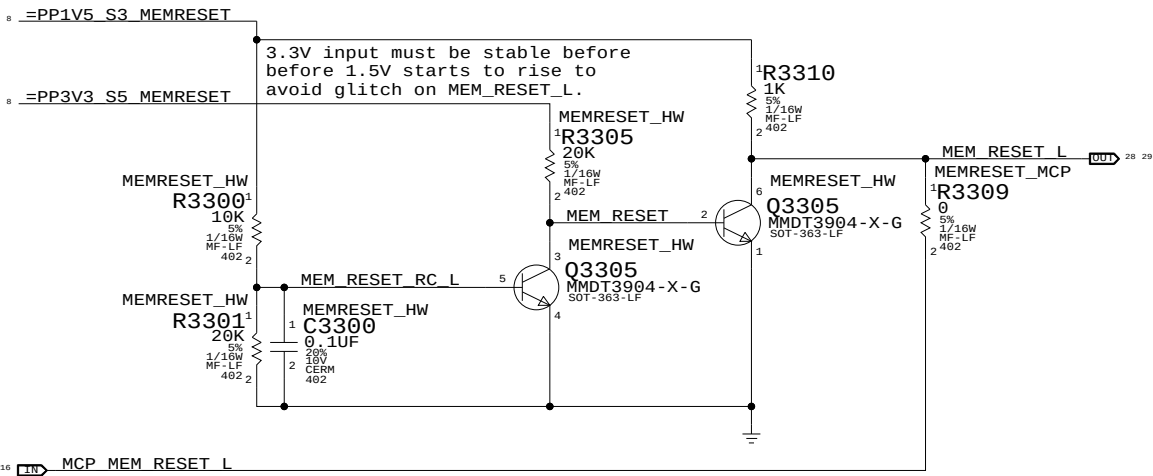
APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	29	96



DDR3 RESET Support

MCP79 cannot control this signal directly since it must be high in sleep and MCP MEM rails are not powered in sleep.



DDR3 Support

SYNC_MASTER=T18_MLB SYNC_DATE=06/18/2008

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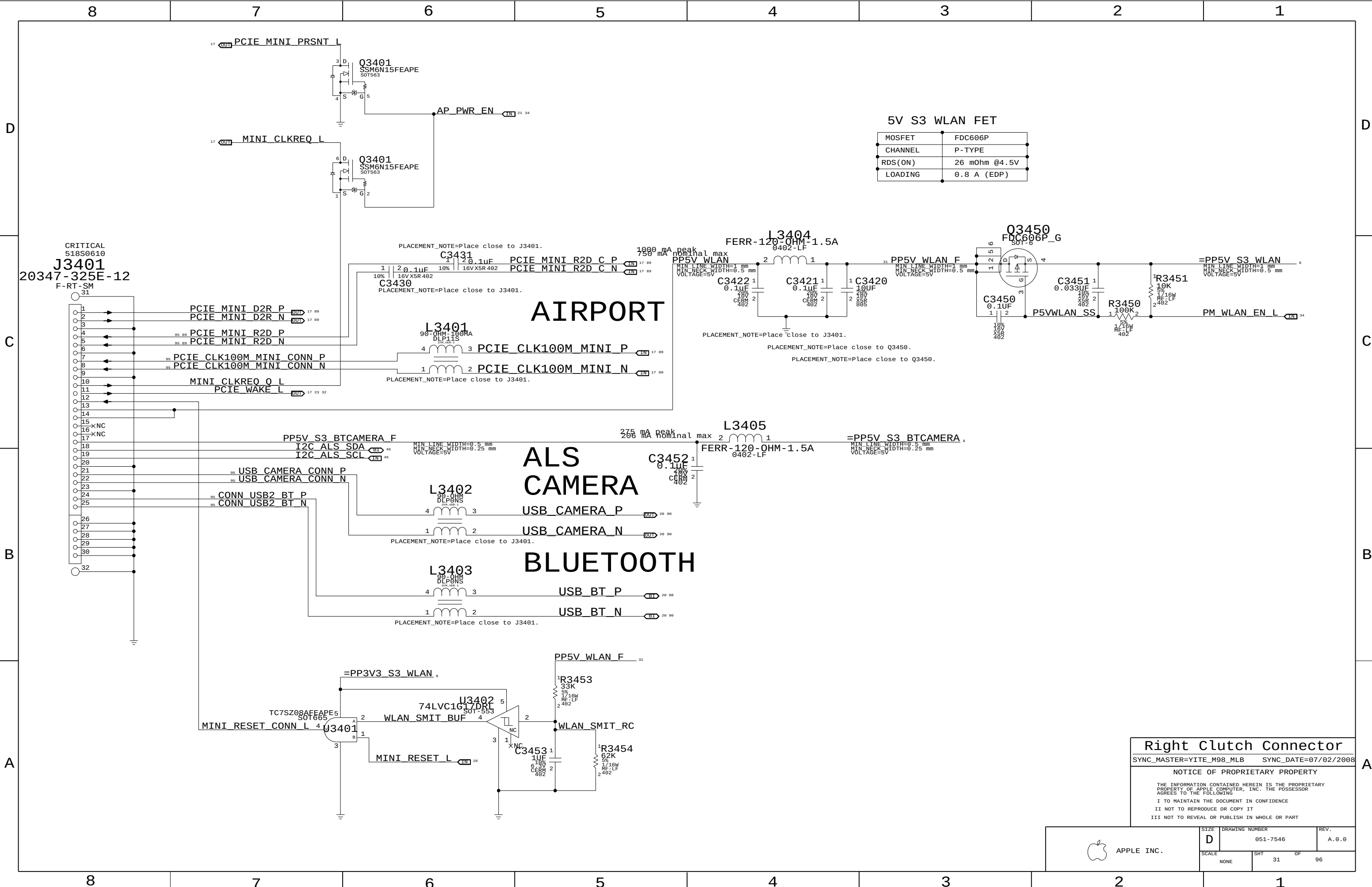
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SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	30	96



5V S3 WLAN FET

MOSFET	FDC606P
CHANNEL	P-TYPE
RDS(ON)	26 mOhm @4.5V
LOADING	0.8 A (EDP)

AIRPORT

ALS
CAMERA

BLUETOOTH

Right Clutch Connector

SYNC_MASTER=YITE_M98_MLB SYNC_DATE=07/02/2008

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APPLE INC.

SIZE

D

DRAWING NUMBER

051-7546

REV.

A.0.0

SCALE

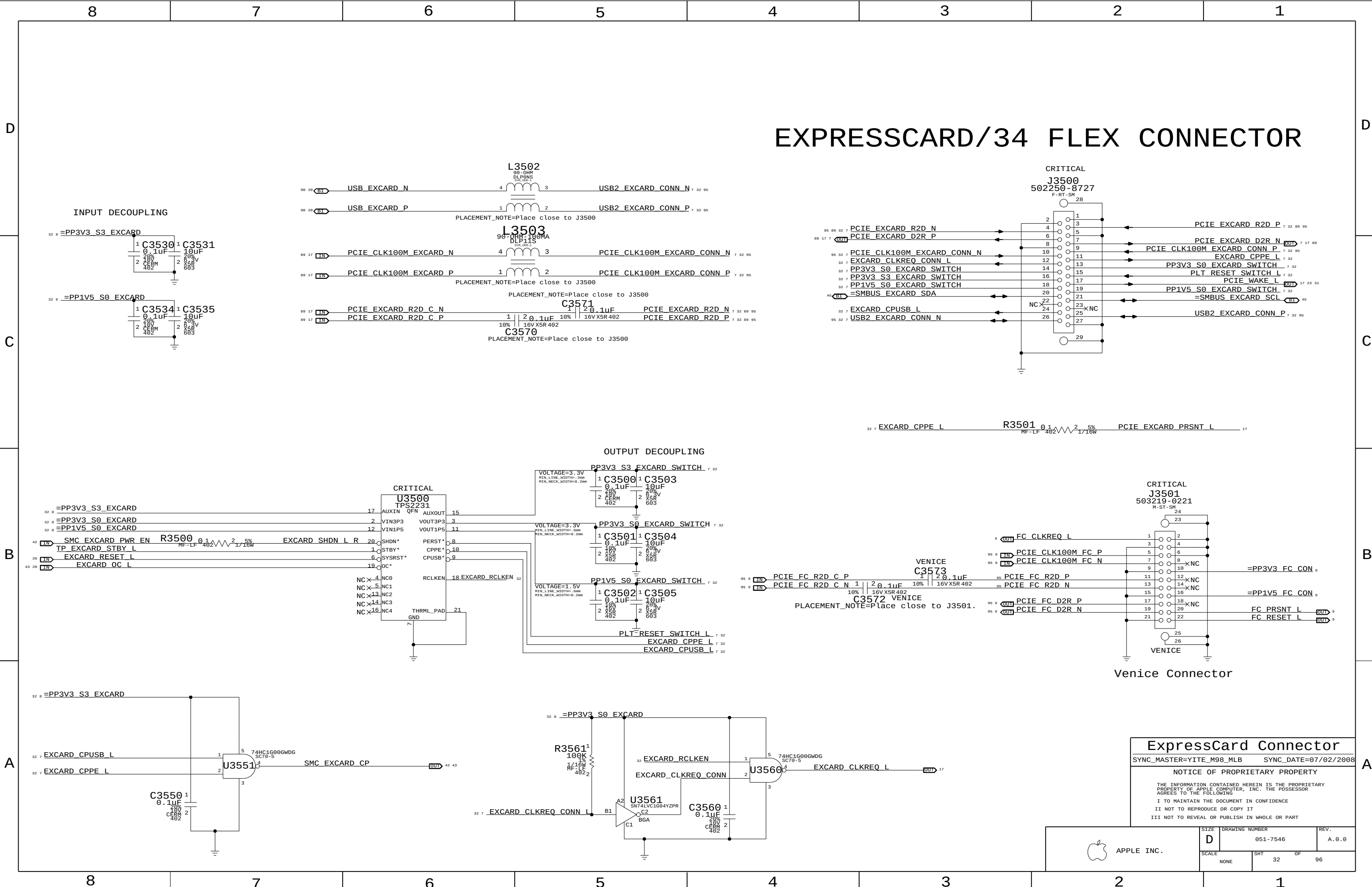
NONE

SHT

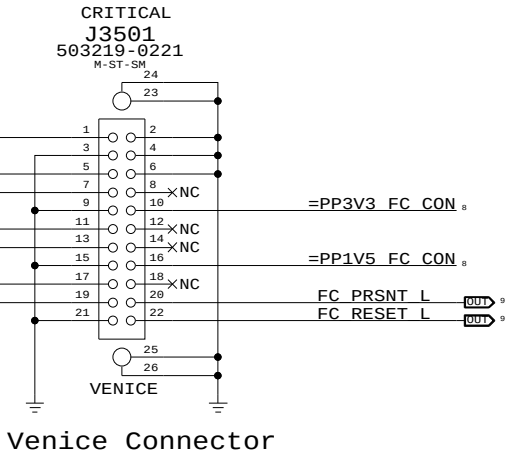
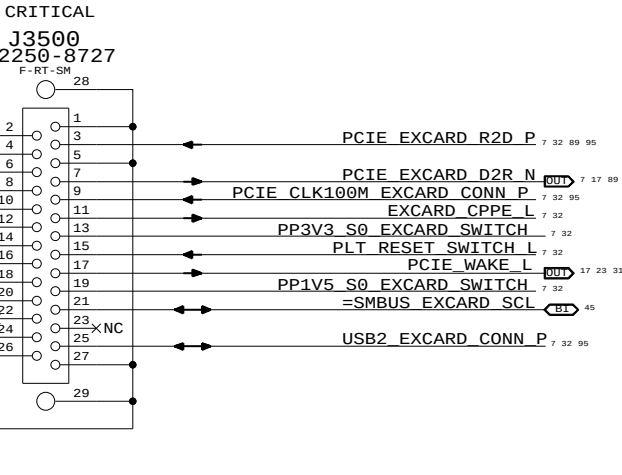
31

OF

96



EXPRESSCARD/34 FLEX CONNECTOR



ExpressCard Connector

SYNC_MASTER=YITE_M98_MLB SYNC_DATE=07/02/2008

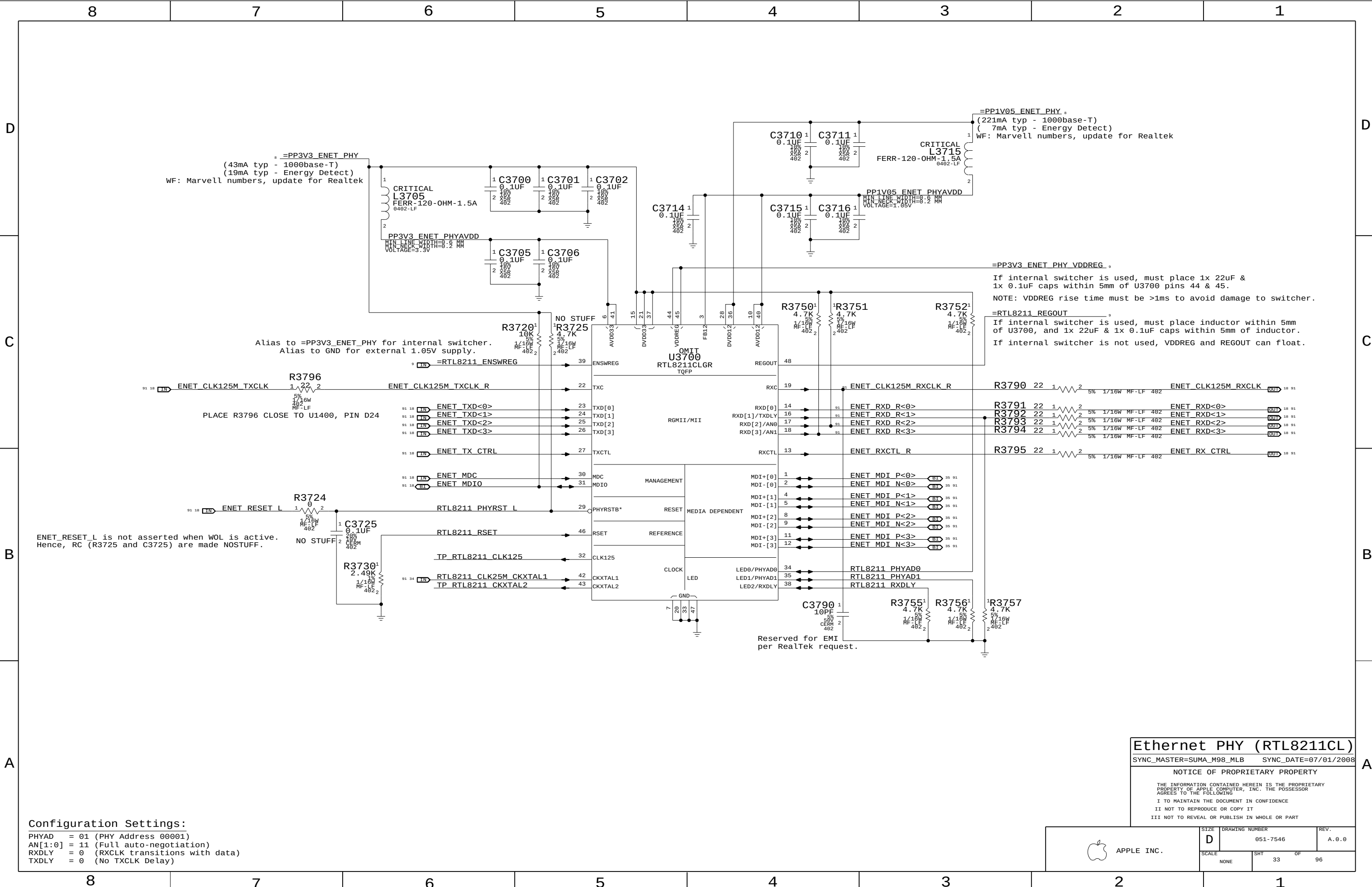
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SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	32	96



Configuration Settings:
PHYAD = 01 (PHY Address 00001)
AN[1:0] = 11 (Full auto-negotiation)
RXDLY = 0 (RXCLK transitions with data)
TXDLY = 0 (No TXCLK Delay)

Ethernet PHY (RTL8211CL)

SYNC_MASTER=SUMA_M98_MLB SYNC_DATE=07/01/2008

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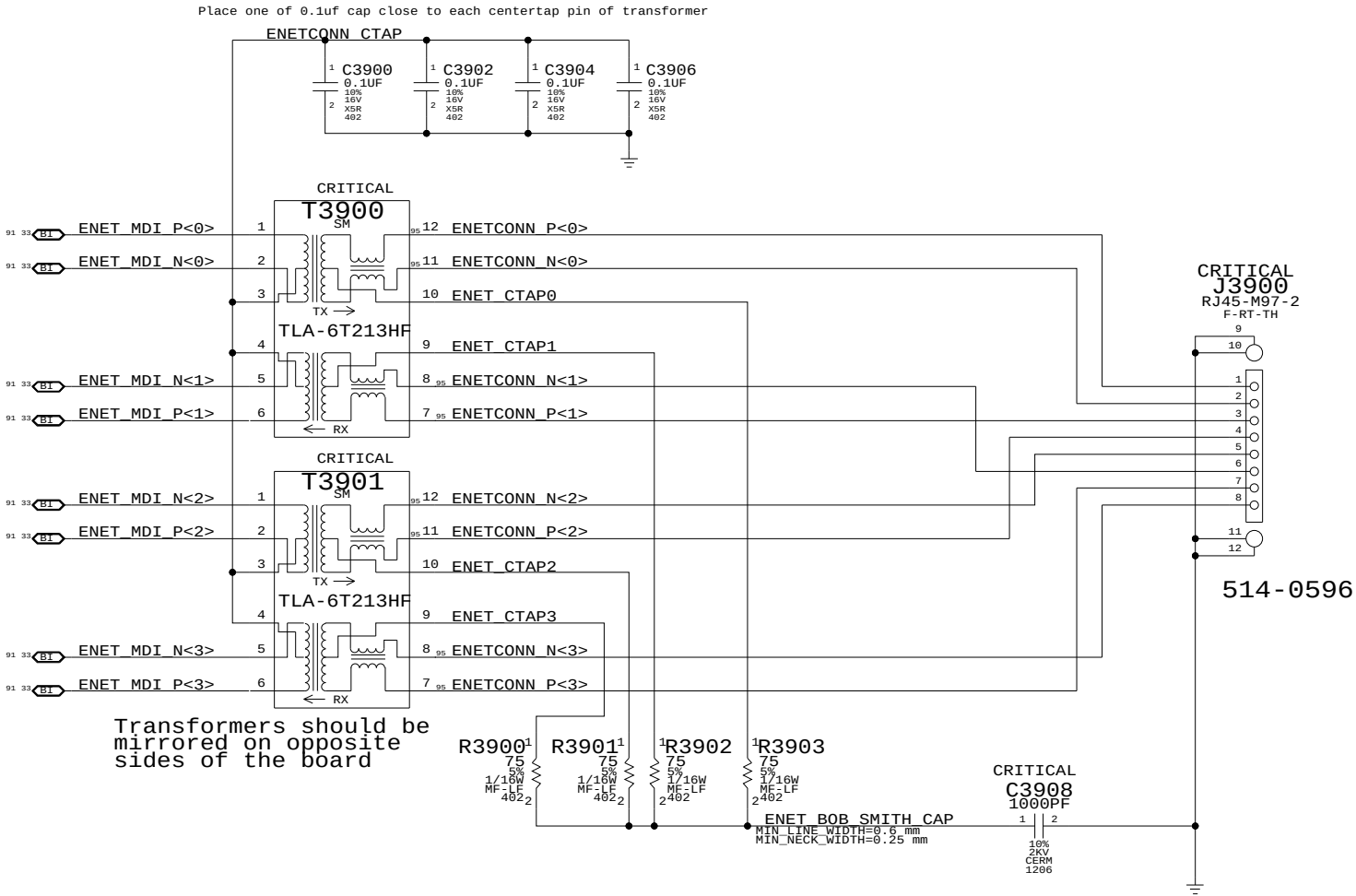
SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	33	96

Page Notes

Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



Ethernet Connector

SYNC_MASTER=SUMA_M98_MLB SYNC_DATE=07/01/2008

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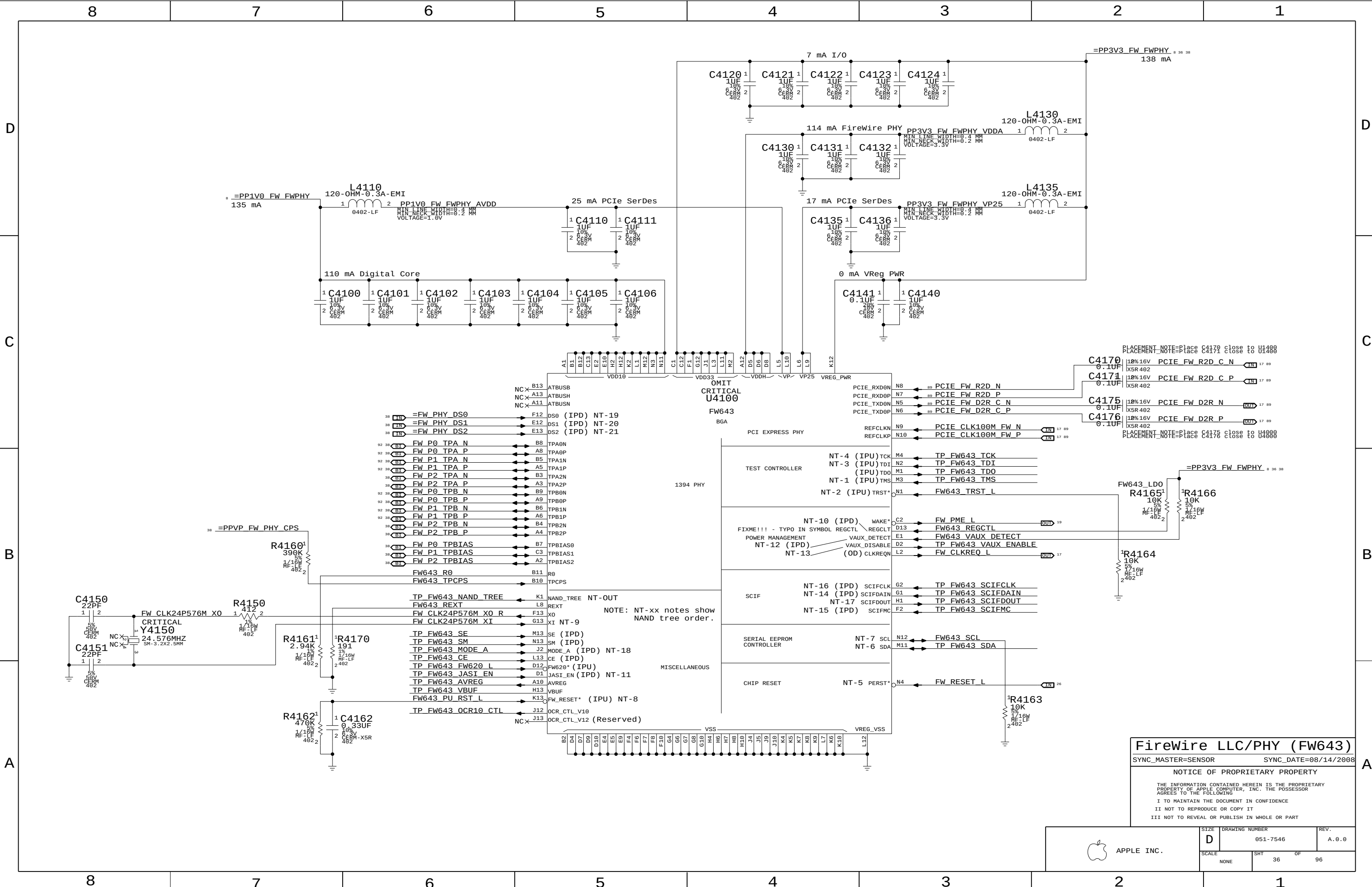
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SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	35	96



FireWire LLC/PHY (FW643)

SYNC_MASTER=SENSOR SYNC_DATE=08/14/2008

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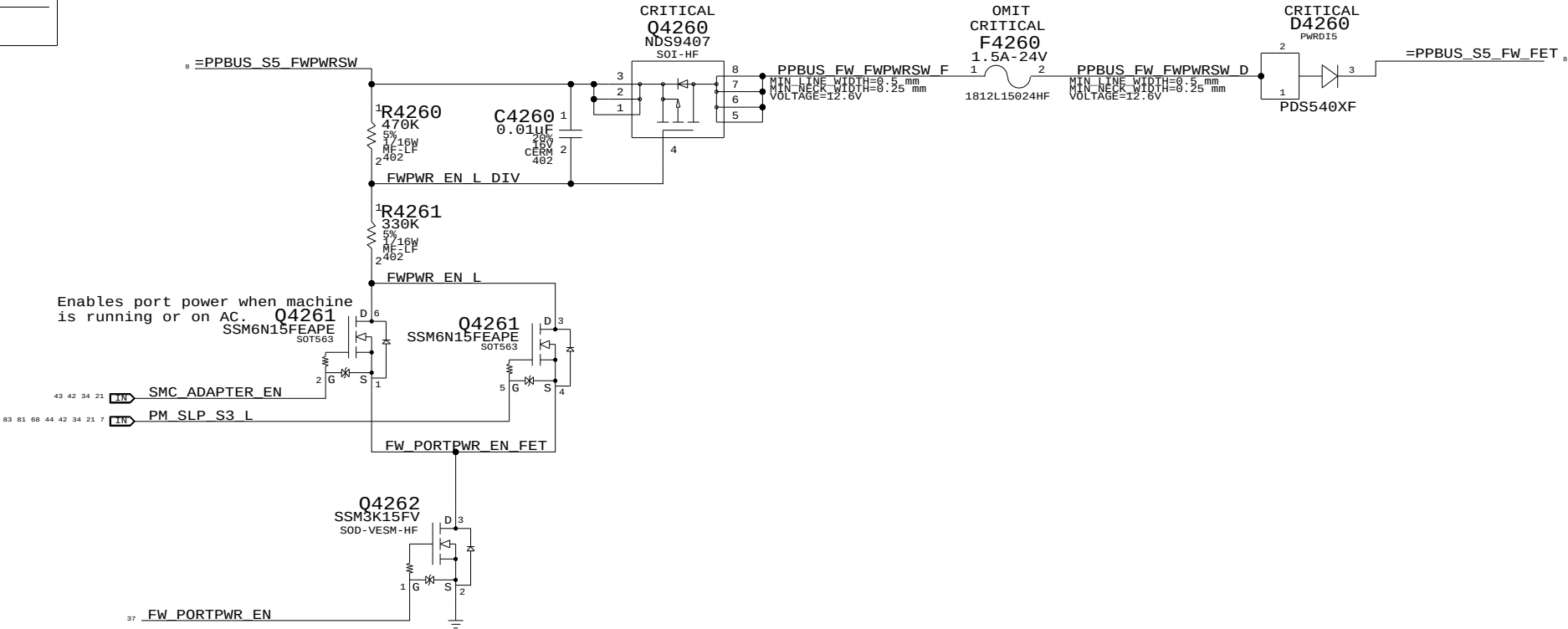
APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	36	96

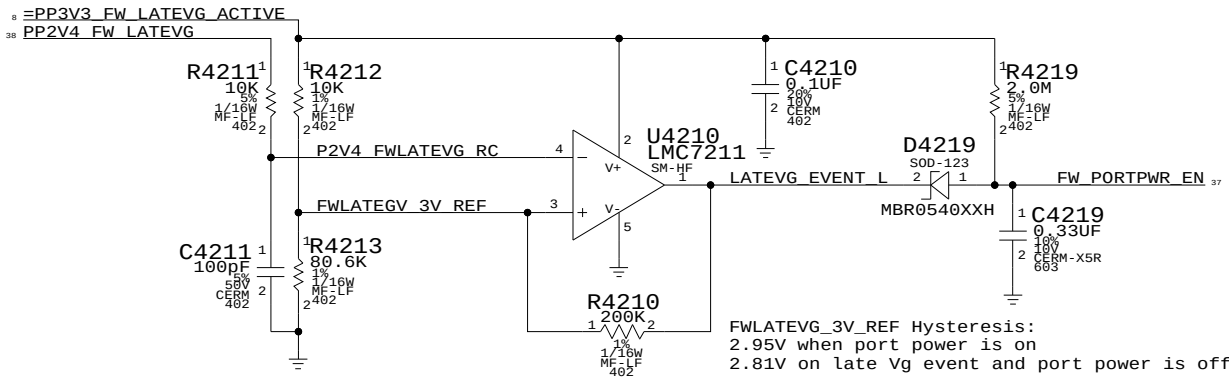
Page Notes

Power aliases required by this page:
- =PPBUS_S5_FWPWSW (system supply for bus power)
- =PP3V3_FW_LATEVG_ACTIVE
- =PPVP_FW_SUMNODE (power passthru summation node)
Signal aliases required by this page:
(NONE)
BOM options provided by this page:
- FW_PORT_FAULT_PU

FireWire Port Power Switch



Late-VG Event Detection



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
740S0080	1	LITTLEFUSE, 1.5A RESETTABLE 24V	F4260	CRITICAL	

FireWire Port Power

SYNC_MASTER=SENSOR SYNC_DATE=08/14/2008

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SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	37	96

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

D

atlas pages required by this page.
(NONE)
NOTE: This page is expected to contain

Filewire PHY config straps

Configure PHY for:

FWPHY_DS2 == FW_PHY_DS2 36
MAKE_BASE=TRUE

D

C C



B

R4362 56.2k R4363 56.2k

(GND_FW_PORT1_VG)

NC Vg

Vg

B

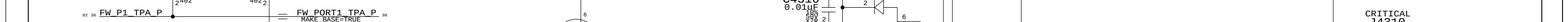


R4360 R4361
56k 56k
MF MF
1/18W 1/18W
402 402

CBS EN I

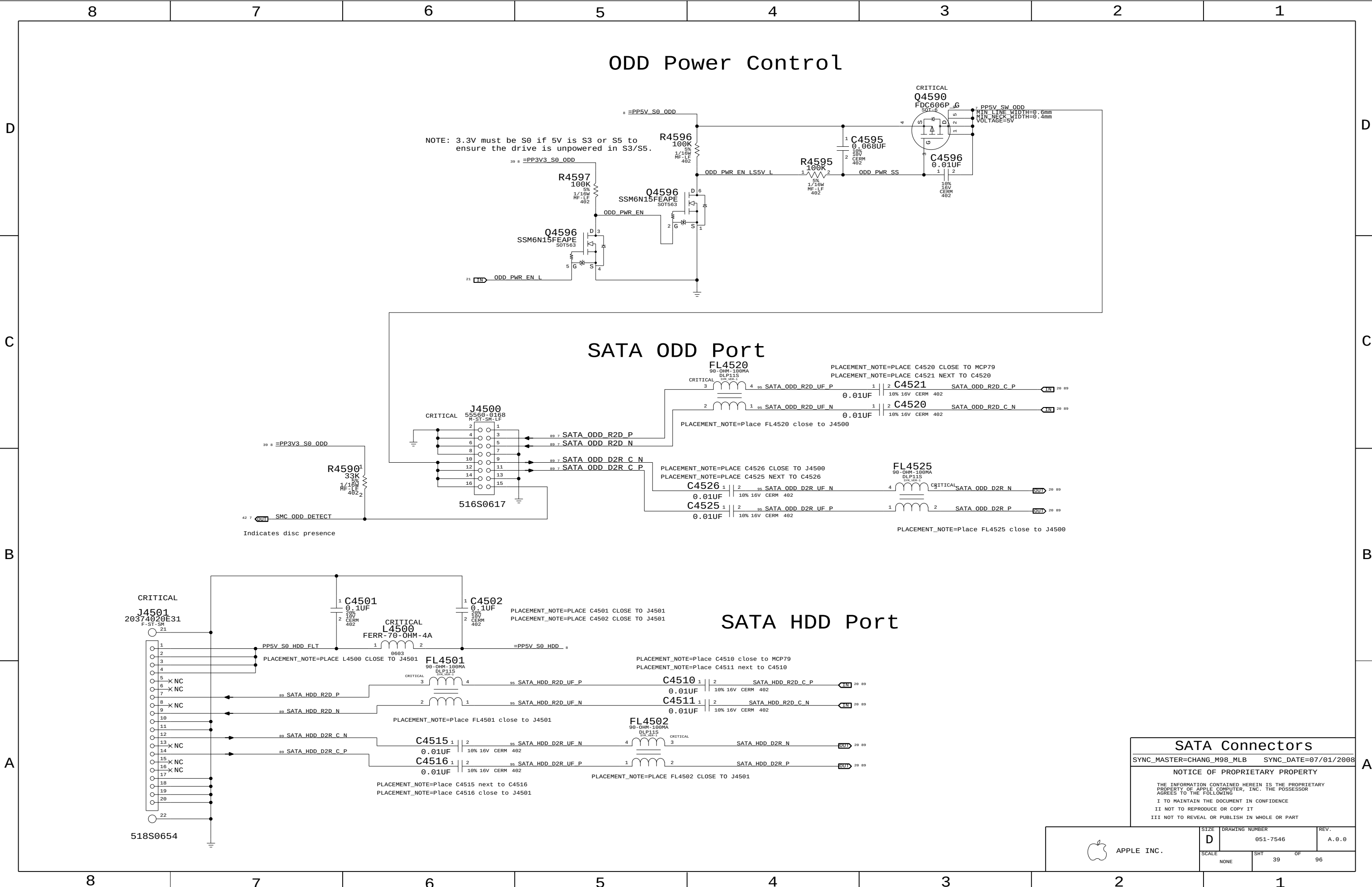
DP4310
8 6 4
402 402

BILINGUAL



A R4390

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---



NOTE: 3.3V must be S0 if 5V is S3 or S5 to ensure the drive is unpowered in S3/S5.

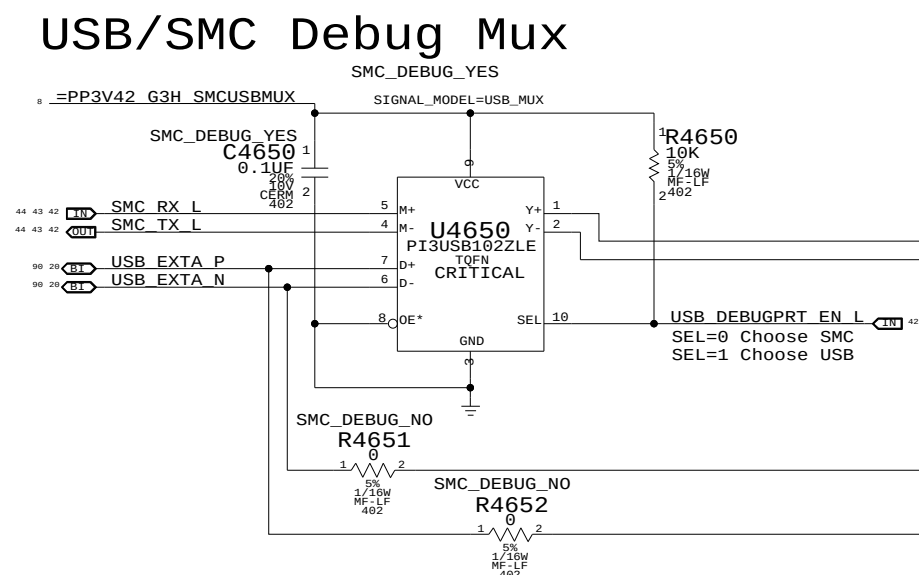
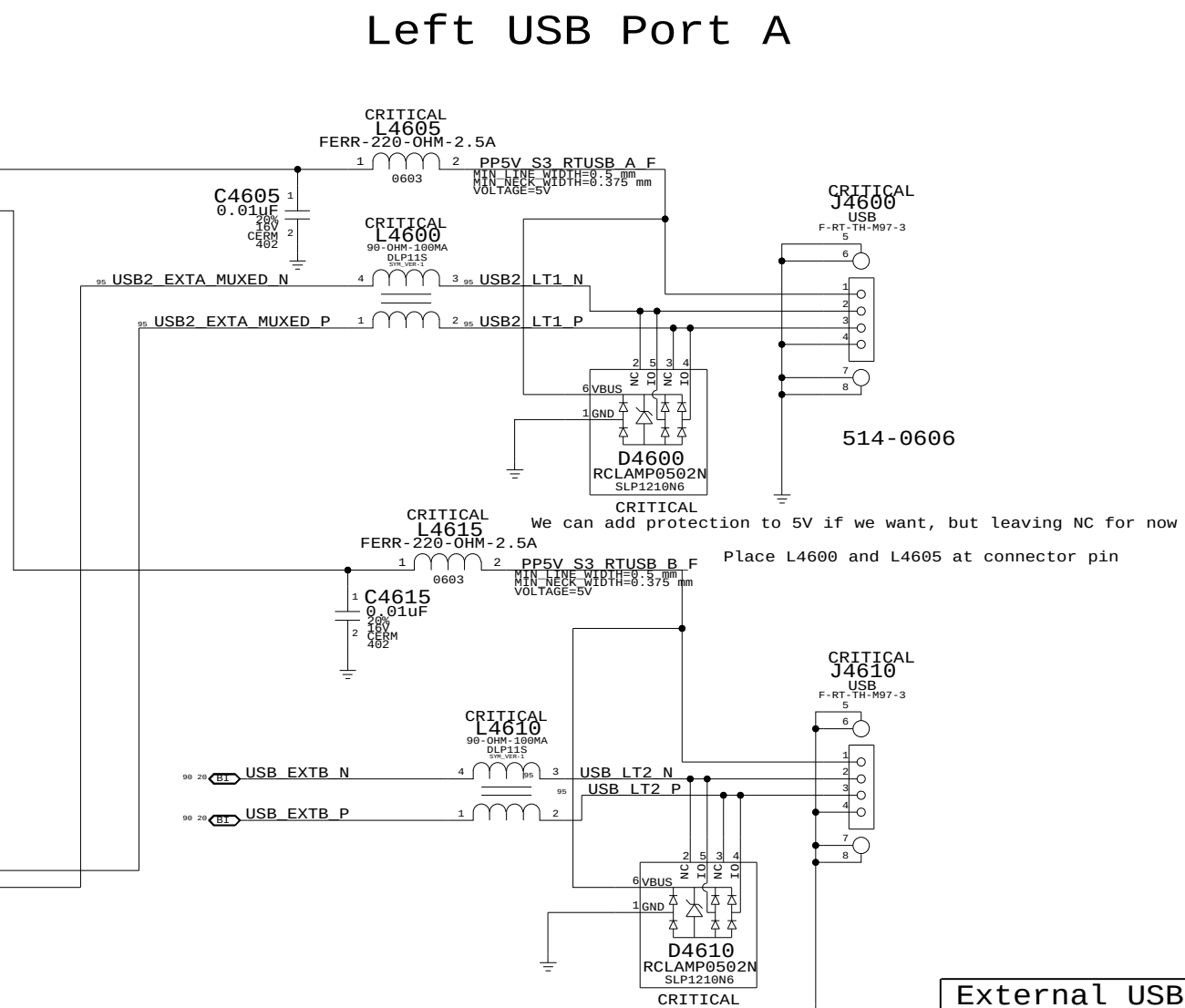
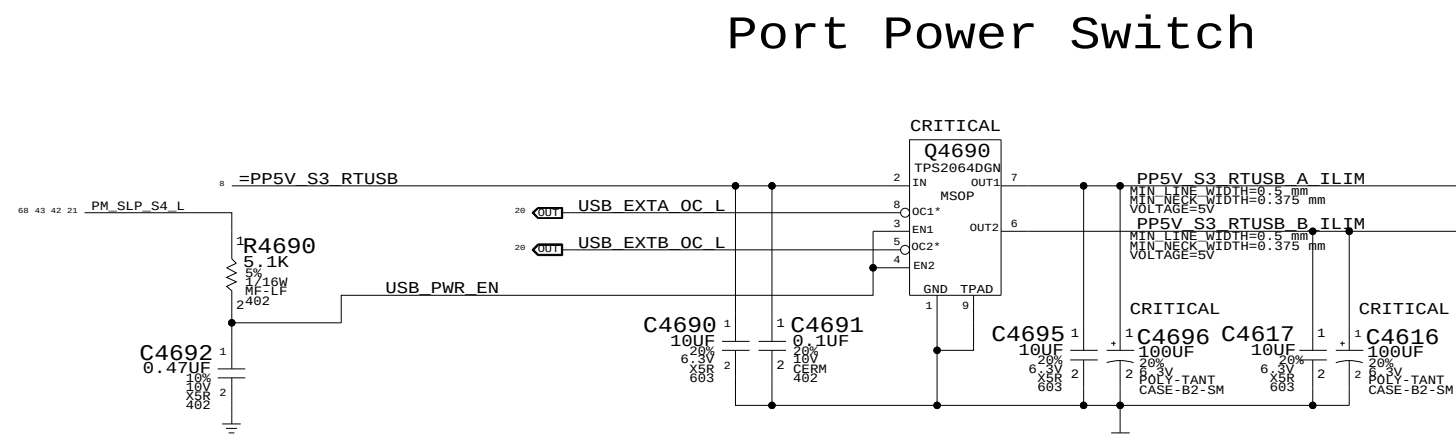
ODD Power Control

SATA ODD Port

SATA HDD Port

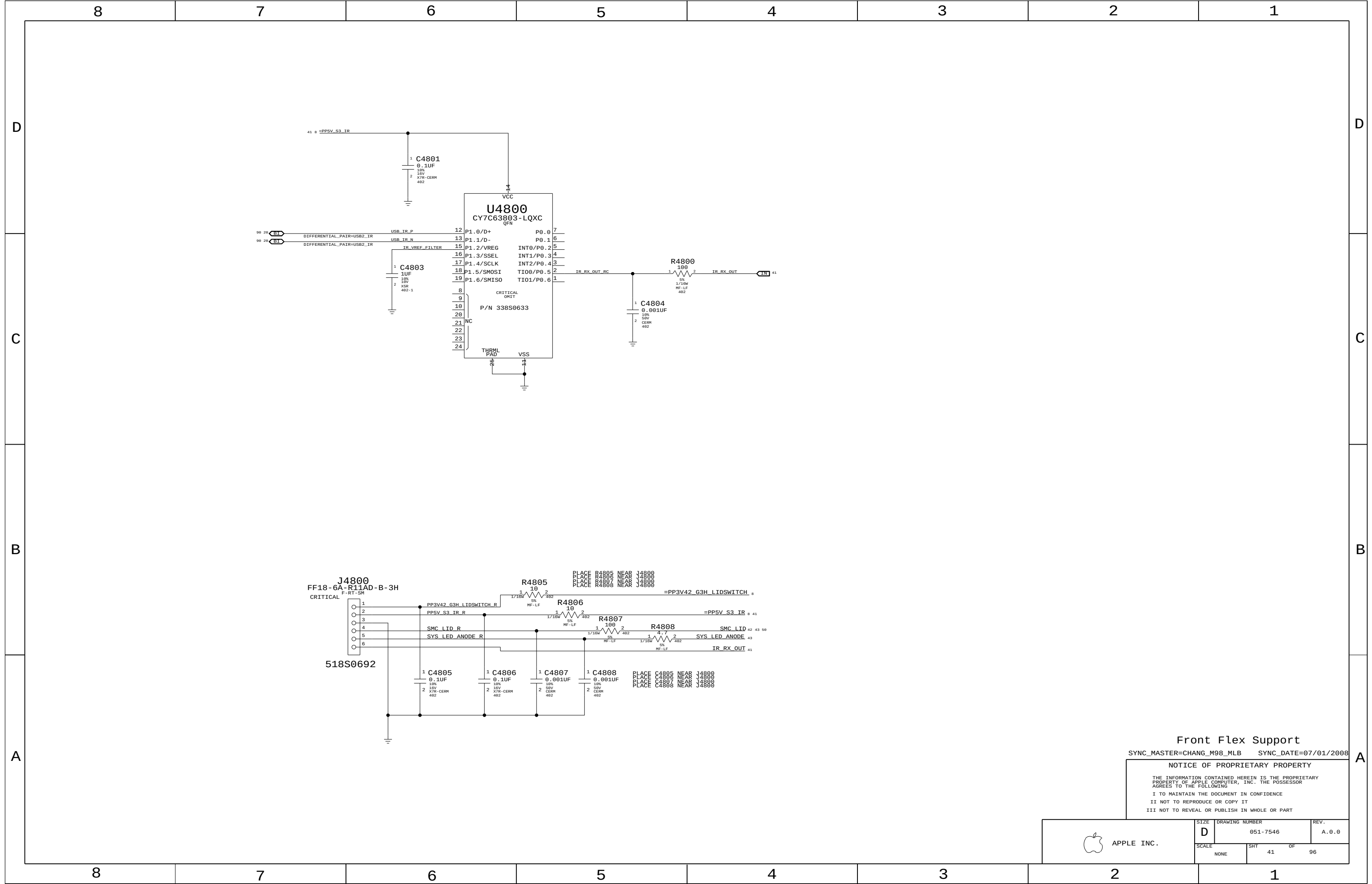
SATA Connectors		
SYNC_MASTER=CHANG_M98_MLB SYNC_DATE=07/01/2008		
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	SCALE NONE	SHT 39	OF 96



External USB Connectors	
SYNC_MASTER=AMASON_M98_MLB	SYNC_DATE=07/02/2008
A	
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	SCALE NONE	SHT 40	OF 96



Front Flex Support
SYNC_MASTER=CHANG_M98_MLB SYNC_DATE=07/01/2008

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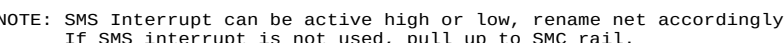
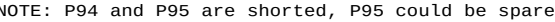
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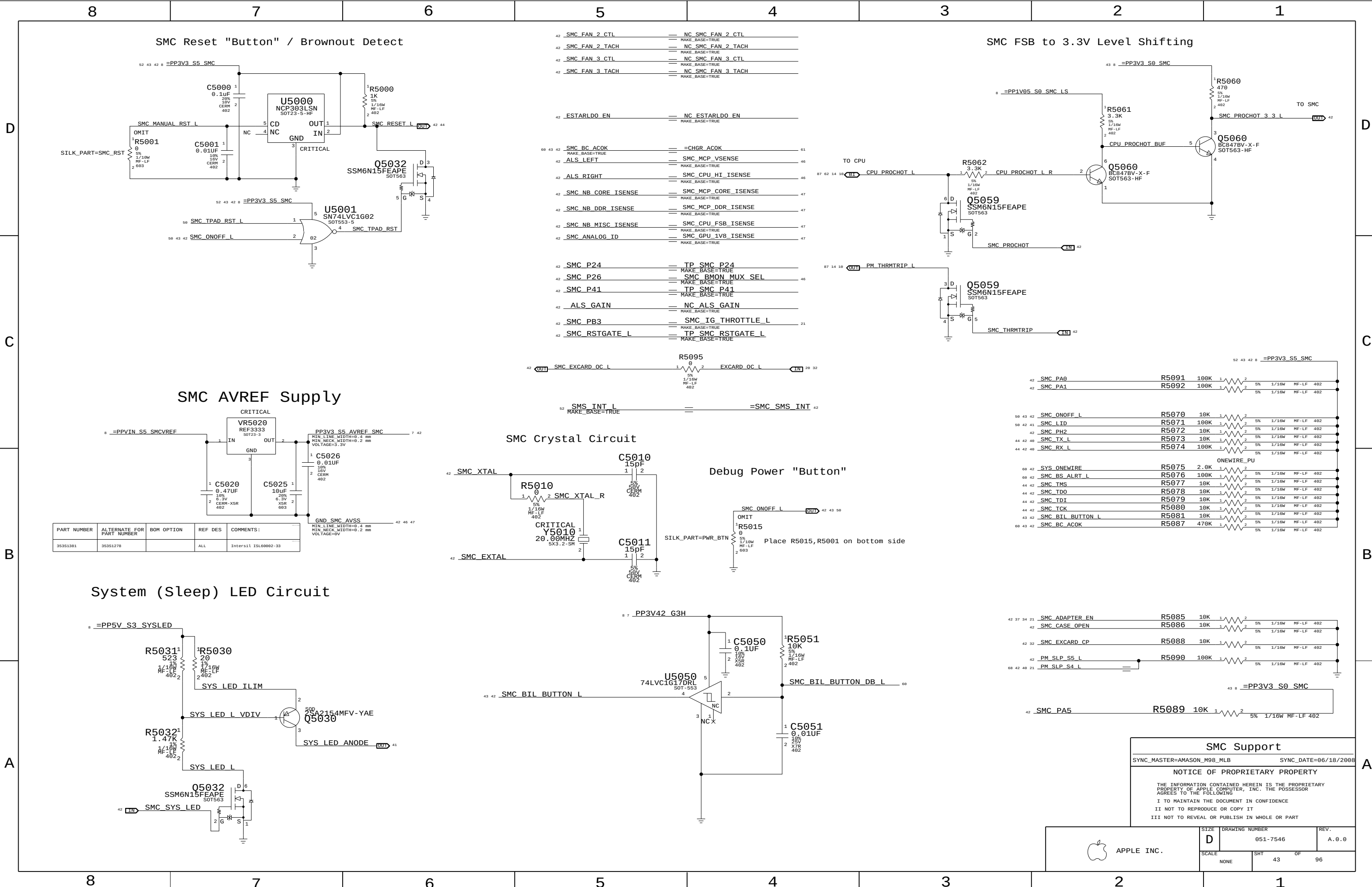
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	SCALE NONE	SHT 41	OF 96

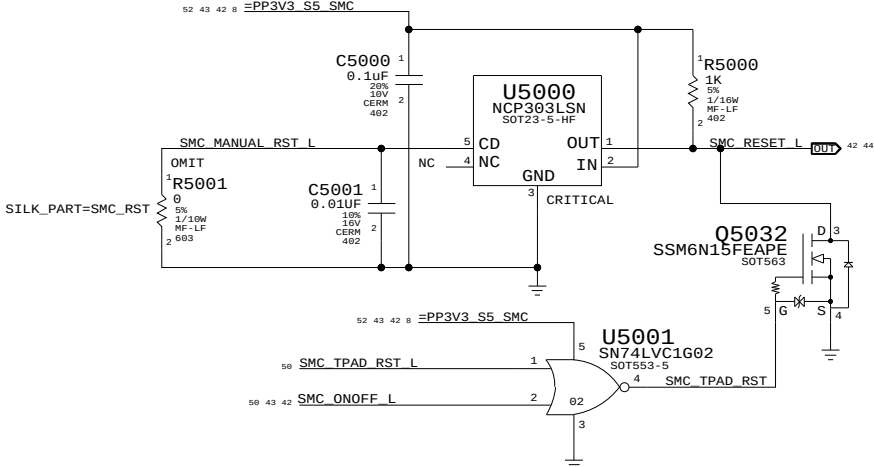
A

 APPLE INC.

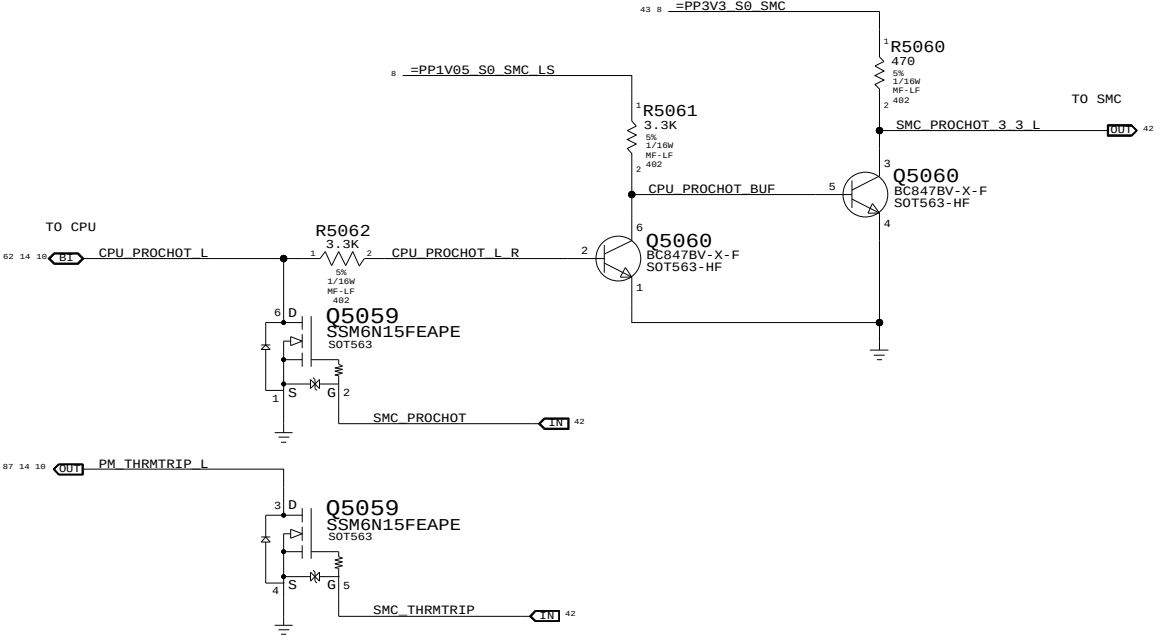
SIZE D	DRAWING NUMBER 051-7546	REV. A.0.
SCALE NONE	SHT 42	OF 96



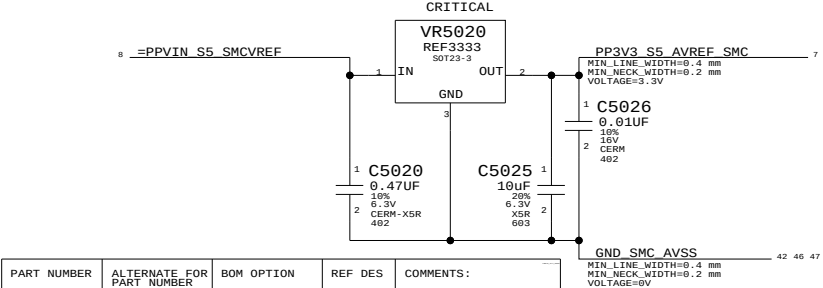
SMC Reset "Button" / Brownout Detect



SMC FSB to 3.3V Level Shifting

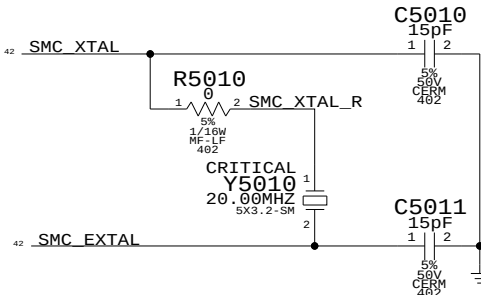


SMC AVREF Supply

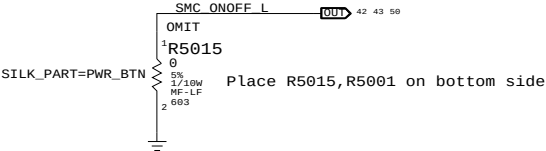


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S1381	353S1278		ALL	Intersil ISL60002-33

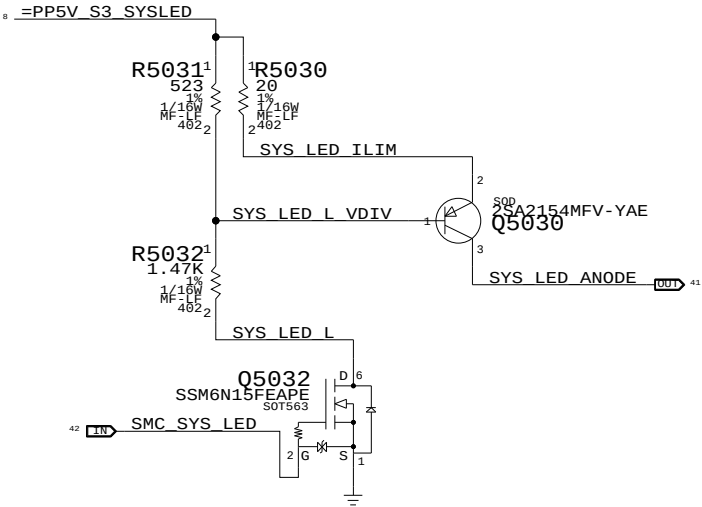
SMC Crystal Circuit



Debug Power "Button"



System (Sleep) LED Circuit



SMC PA0	R5091	100K	1	2	5%	1/16W	MF-LF	402
SMC PA1	R5092	100K	1	2	5%	1/16W	MF-LF	402
SMC ONOFF_L	R5070	10K	1	2	5%	1/16W	MF-LF	402
SMC LID	R5071	100K	1	2	5%	1/16W	MF-LF	402
SMC PH2	R5072	10K	1	2	5%	1/16W	MF-LF	402
SMC TX_L	R5073	10K	1	2	5%	1/16W	MF-LF	402
SMC RX_L	R5074	100K	1	2	5%	1/16W	MF-LF	402
SYS ONEWIRE	R5075	2.0K	1	2	5%	1/16W	MF-LF	402
SMC BS_ALERT_L	R5076	100K	1	2	5%	1/16W	MF-LF	402
SMC TMS	R5077	10K	1	2	5%	1/16W	MF-LF	402
SMC TDO	R5078	10K	1	2	5%	1/16W	MF-LF	402
SMC TDI	R5079	10K	1	2	5%	1/16W	MF-LF	402
SMC TCK	R5080	10K	1	2	5%	1/16W	MF-LF	402
SMC BIL_BUTTON_L	R5081	10K	1	2	5%	1/16W	MF-LF	402
SMC BC_ACOK	R5087	470K	1	2	5%	1/16W	MF-LF	402

SMC_ADAPTER_EN	R5085	10K	1	2	5%	1/16W	MF-LF	402
SMC_CASE_OPEN	R5086	10K	1	2	5%	1/16W	MF-LF	402
SMC_EXCARD_CP	R5088	10K	1	2	5%	1/16W	MF-LF	402
PM_SLP_S5_L	R5090	100K	1	2	5%	1/16W	MF-LF	402
PM_SLP_S4_L								
SMC_PA5	R5089	10K	1	2	5%	1/16W	MF-LF	402

SMC Support

SYNC_MASTER=AMASON_M98_MLB SYNC_DATE=06/18/2008

NOTICE OF PROPRIETARY PROPERTY

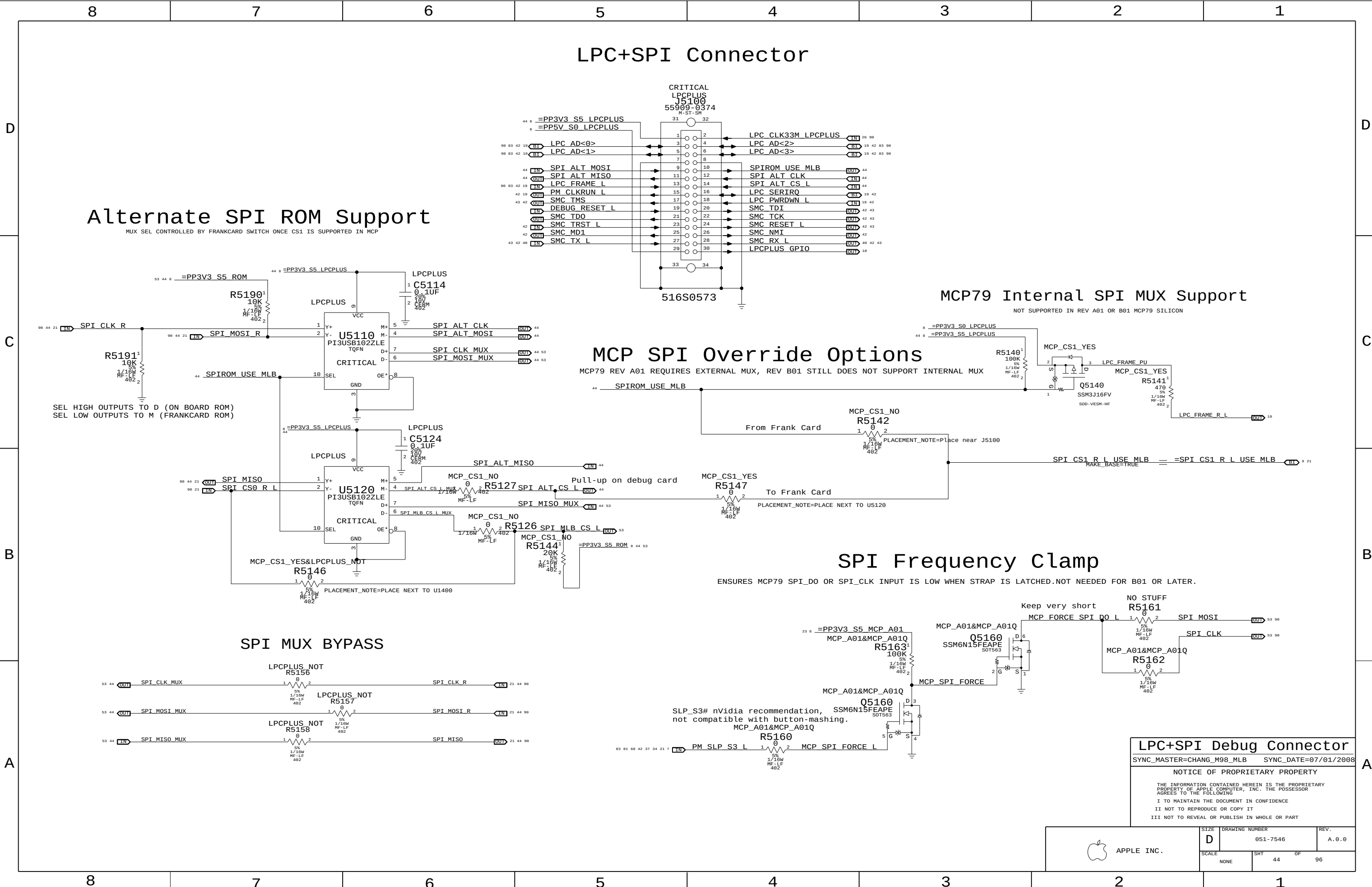
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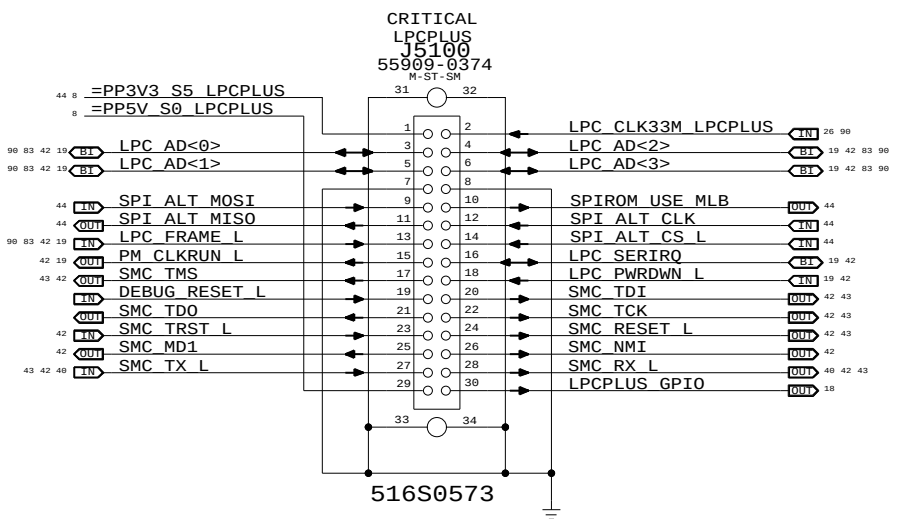
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APPLE INC.	SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 43 OF 96	

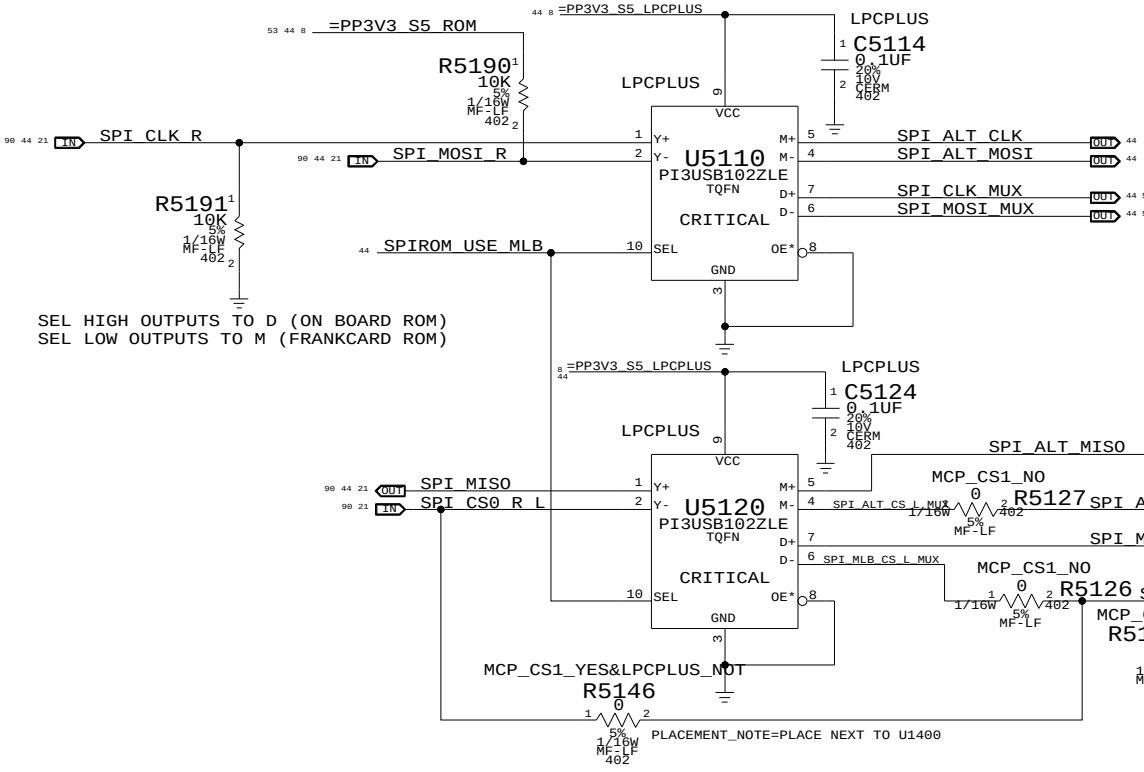


LPC+SPI Connector



Alternate SPI ROM Support

MUX SEL CONTROLLED BY FRANKCARD SWITCH ONCE CS1 IS SUPPORTED IN MCP

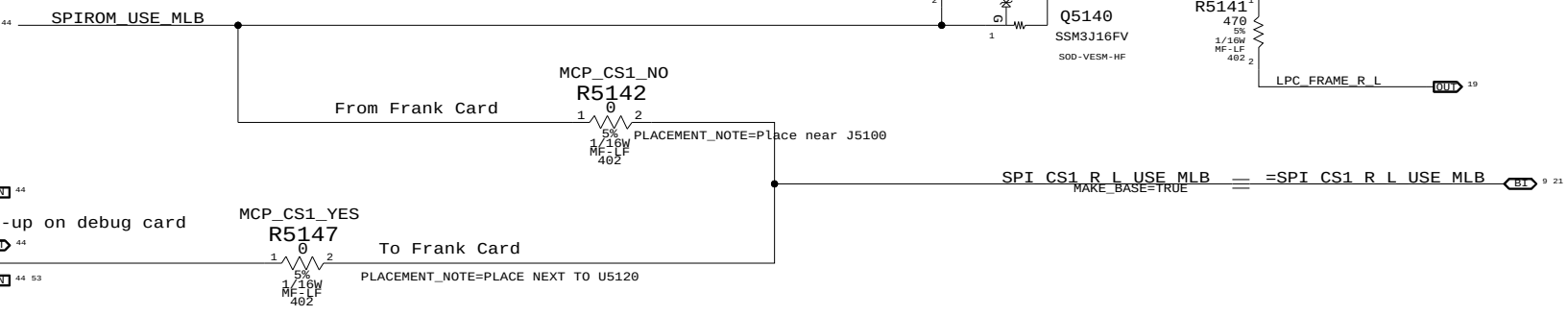


MCP79 Internal SPI MUX Support

NOT SUPPORTED IN REV A01 OR B01 MCP79 SILICON

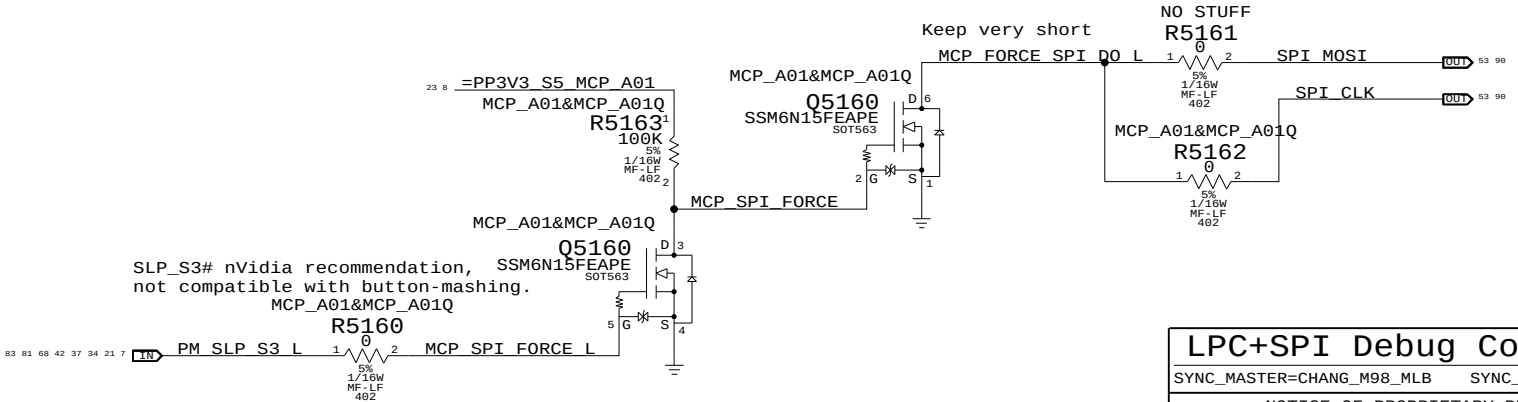
MCP SPI Override Options

MCP79 REV A01 REQUIRES EXTERNAL MUX, REV B01 STILL DOES NOT SUPPORT INTERNAL MUX

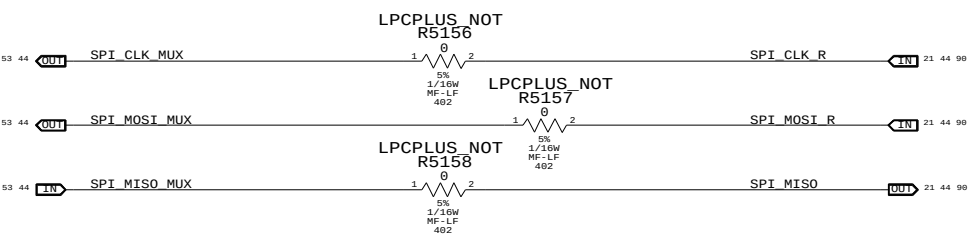


SPI Frequency Clamp

ENSURES MCP79 SPI_D0 OR SPI_CLK INPUT IS LOW WHEN STRAP IS LATCHED. NOT NEEDED FOR B01 OR LATER.



SPI MUX BYPASS



LPC+SPI Debug Connector

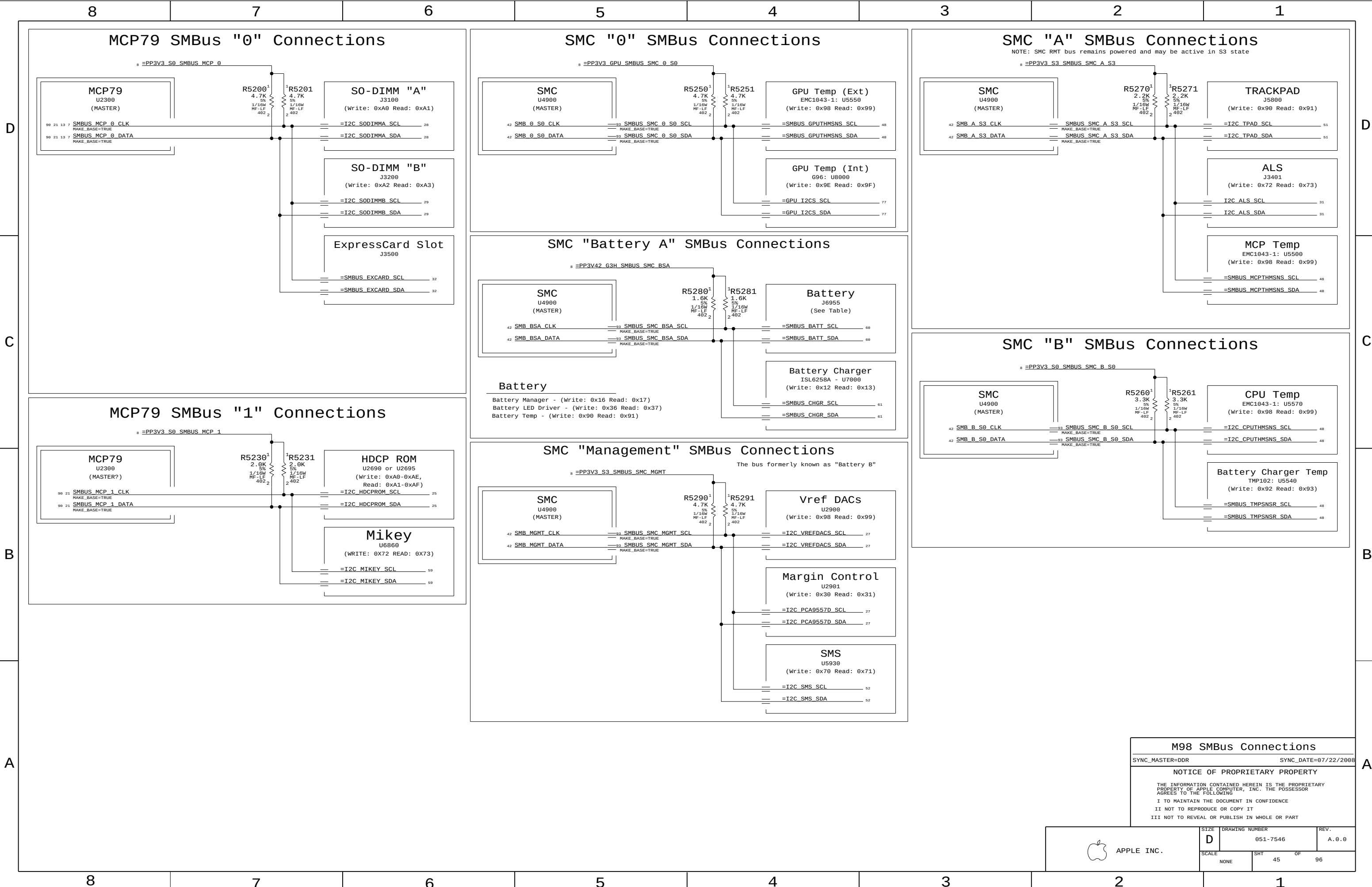
SYNC_MASTER=CHANG_M98_MLB SYNC_DATE=07/01/2008

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SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	44	96



M98 SMBus Connections

SYNC_MASTER=DDR SYNC_DATE=07/22/2008

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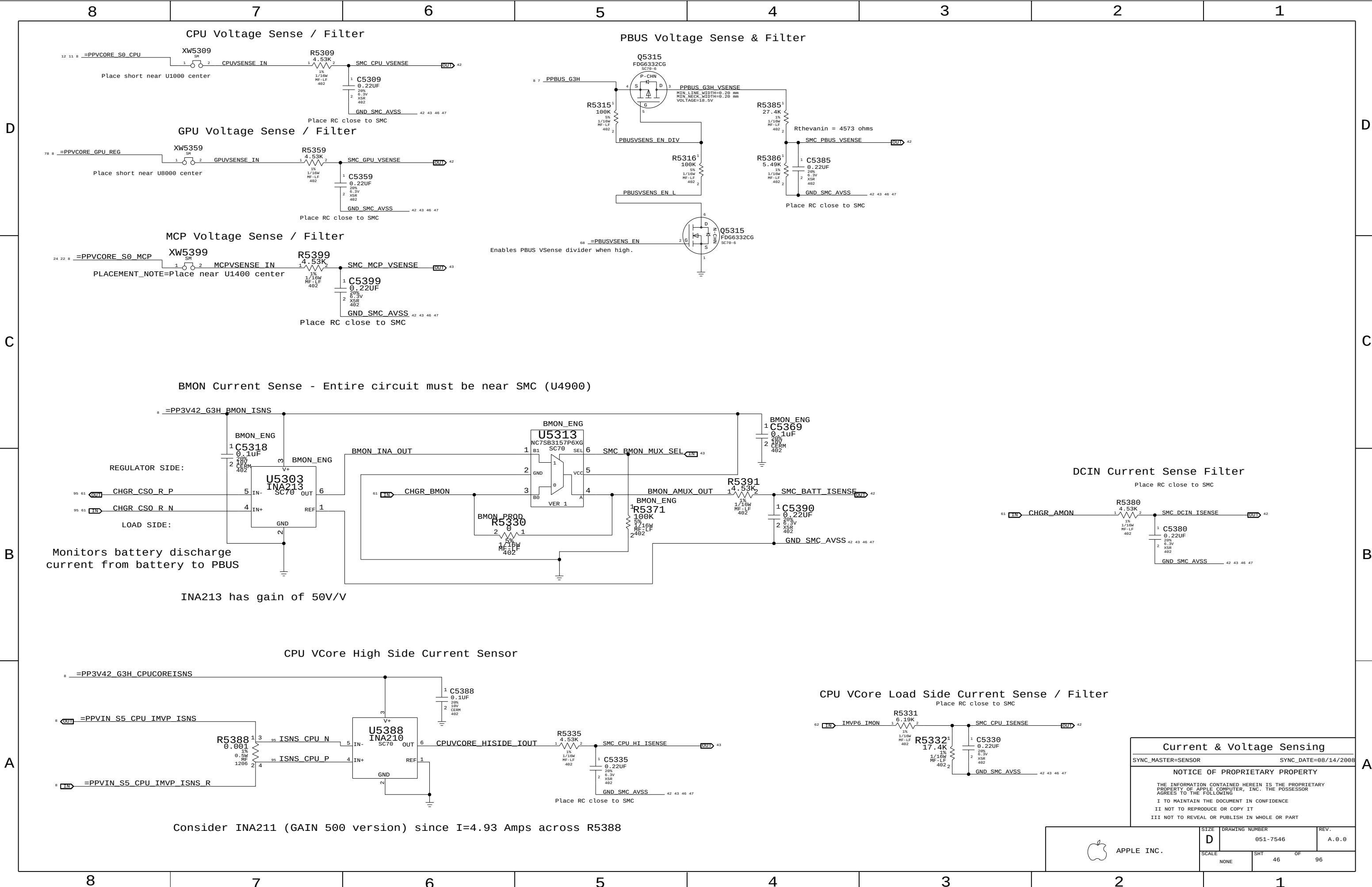
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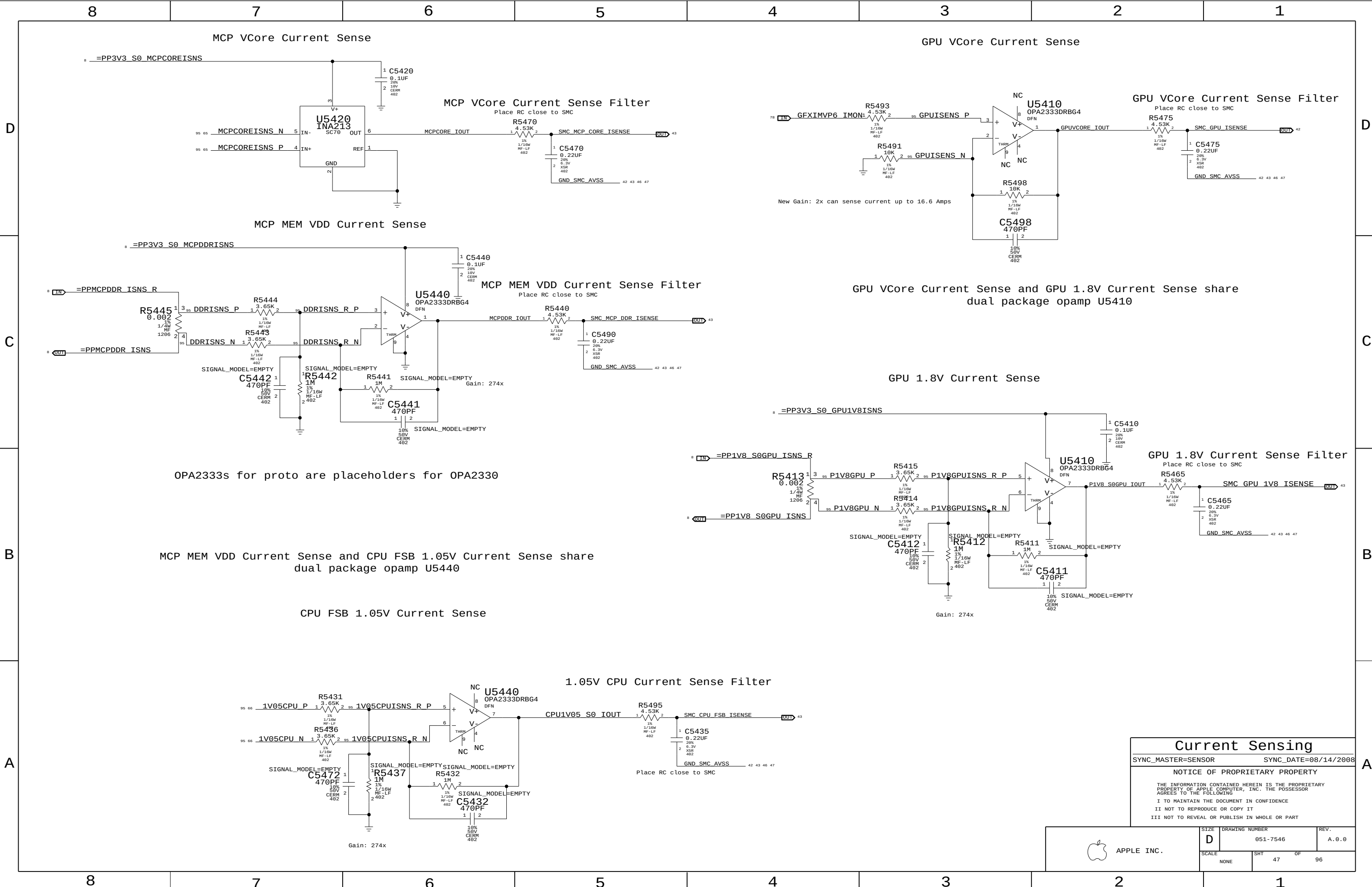
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.	SIZE	DRAWING NUMBER	REV.
	D	051-7546	A.0.0
SCALE	NONE	SHT	45 OF 96



Current & Voltage Sensing		
SYNC_MASTER=SENSOR		SYNC_DATE=08/14/2008
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APPLE INC.	SIZE	DRAWING NUMBER	REV.
	D	051-7546	A.0.0
SCALE		SHT	OF
NONE		46	96



MCP VCore Current Sense

GPU VCore Current Sense

MCP VCore Current Sense Filter

GPU VCore Current Sense Filter

MCP MEM VDD Current Sense

MCP MEM VDD Current Sense Filter

GPU VCore Current Sense and GPU 1.8V Current Sense share dual package opamp U5410

GPU 1.8V Current Sense

GPU 1.8V Current Sense Filter

OPA2333s for proto are placeholders for OPA2330

MCP MEM VDD Current Sense and CPU FSB 1.05V Current Sense share dual package opamp U5440

CPU FSB 1.05V Current Sense

1.05V CPU Current Sense Filter

Current Sensing

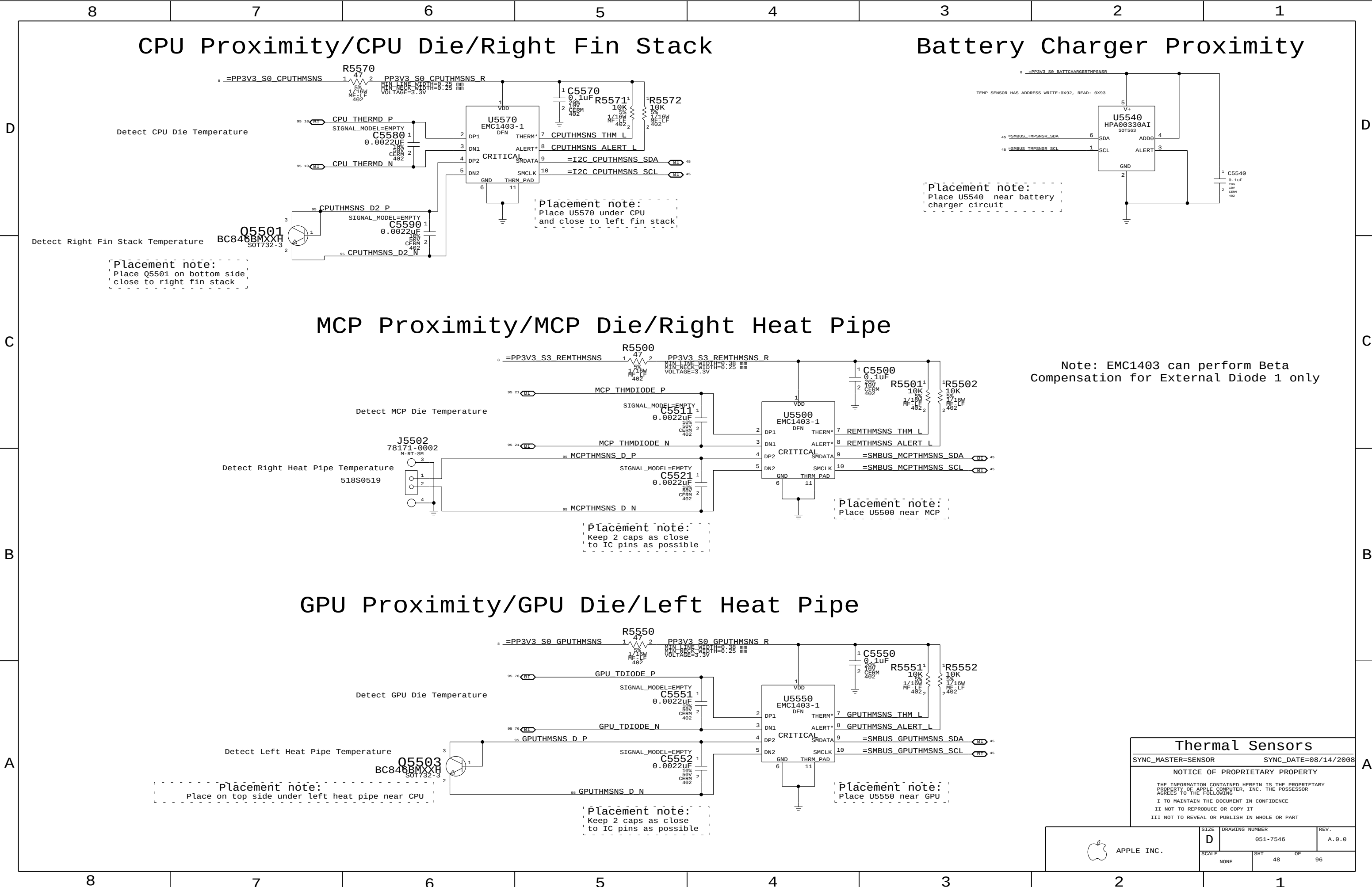
SYNC_MASTER=SENSOR SYNC_DATE=08/14/2008

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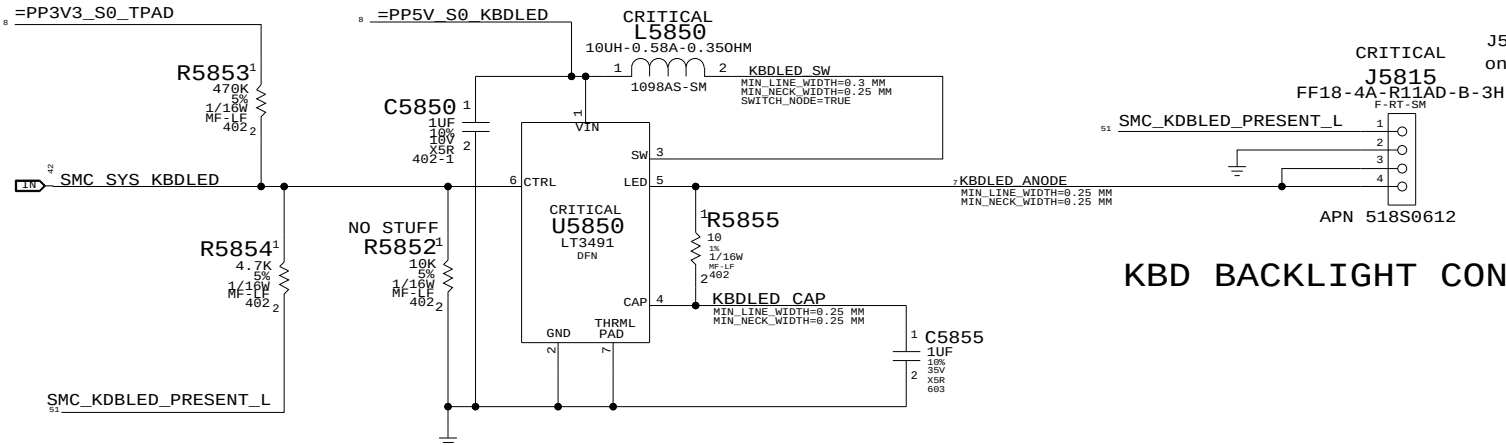
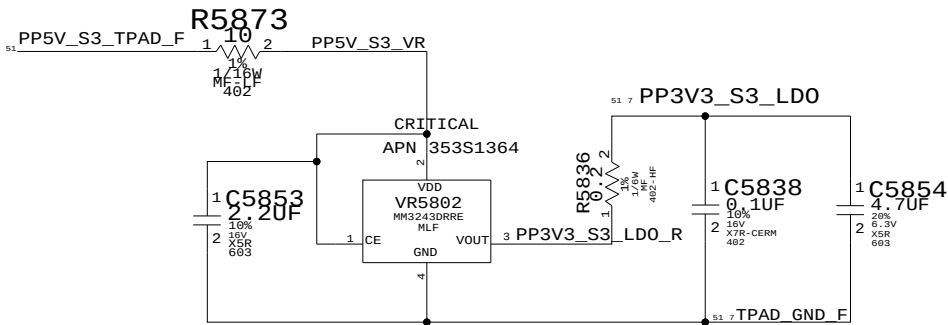
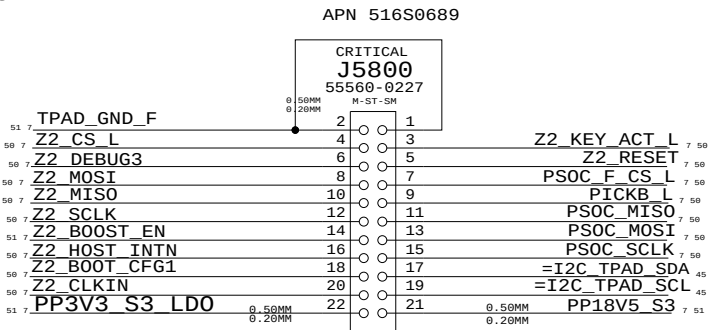
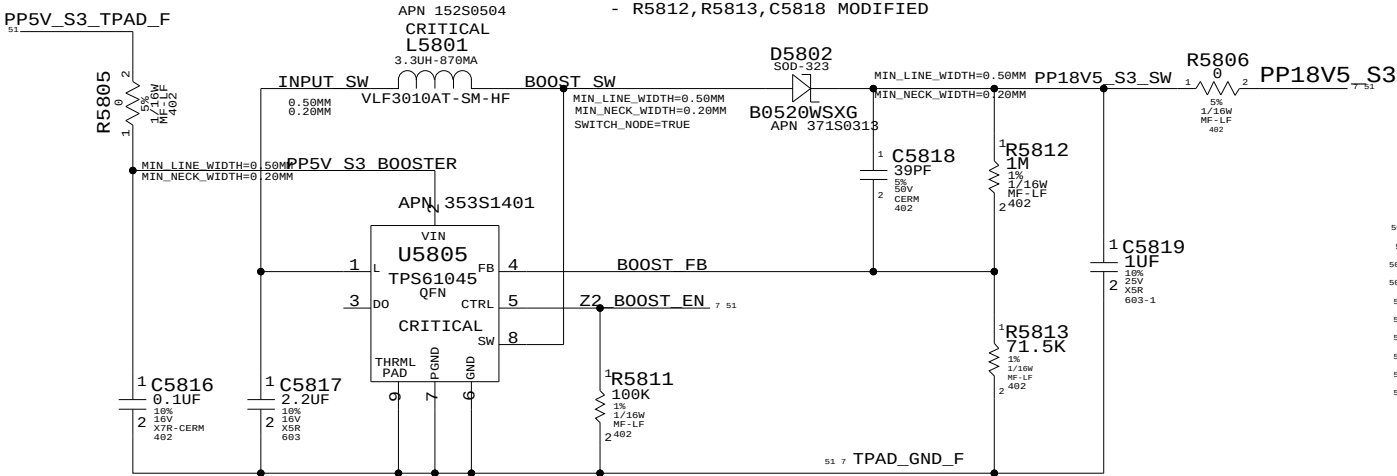
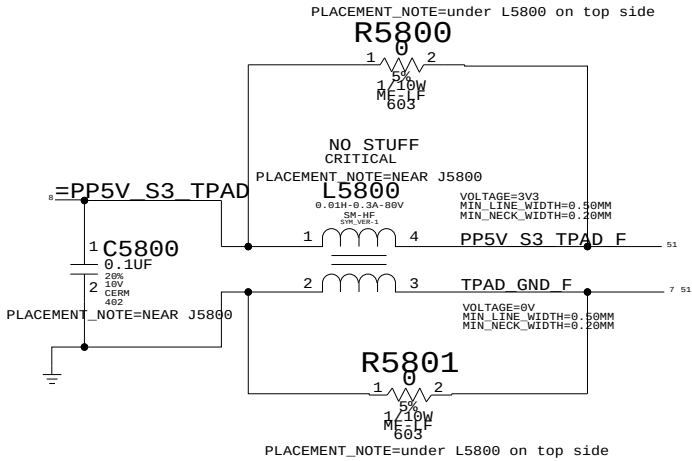
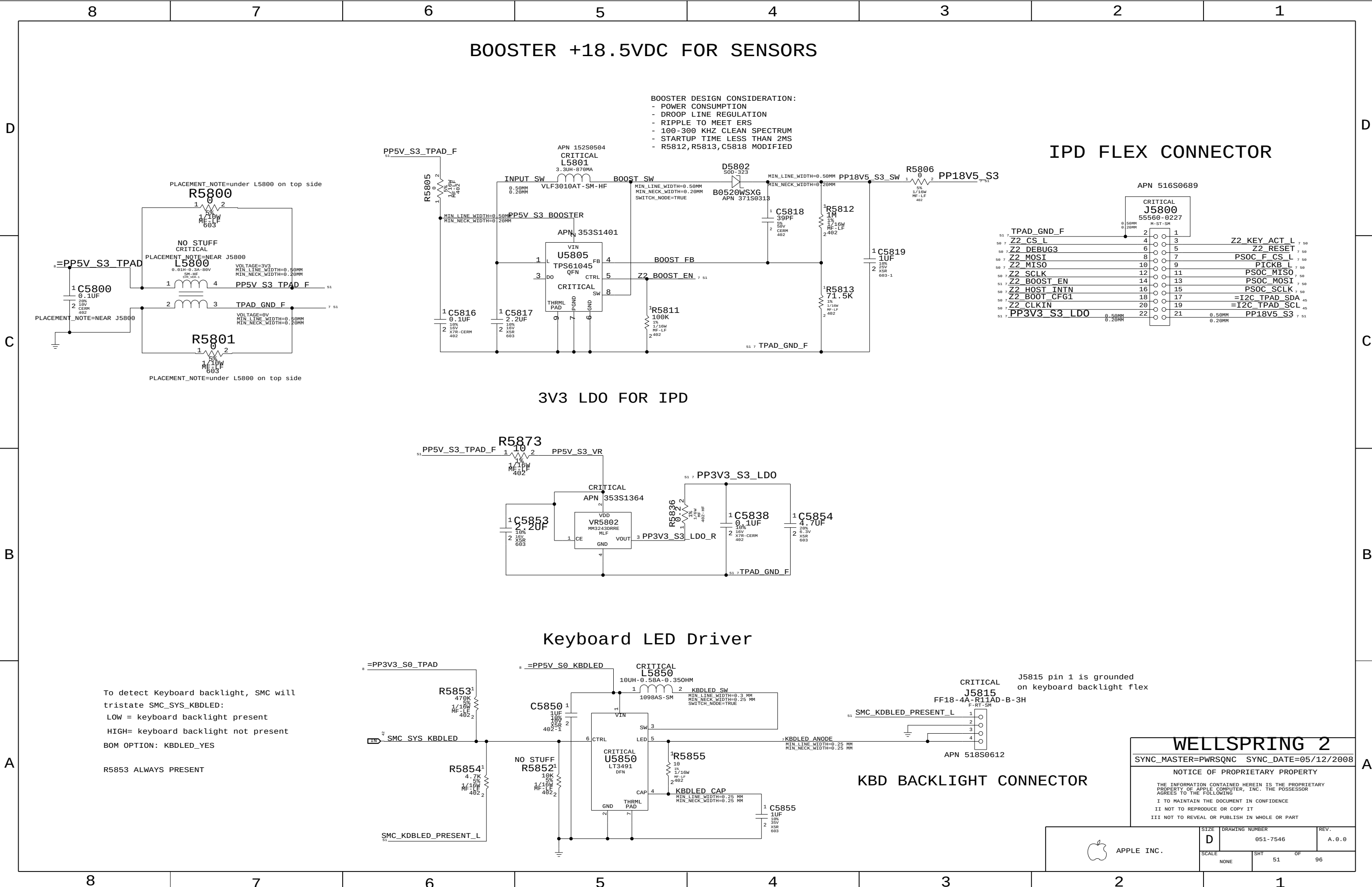
SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	47	96



Note: EMC1403 can perform Beta Compensation for External Diode 1 only

Thermal Sensors		
SYNC_MASTER=SENSOR		SYNC_DATE=08/14/2008
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APPLE INC.	SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 48	OF 96



To detect Keyboard backlight, SMC will tristate SMC_SYS_KBDLED:
LOW = keyboard backlight present
HIGH= keyboard backlight not present
BOM OPTION: KBDLED_YES
R5853 ALWAYS PRESENT

WELLSPRING 2		
SYNC_MASTER=PWRSQNC SYNC_DATE=05/12/2008		
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APPLE INC.	SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 51	OF 96

8	7	6	5	4	3	2	1
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D



C

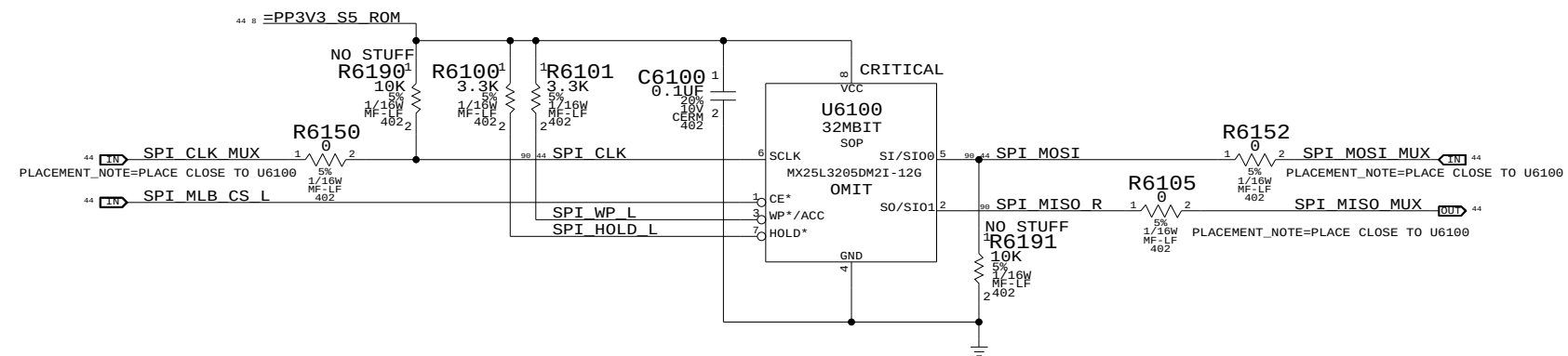
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B



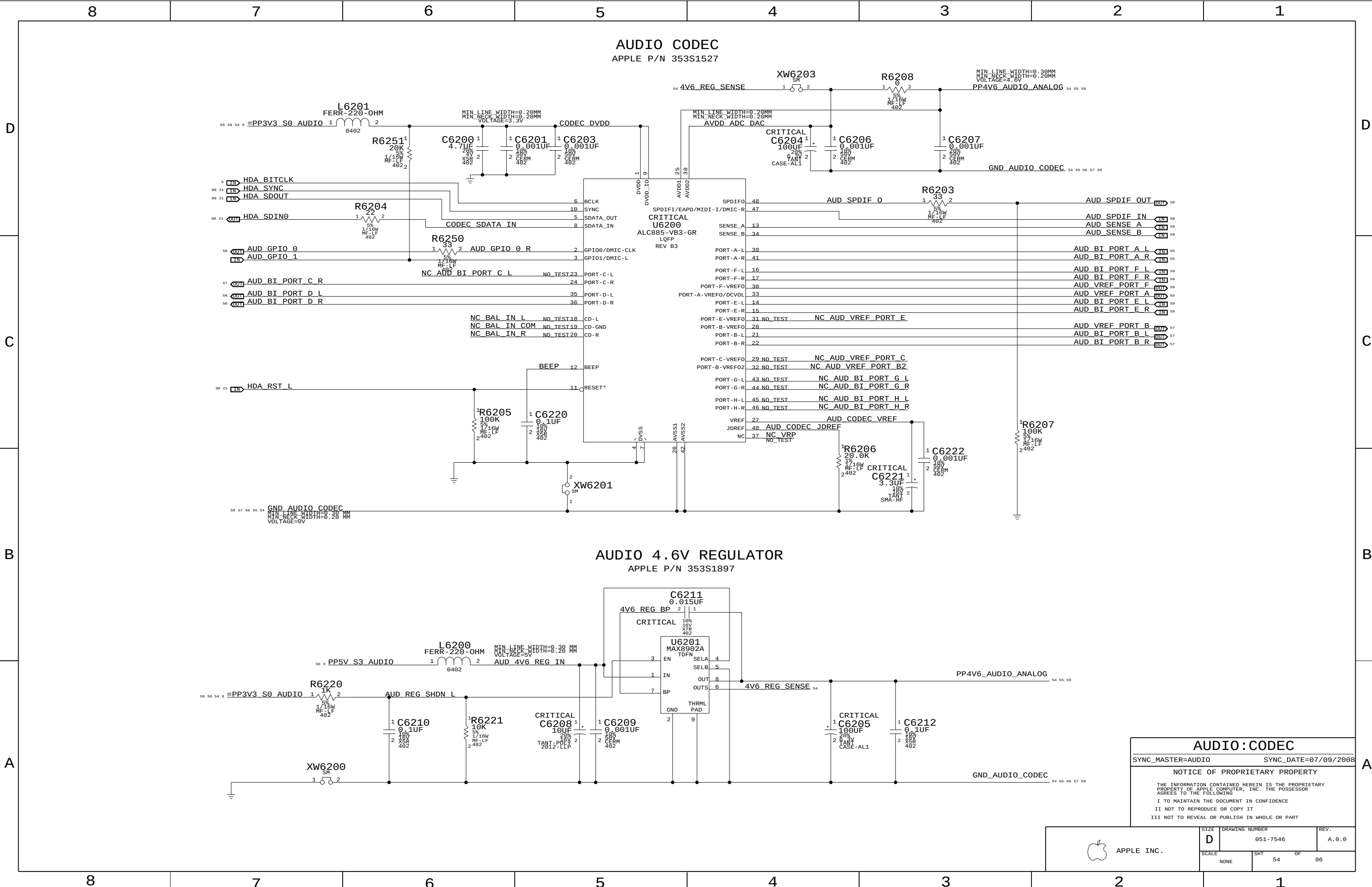
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8	7	6	5	4	3	2	1
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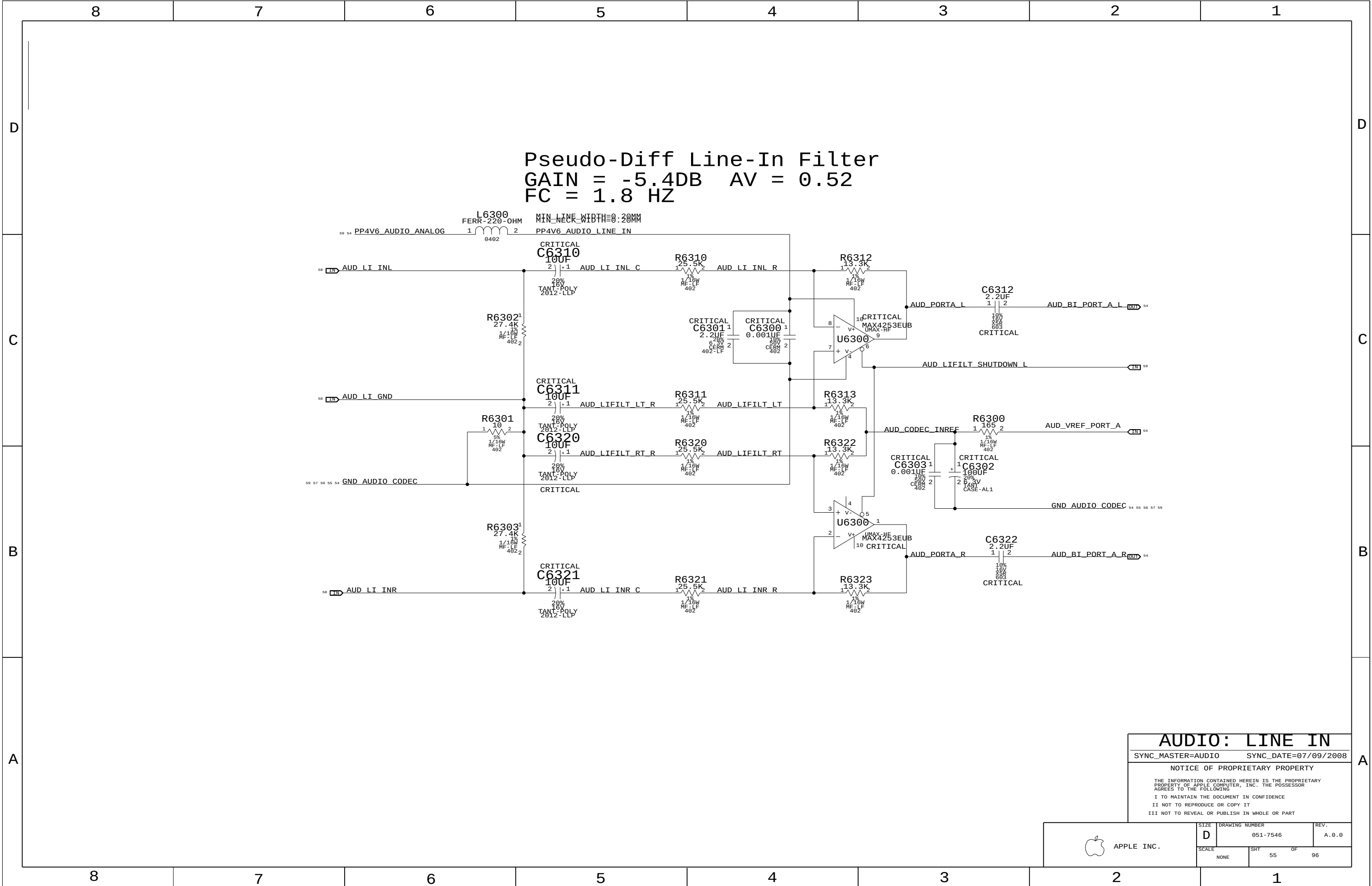
MCP79 SPI Frequency Select		
Frequency	SPI_M0SI	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

25MHz is selected with R5190 and R5191
Any of the 4 frequencies can be selected
with R6190, R6191, R5190 and R5191



AUDIO:CODEC
SYNC_MASTER=AUDIO SYNC_DATE=07/09/2008
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APPLE INC.	SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 54	OF 96



AUDIO: LINE IN

SYNC_MASTER=AUDIO SYNC_DATE=07/09/2008

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APPLE INC.

SIZE

D

DRAWING NUMBER

051-7546

REV.

A.0.0

SCALE

NONE

SHT

55

OF

96

D



C

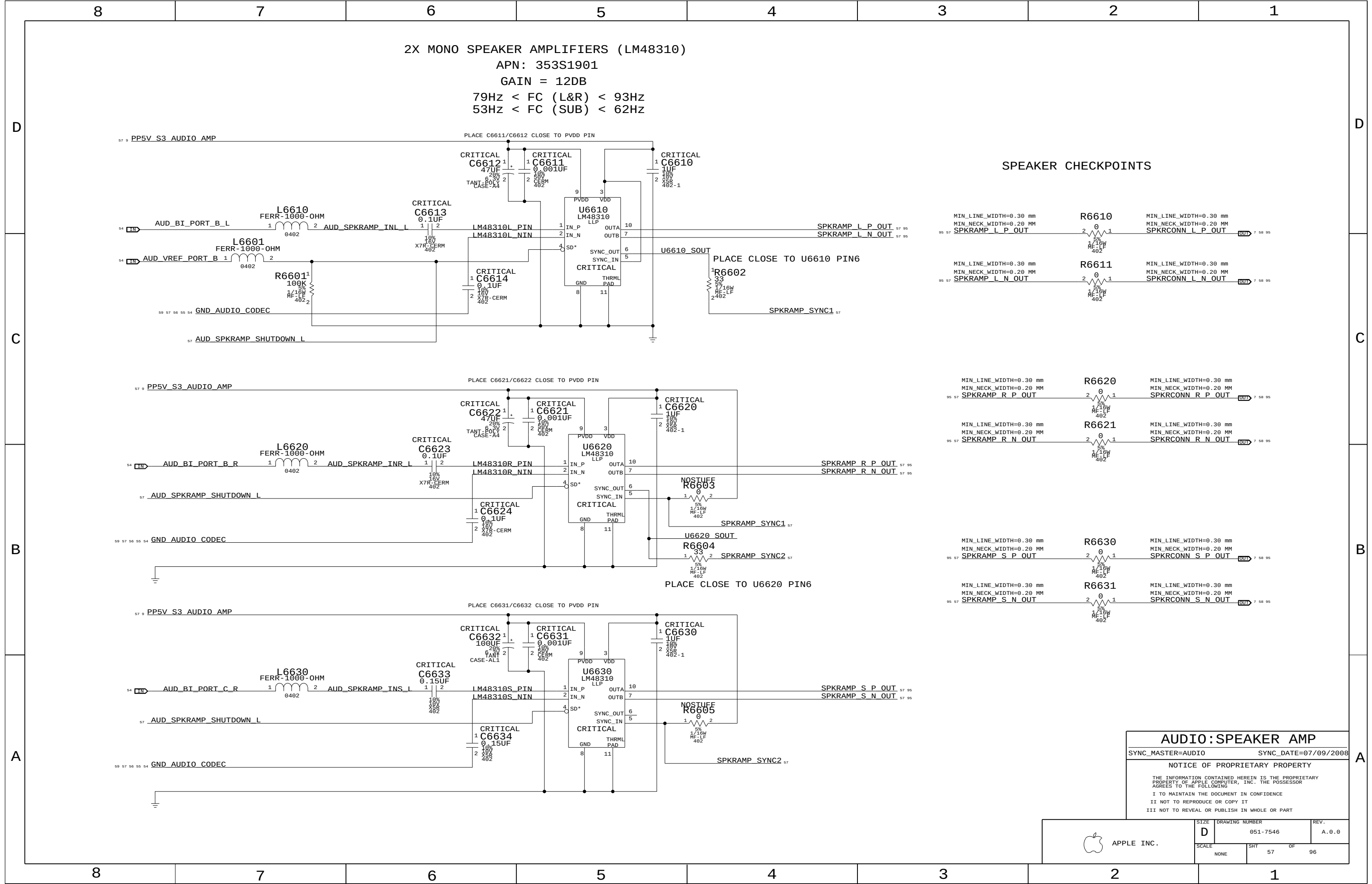


B

A



SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
SCALE NONE	SHT 56	OF 96



SPEAKER CHECKPOINTS

AUDIO:SPEAKER AMP

SYNC_MASTER=AUDIO SYNC_DATE=07/09/2008

NOTICE OF PROPRIETARY PROPERTY

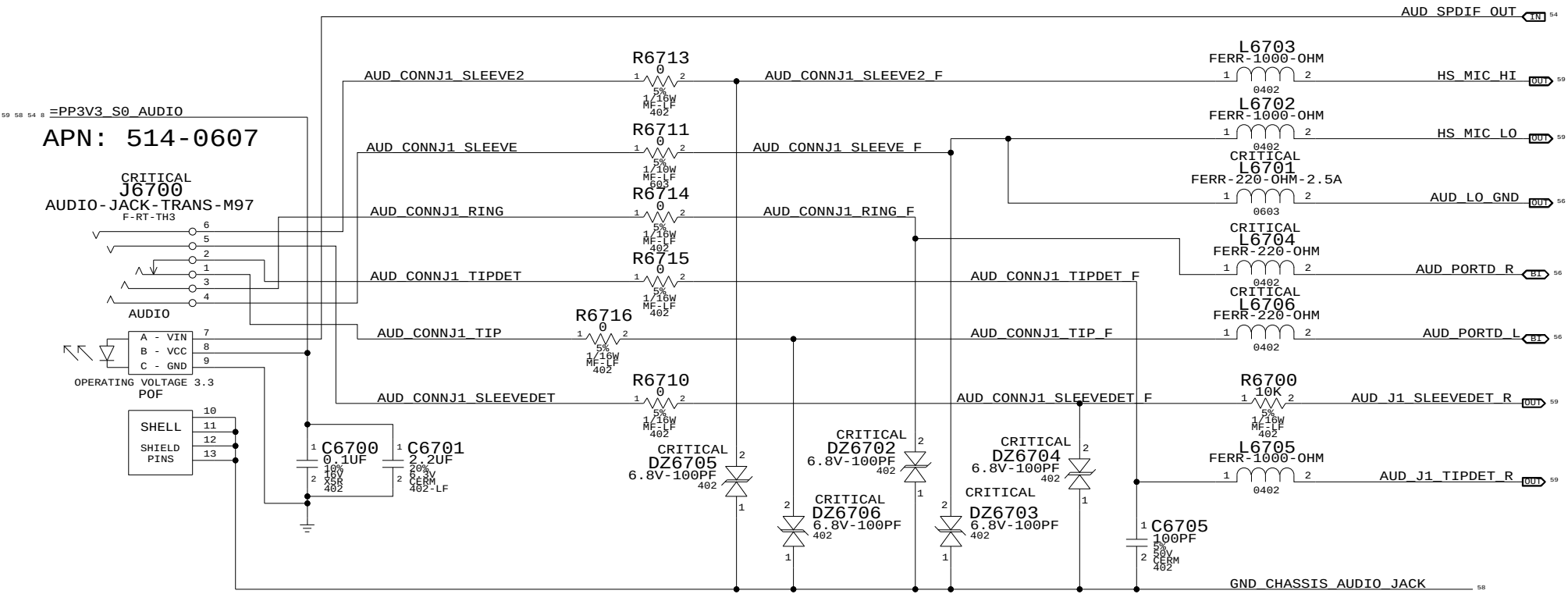
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APPLE INC.

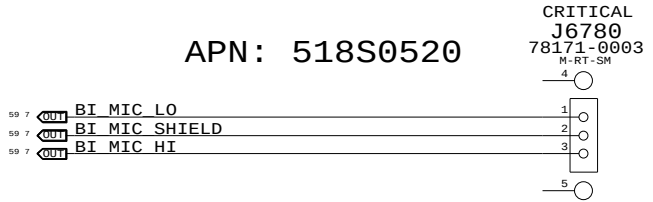
SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	57	96

AUDIO JACK 1 LO/HP JACK, SPDIF TX



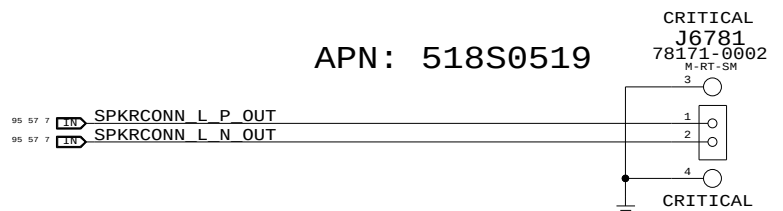
MIC CONNECTOR

APN: 518S0520

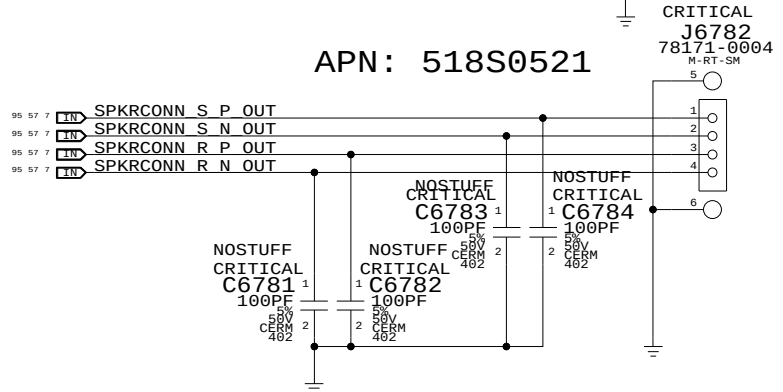


SPEAKER CONNECTOR

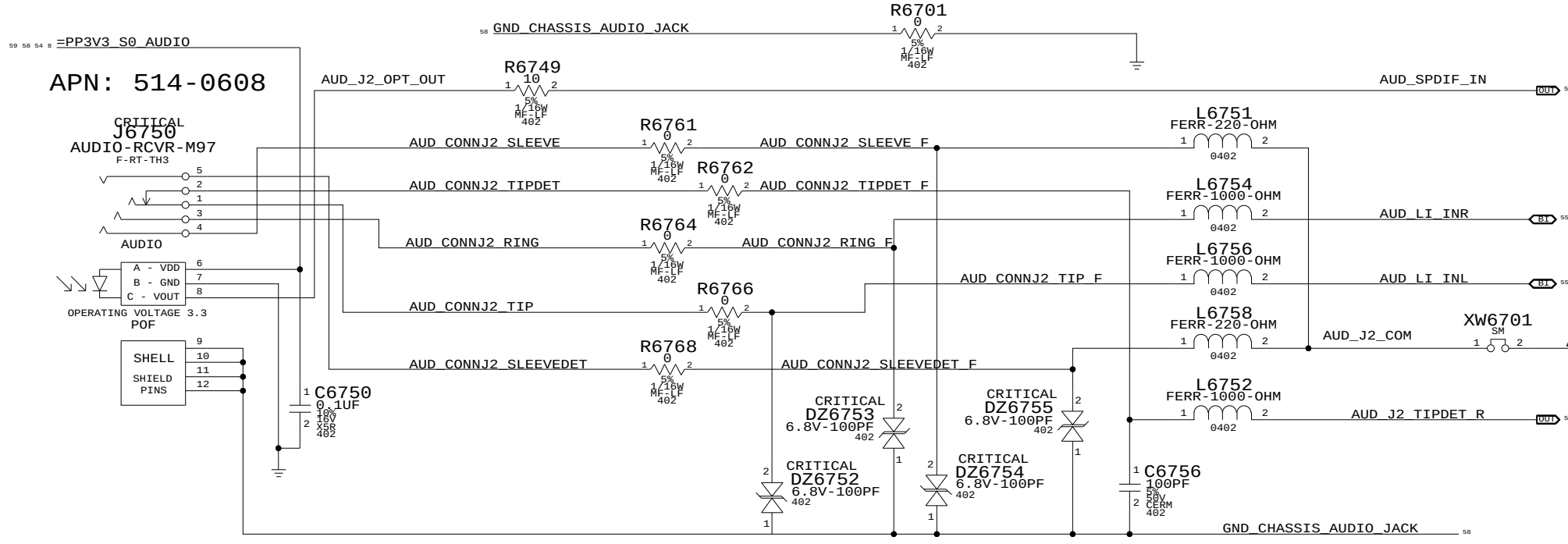
APN: 518S0519



APN: 518S0521



RETURN FOR HF NOISE



AUDIO JACK 2 LINE IN JACK, SPDIF RX

AUDIO: JACKS
SYNC_MASTER=AUDIO SYNC_DATE=07/09/2008
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APPLE INC.	SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 58	OF 96

D

C

B

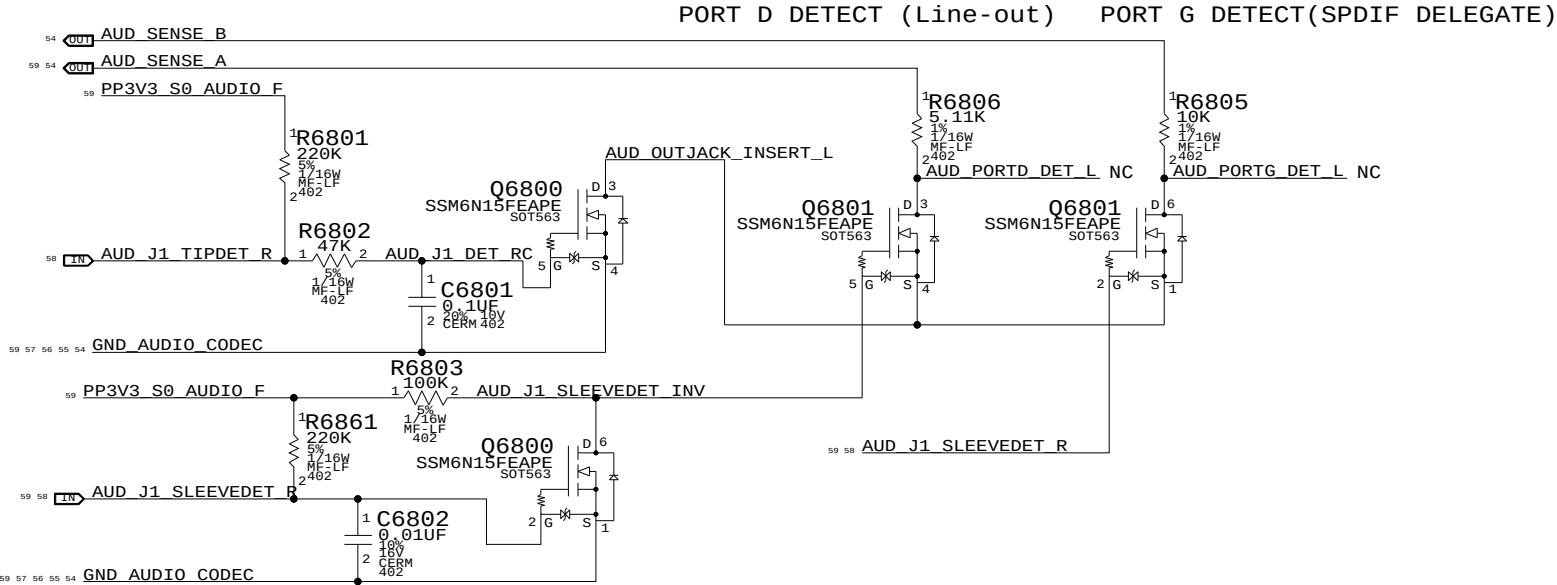
A

CODEC OUTPUT SIGNAL PATHS

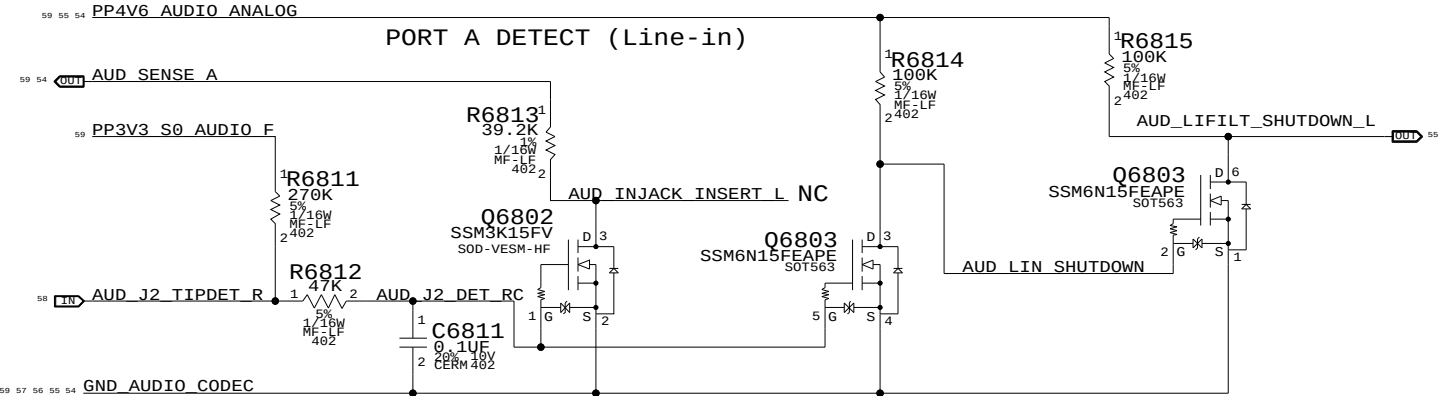
FUNCTION	VOLUME	CONVERTER	MIXER(OUTPUT)	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X0C (12)	0X02 (2)	0X0C (12)	0X14 (20,D)	GPIO_0	0X14 (20,D)
SATELLITES	0X0D (13)	0X03 (3)	0X0D (13)	0X18 (24,B)	VREF_B (100%)	N/A
SUB	0X0F (15)	0X05 (05)	0X0F (15)	0X1A (26,C)	VREF_B (100%)	N/A
SPDIF OUT	N/A	0X06 (6)		0X1E (SPDIF OUT)	N/A	0X16 (22,G)

CODEC INPUT SIGNAL PATHS

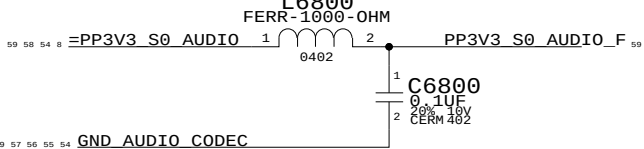
FUNCTION	MIXER(INPUT)	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
LINE IN	0X23 (35)	0X08 (8)	0X15 (21,A)	VREF_A (50%)	0X15 (21,A)
SPDIF IN	N/A	0X0A (10)	0X1F (SPDIF IN)	N/A	N/A
BUILT-IN MIC	0X24 (36)	0X07 (7)	0X19 (25,F)	VREF_F (100%)	N/A
HEADSET MIC	0X24 (36)	0X07 (7)	0X1B (27,E)	MIKEY	MIKEY



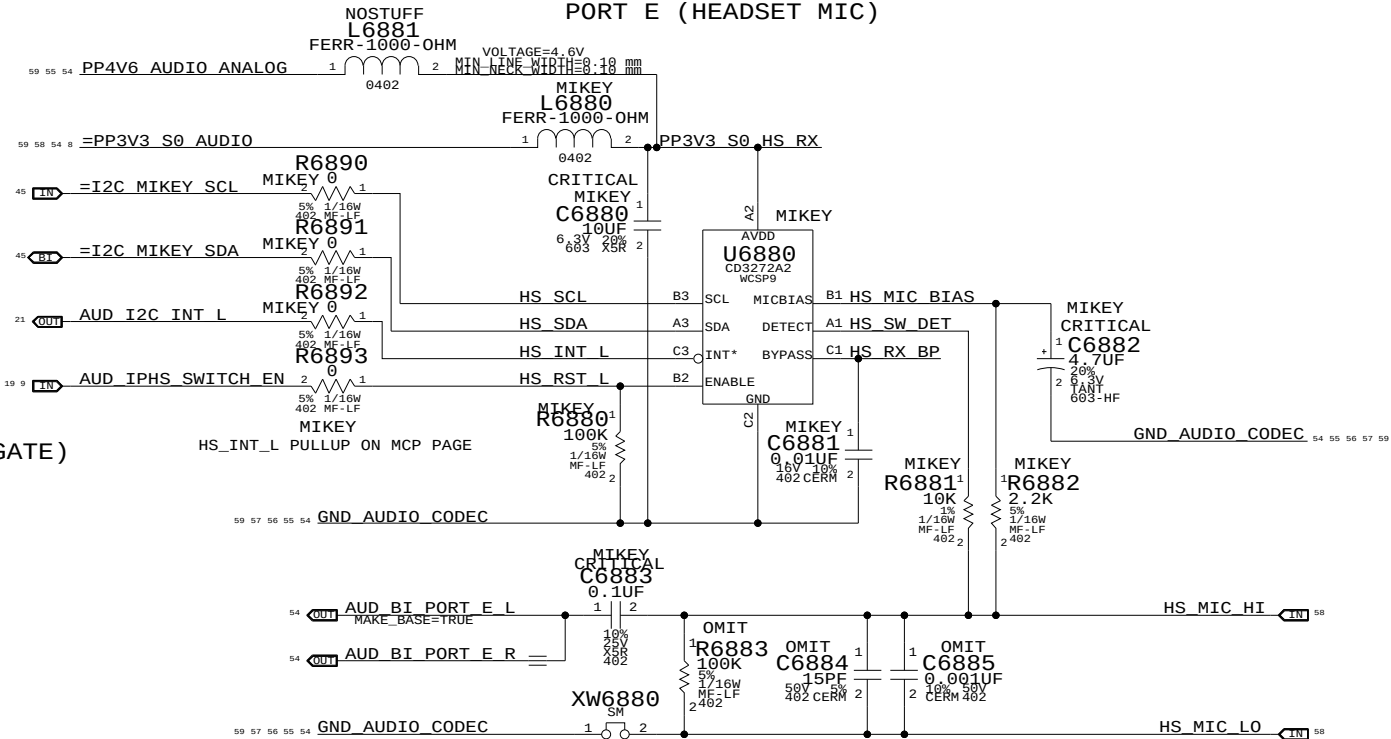
LINE_IN AMP SHUTDOWN CONTROL



PLACE L6800/C6800 CLOSE TO Q6800/01/02

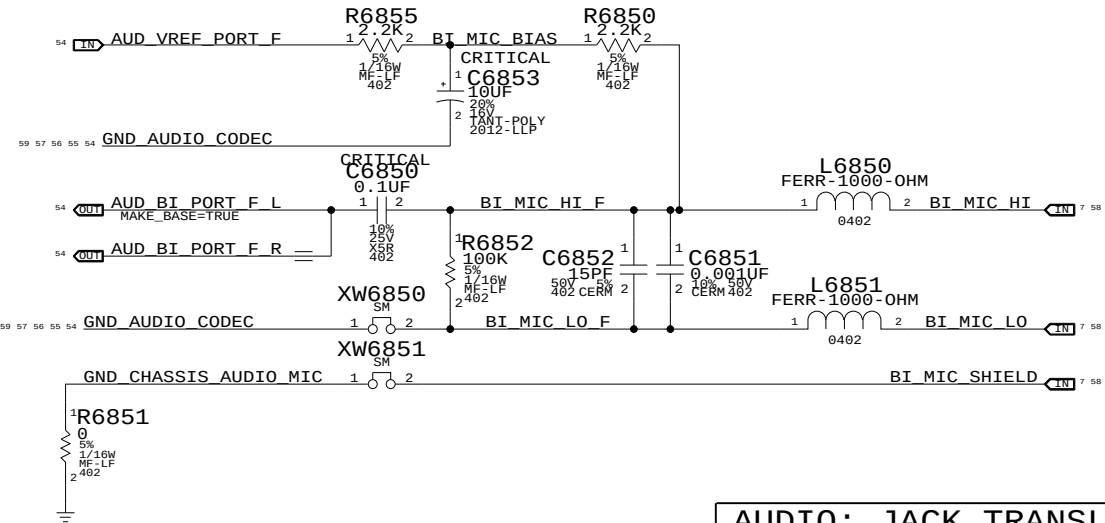


PORT E (HEADSET MIC)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0114	1	100K 5% 0402 RESISTOR	R6883	MIKEY
131S1513	1	15PF 5% 0402 CAPACITOR	C6884	MIKEY
132S0045	1	100PF 10% 0402 CAPACITOR	C6885	MIKEY
116S0004	1	0 OHMS 5% 0402 RESISTOR	R6883	NOMIKEY
116S0004	1	0 OHMS 5% 0402 RESISTOR	C6884	NOMIKEY
116S0004	1	0 OHMS 5% 0402 RESISTOR	C6885	NOMIKEY

PORT F (BUILT-IN MIC)



AUDIO: JACK TRANSLATORS

SYNC_MASTER=AUDIO SYNC_DATE=07/09/2008

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APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	59	96

8	7	6	5	4	3	2	1
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C

B

A



C

B

A

SYNC_MASTER=T18_MLB SYNC_DATE=12/06/2007

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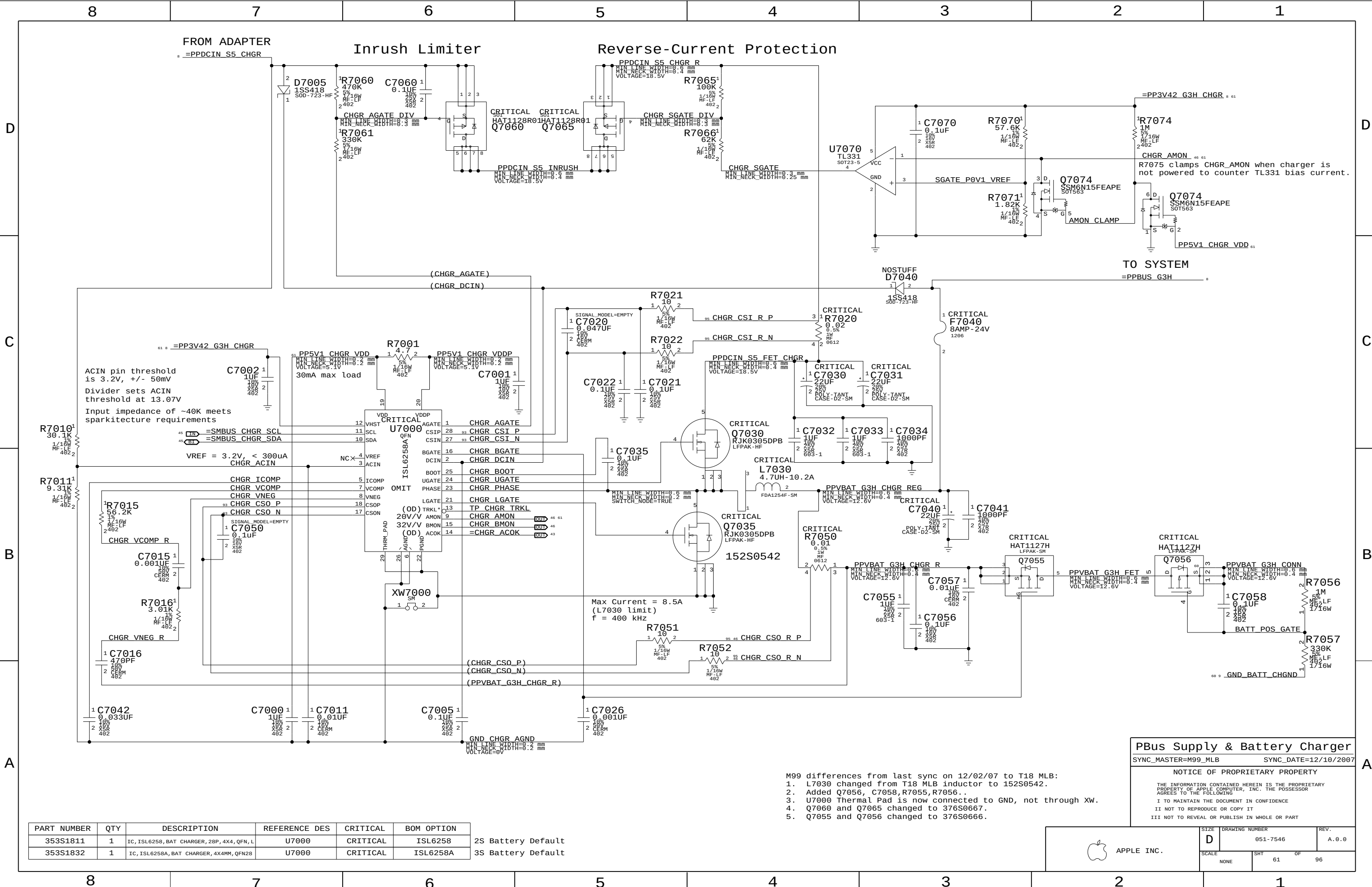


D	051-7546	A.O.
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SCALE	SHI	OF
NONE	60	96

1

	I
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S1811	1	IC, ISL6258, BAT CHARGER, 28P, 4X4, QFN, L	U7000	CRITICAL	ISL6258
353S1832	1	IC, ISL6258A, BAT CHARGER, 4X4MM, QFN28	U7000	CRITICAL	ISL6258A

2S Battery Default
3S Battery Default

PBus Supply & Battery Charger

SYNC_MASTER=M99_MLB

SYNC_DATE=12/10/2007

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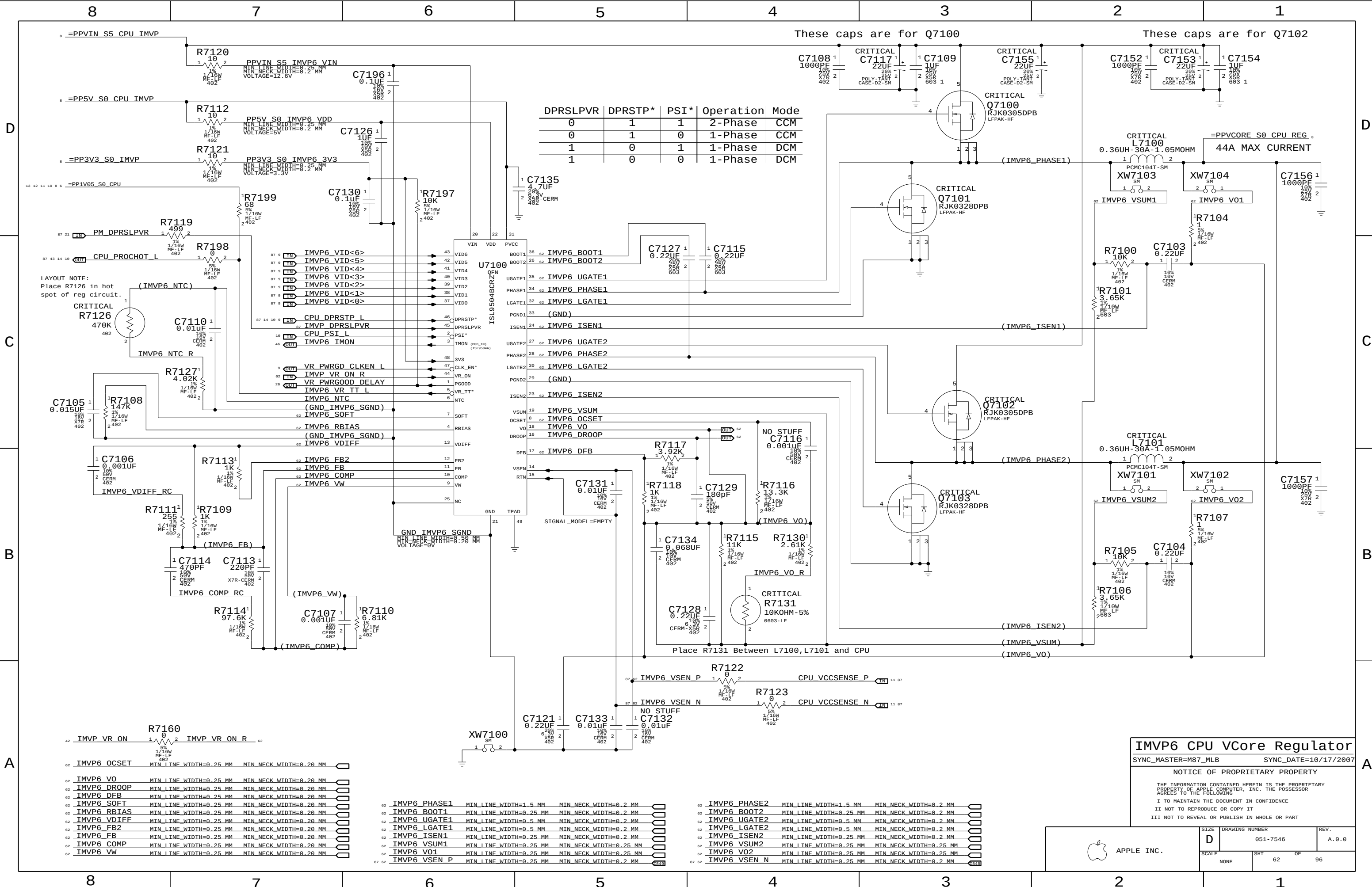
APPLE INC.

SCALE: NONE

DRAWING NUMBER: 051-7546

SHT: 61 OF 96

REV.: A.0.0



DPRSLPVR	DPRSTP*	PSI*	Operation	Mode
0	1	1	2-Phase	CCM
0	1	0	1-Phase	CCM
1	0	1	1-Phase	DCM
1	0	0	1-Phase	DCM

IMVP6 CPU VCore Regulator

SYNC_MASTER=M87_MLB SYNC_DATE=10/17/2007

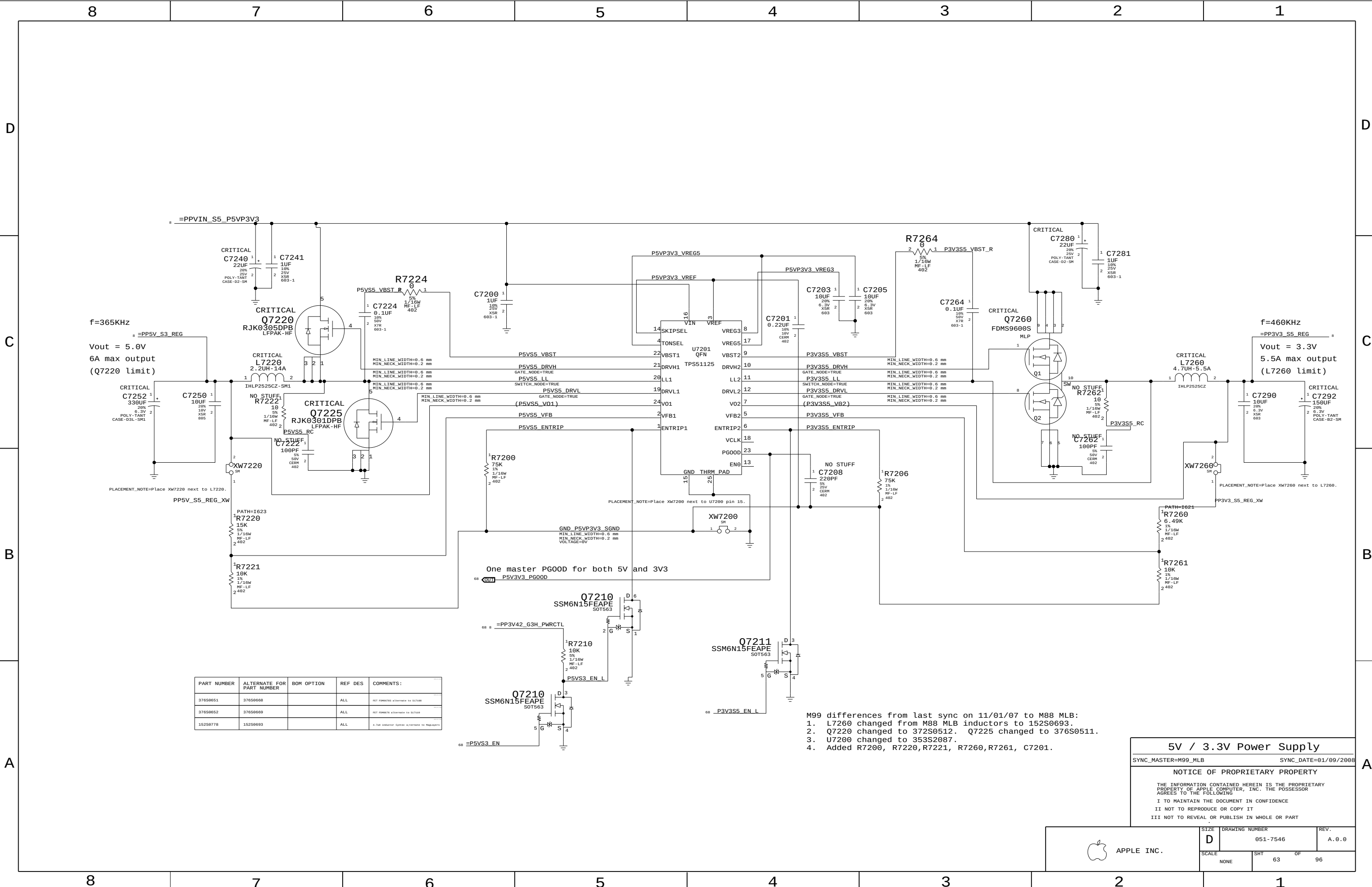
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APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	62	96

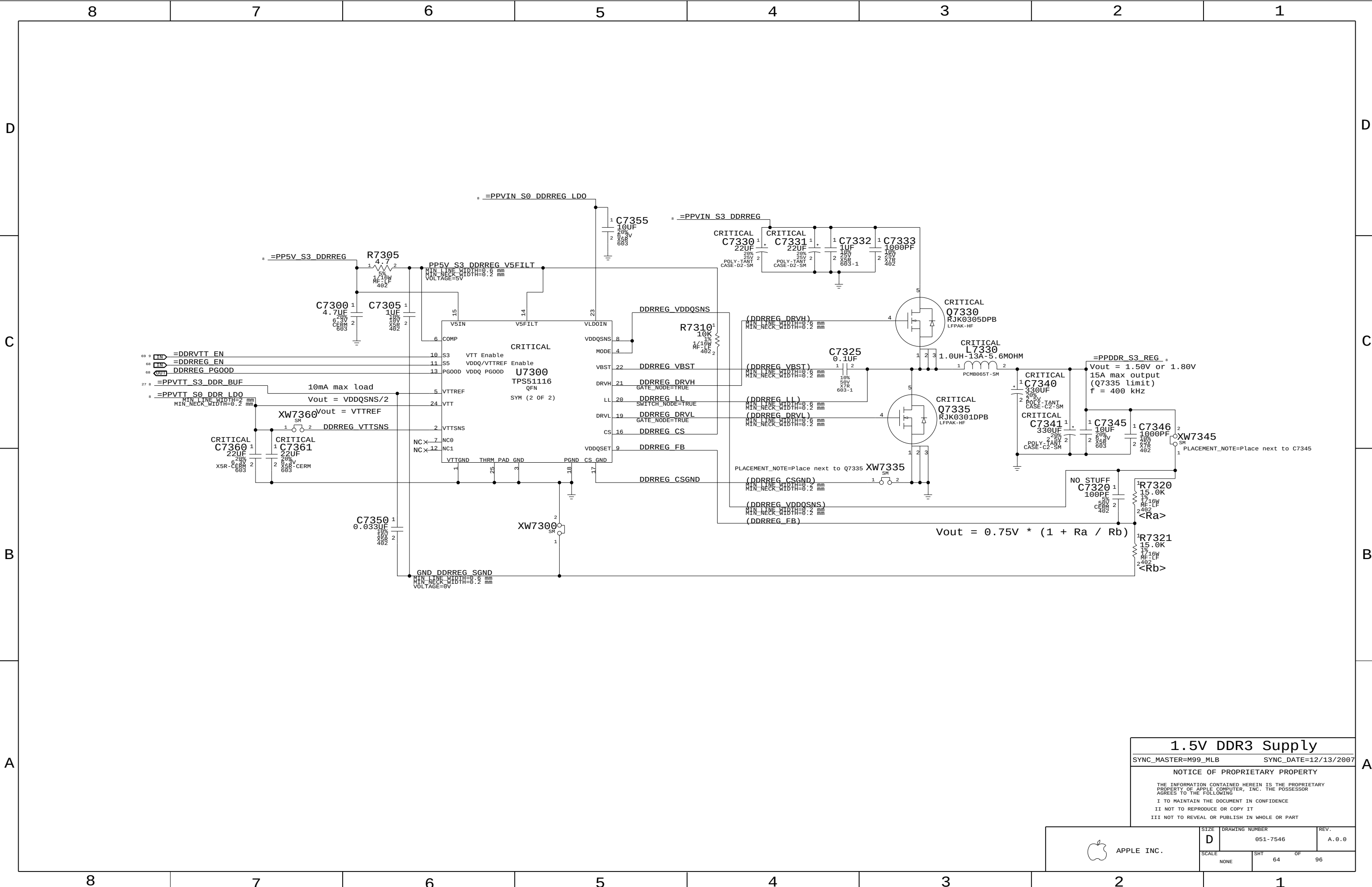


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
37650651	37650668		ALL	EXT FORMERLY ALTERNATE TO S1700
37650652	37650669		ALL	EXT FORMERLY ALTERNATE TO S1700
15250778	15250693		ALL	A.Tad Inductor Option alternate to Mag-Apple's

M99 differences from last sync on 11/01/07 to M88 MLB:
1. L7260 changed from M88 MLB inductors to 152S0693.
2. Q7220 changed to 372S0512. Q7225 changed to 376S0511.
3. U7200 changed to 353S2087.
4. Added R7200, R7220, R7221, R7260, R7261, C7201.

5V / 3.3V Power Supply	
SYNC_MASTER=M99_MLB	SYNC_DATE=01/09/2008
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	D	051-7546	A.0.0
SCALE		SHT	OF
NONE		63	96



1.5V DDR3 Supply

SYNC_MASTER=M99_MLB

SYNC_DATE=12/13/2007

NOTICE OF PROPRIETARY PROPERTY

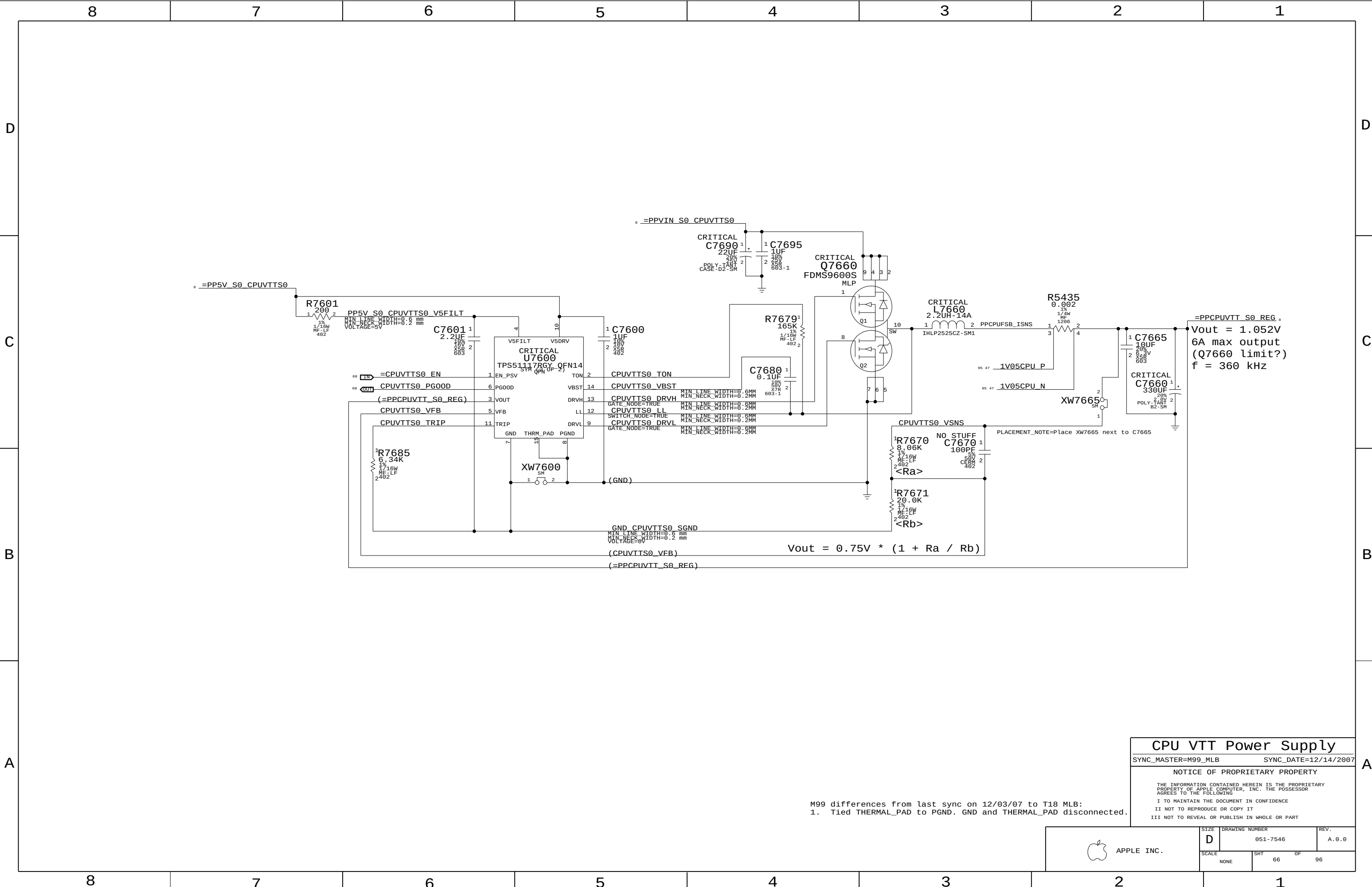
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	D	051-7546	A.0.0
SCALE		SHT	OF
NONE		64	96



M99 differences from last sync on 12/03/07 to T18 MLB:
1. Tied THERMAL_PAD to PGND. GND and THERMAL_PAD disconnected.

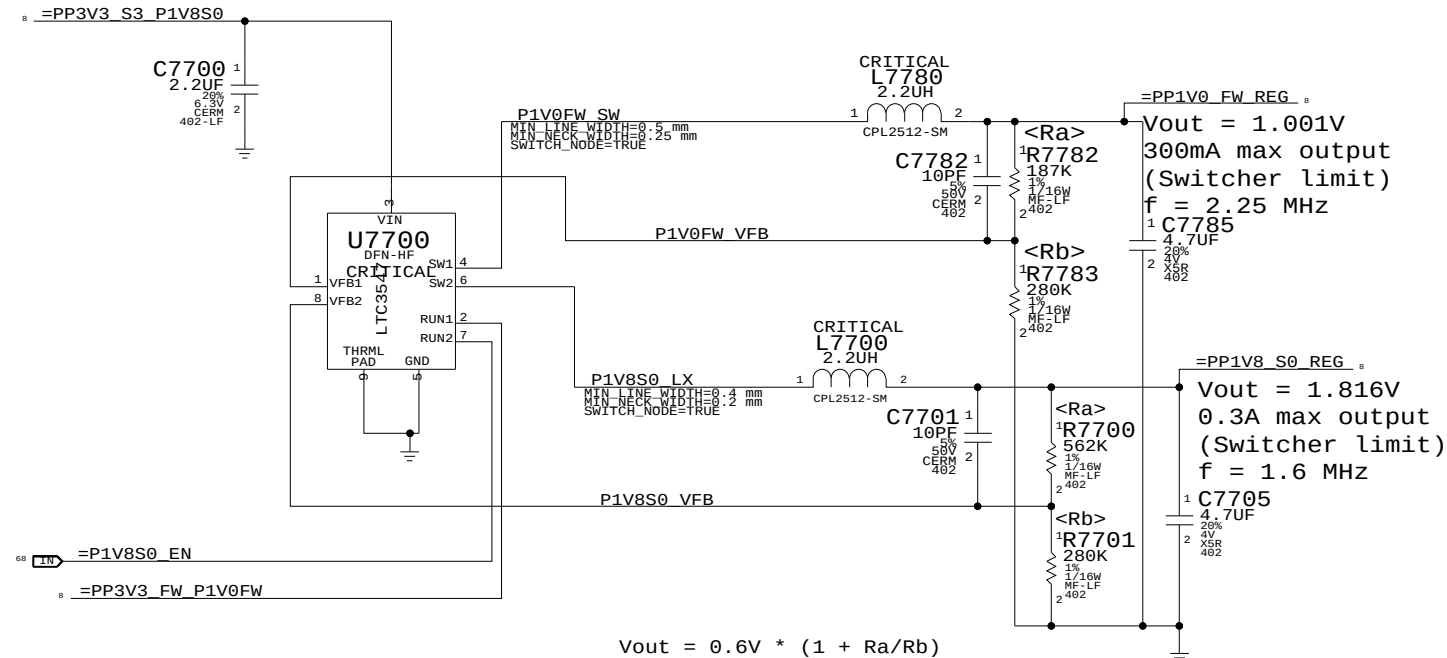
CPU VTT Power Supply		
SYNC_MASTER=M99_MLB		SYNC_DATE=12/14/2007
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SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 66 OF 96



APPLE INC.

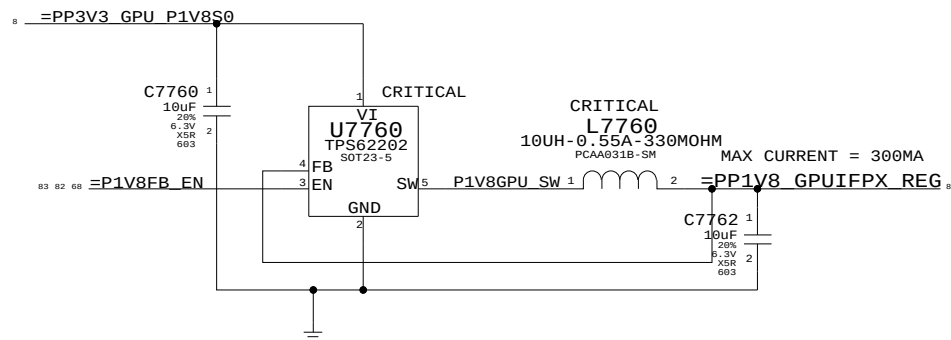
1.8V S0 Switcher / 1.0VFW SWITCHER

S5 power required for output discharge feature

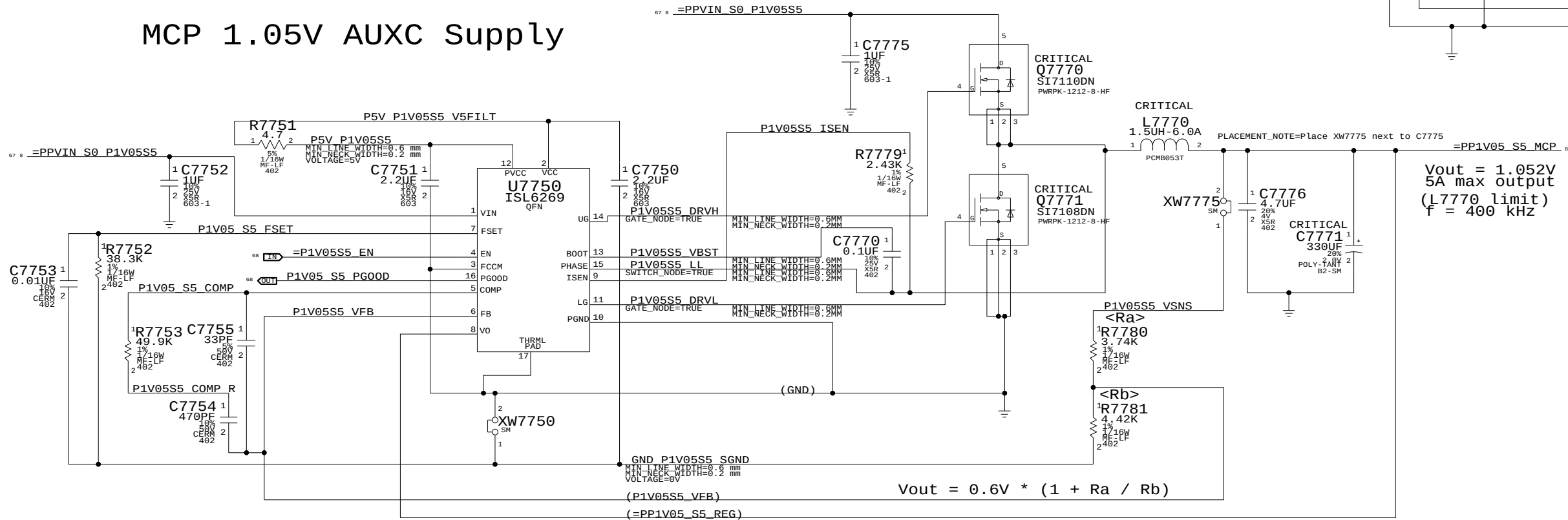


1.8V S0 Switcher

INPUT RAIL IS 3.3V S0



MCP 1.05V AUXC Supply



Misc Power Supplies

SYNC_MASTER=M99_MLB SYNC_DATE=12/14/2007

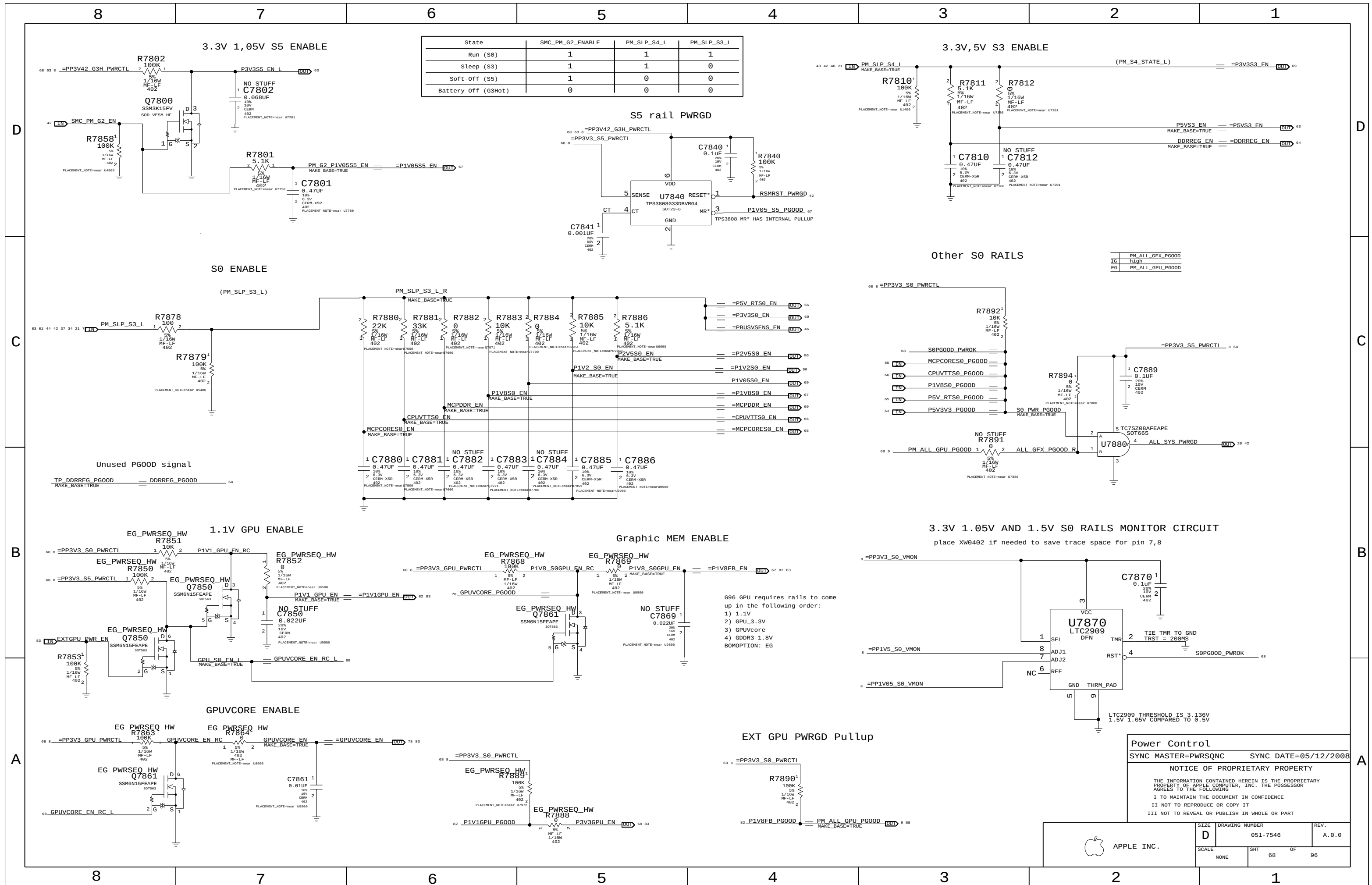
NOTICE OF PROPRIETARY PROPERTY

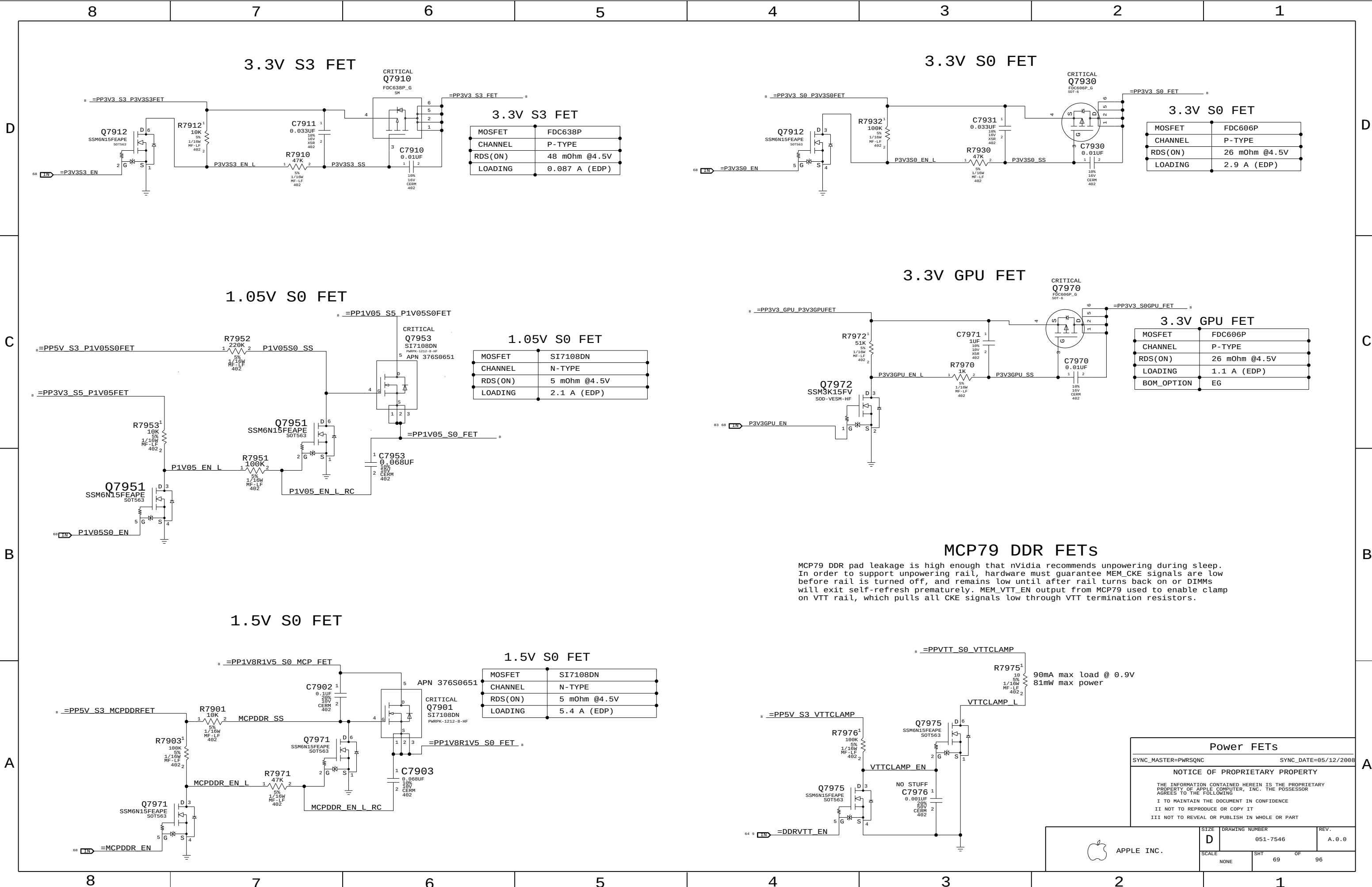
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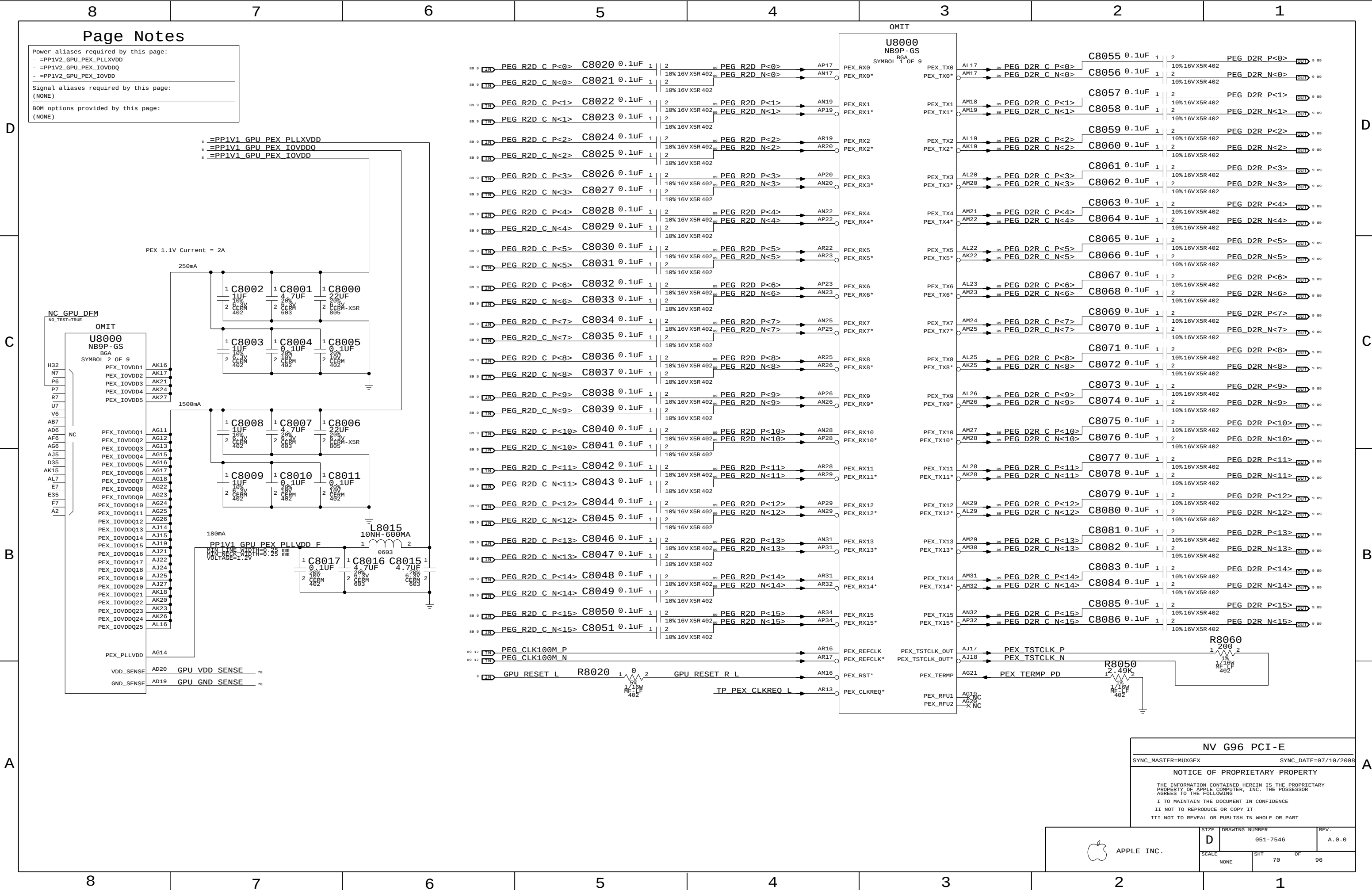


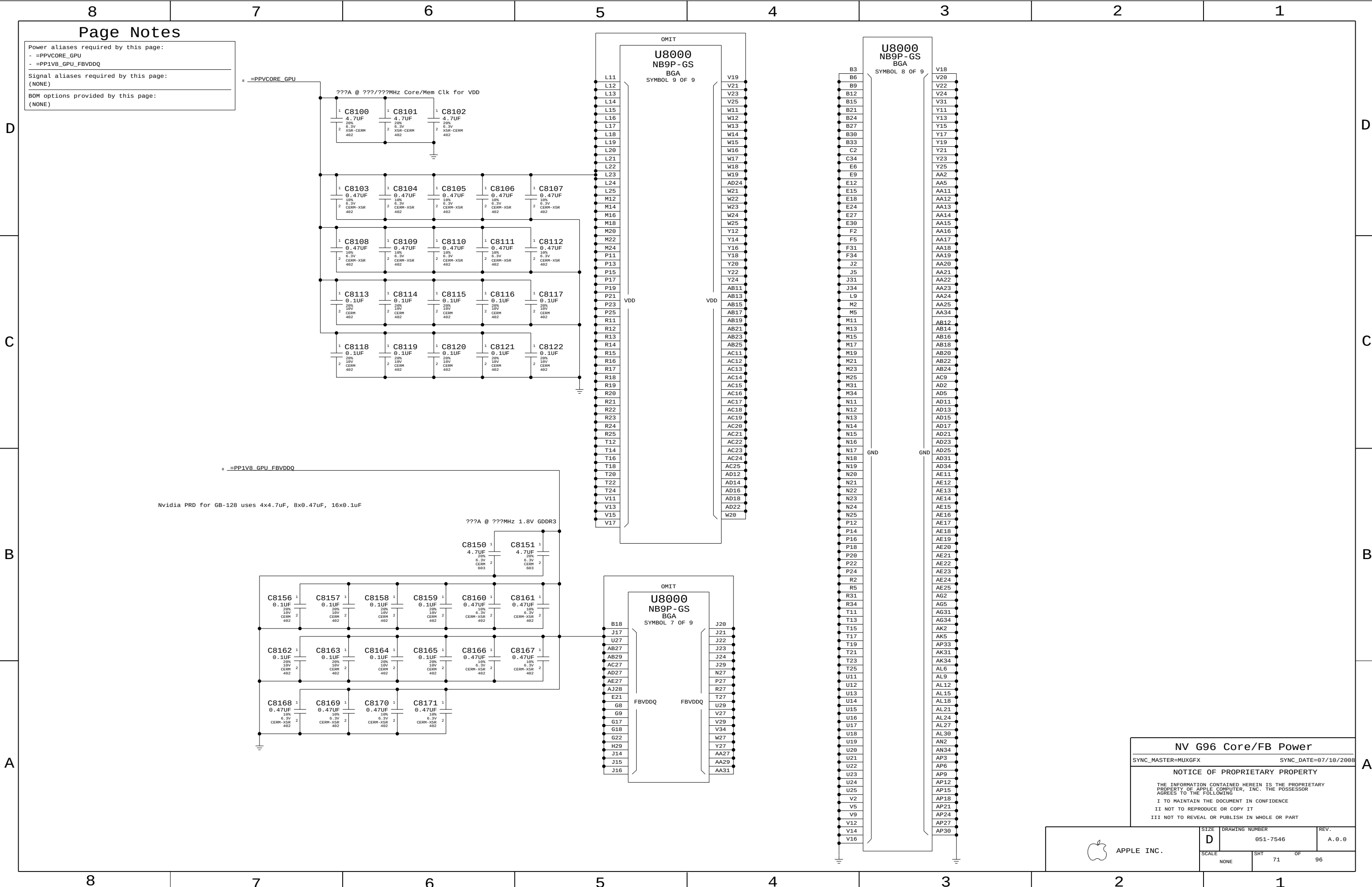
APPLE INC.

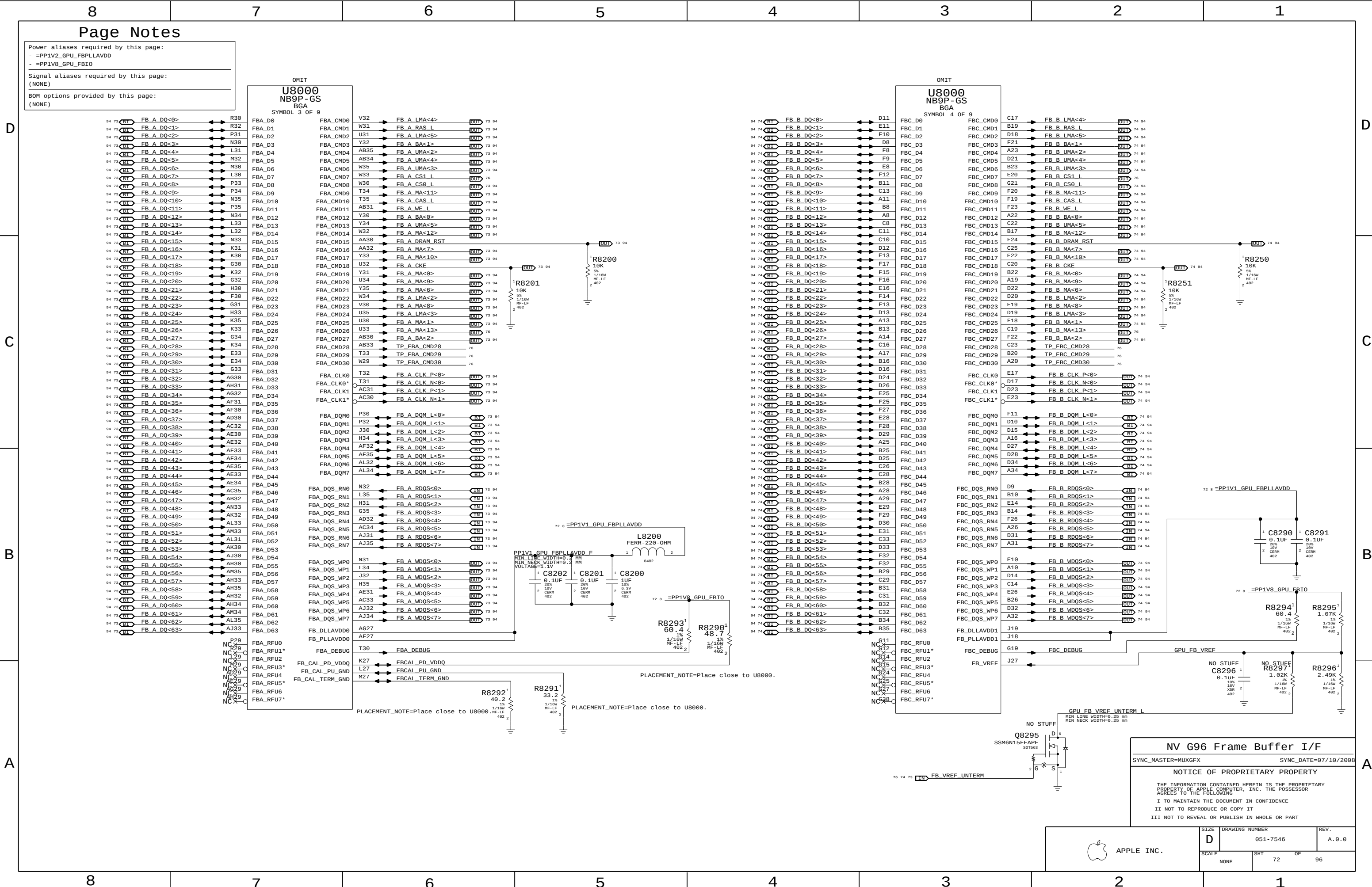
SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	67	96











Page Notes

Power aliases required by this page:
- =PP1V2_GPU_FBLLAVDD
- =PP1V8_GPU_FBIO

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

OMIT
U8000
NB9P-GS
BGA
SYMBOL 3 OF 9

OMIT
U8000
NB9P-GS
BGA
SYMBOL 4 OF 9

NV G96 Frame Buffer I/F

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

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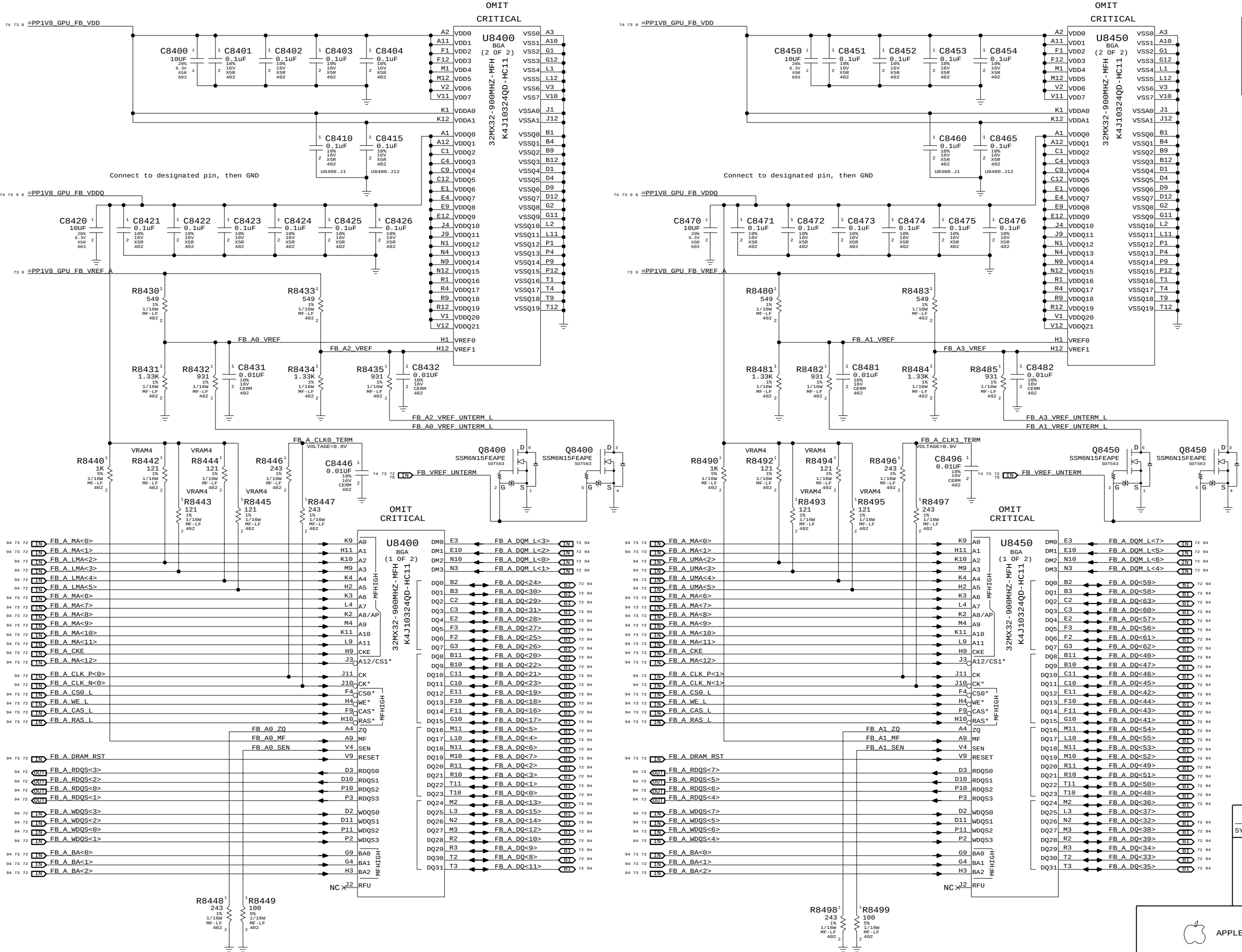
APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	72	96

Power aliases required by this page:
 - =PP1V8_S0_FB_VDD
 - =PP1V8_S0_FB_VREFA

Signal aliases required by this page:
 (NONE)

BOM options provided by this page:
 VRAM4



GDDR3 Frame Buffer A (Top)

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

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SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	73	96

8

7

6

5

4

3

2

1

Page Notes

Power aliases required by this page:

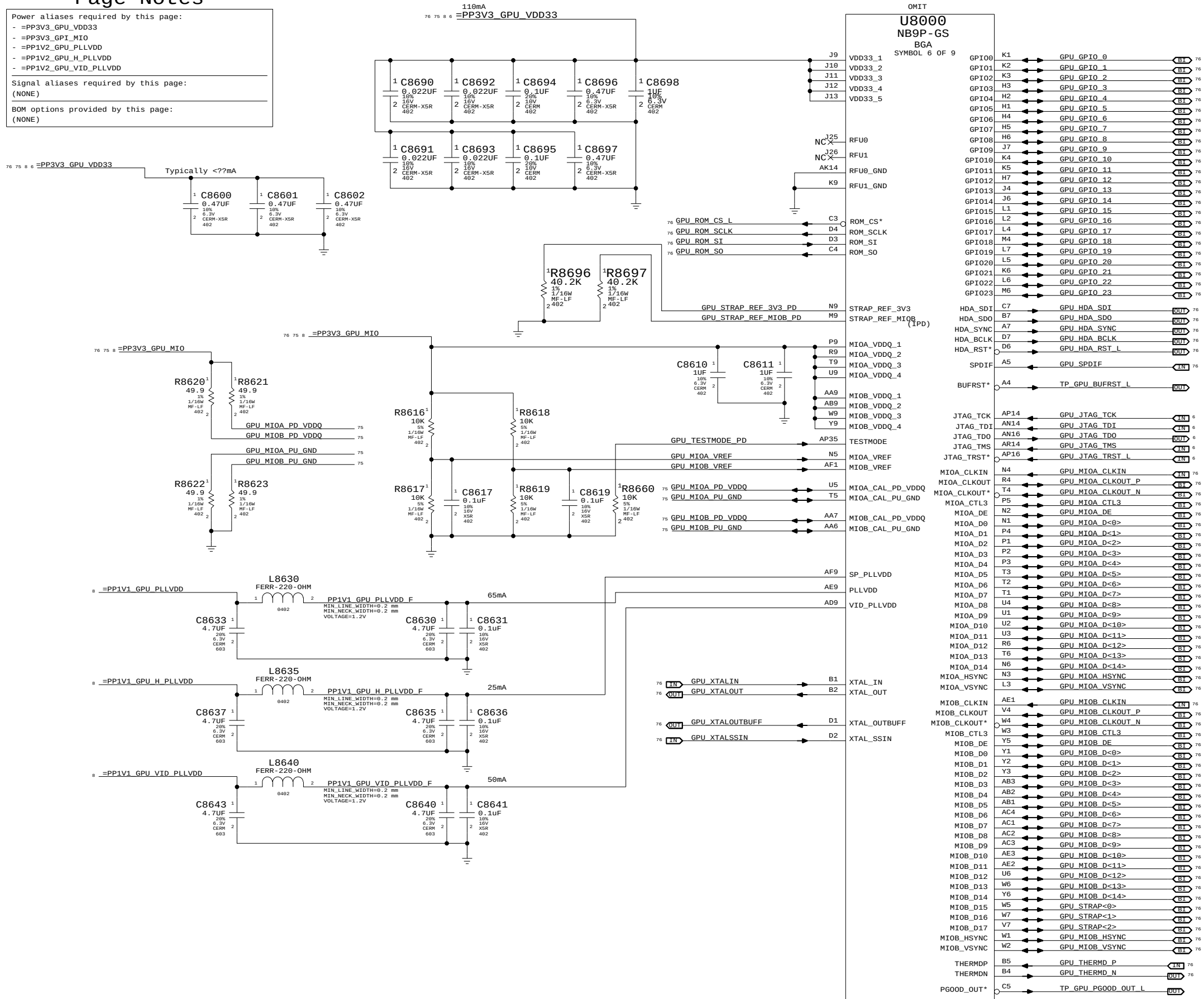
- =PP3V3_GPU_VDD33
- =PP3V3_GPU_MIO
- =PP1V2_GPU_PLLVDD
- =PP1V2_GPU_H_PLLVDD
- =PP1V2_GPU_VID_PLLVDD

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)



NV G96 GPIO/MIO/Misc

SYNC_MASTER=MUXGFX

SYNC_DATE=07/10/2008

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SIZE

D

DRAWING NUMBER

051-7546

REV.

A.0.0

SCALE

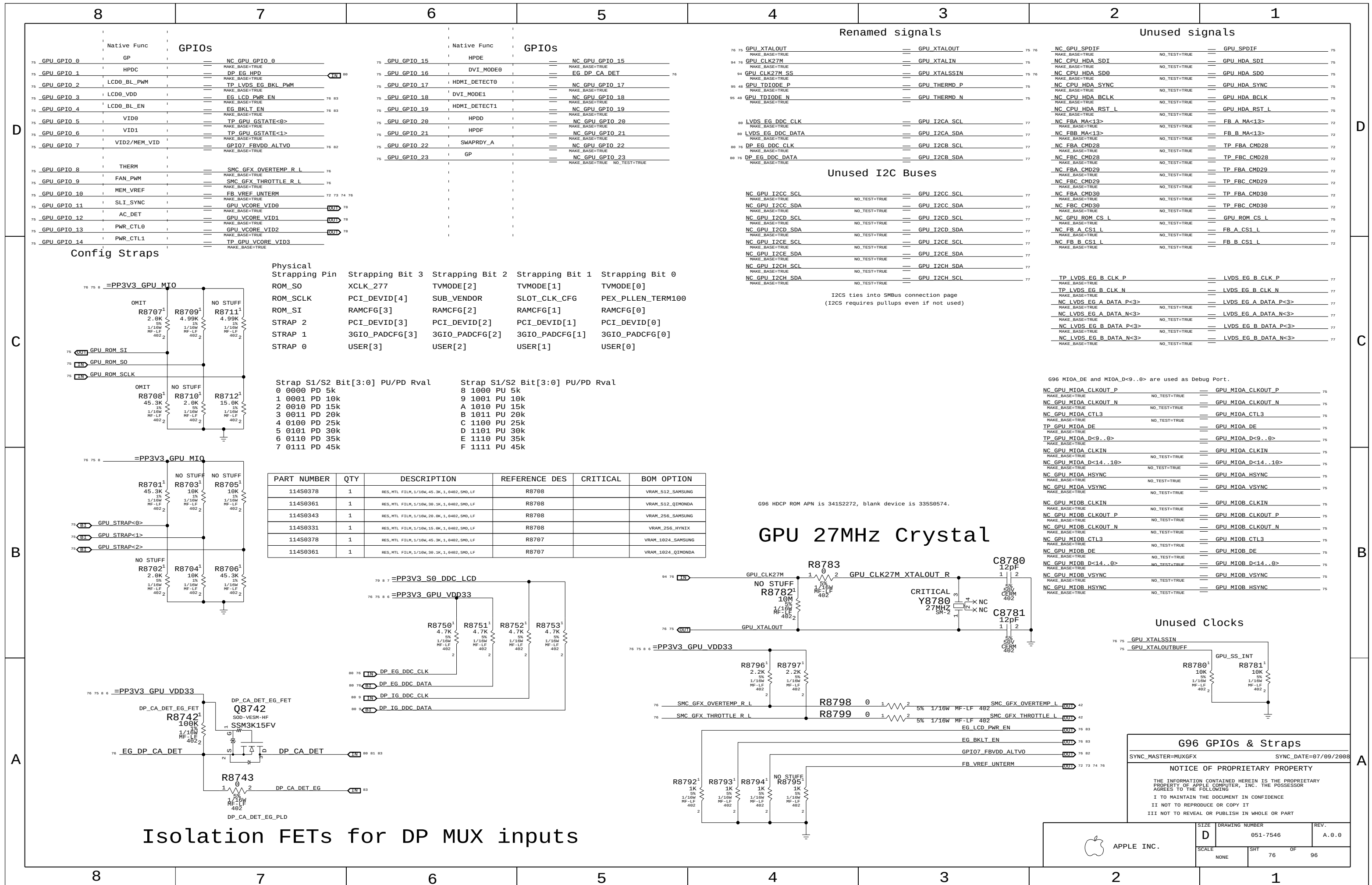
NONE

SHT

75

OF

96



Page Notes

Power aliases required by this page:
- =PP1V8_GPU_IFPX
- =PP3V3_GPU_IFPCD_IOVDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Sum of peak currents: 240mA

=PP1V8_GPU_IFPX

L8800

FERR-220-OHM

0402

?mA peak per diff pair
?mA peak for all pairs

C8800 4.7UF 20% 6.3V CERM 603
C8801 0.1UF 20% 18V CERM 402
C8803 0.1UF 20% 18V CERM 402

Place at AG9 Place at AG10

PP1V8_GPU_IFPAB_IOVDD F

MIN_LINE_WIDTH=0.4 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=1.0V

L8805

FERR-220-OHM

0402

80mA peak

C8805 4.7UF 20% 6.3V CERM 603
C8806 0.1UF 20% 18V CERM 402

Place at AJ8 Place at AK8

PP1V1_GPU_IFPCD_IOVDD F

MIN_LINE_WIDTH=0.4 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=1.0V

PP1V1_GPU_IFPEF_IOVDD F

PP1V8_GPU_IFPCD_PLLVDD F

PP1V8_GPU_IFPEF_PLLVDD F

GPU_IFPAB_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

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GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

GPU_IFPCD_RSET

GPU_IFPEF_RSET

OMIT

U8000

NB9P-GS

BGA

SYMBOL 5 OF 9

IFPA_TXC

IFPB_TXC

IFPC_TXC

IFPD_TXC

IFPE_TXC

IFPF_TXC

IFPA_TXD0

IFPB_TXD0

IFPC_TXD0

IFPD_TXD0

IFPE_TXD0

IFPF_TXD0

IFPA_TXD1

IFPB_TXD1

IFPC_TXD1

IFPD_TXD1

IFPE_TXD1

IFPF_TXD1

IFPA_TXD2

IFPB_TXD2

IFPC_TXD2

IFPD_TXD2

IFPE_TXD2

IFPF_TXD2

IFPA_TXD3

IFPB_TXD3

IFPC_TXD3

IFPD_TXD3

IFPE_TXD3

IFPF_TXD3

IFPA_TXD4

IFPB_TXD4

IFPC_TXD4

IFPD_TXD4

IFPE_TXD4

IFPF_TXD4

IFPA_TXD5

IFPB_TXD5

IFPC_TXD5

IFPD_TXD5

IFPE_TXD5

IFPF_TXD5

IFPA_TXD6

IFPB_TXD6

IFPC_TXD6

IFPD_TXD6

IFPE_TXD6

IFPF_TXD6

IFPA_TXD7

IFPB_TXD7

IFPC_TXD7

IFPD_TXD7

IFPE_TXD7

IFPF_TXD7

IFPA_TXD8

IFPB_TXD8

IFPC_TXD8

IFPD_TXD8

IFPE_TXD8

IFPF_TXD8

IFPA_TXD9

IFPB_TXD9

IFPC_TXD9

IFPD_TXD9

IFPE_TXD9

IFPF_TXD9

IFPA_TXD10

IFPB_TXD10

IFPC_TXD10

IFPD_TXD10

IFPE_TXD10

IFPF_TXD10

IFPA_TXD11

IFPB_TXD11

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IFPD_TXD11

IFPE_TXD11

IFPF_TXD11

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IFPB_TXD12

IFPC_TXD12

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IFPF_TXD12

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IFPB_TXD13

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IFPD_TXD13

IFPE_TXD13

IFPF_TXD13

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IFPB_TXD14

IFPC_TXD14

IFPD_TXD14

IFPE_TXD14

IFPF_TXD14

IFPA_TXD15

IFPB_TXD15

IFPC_TXD15

IFPD_TXD15

IFPE_TXD15

IFPF_TXD15

IFPA_TXD16

IFPB_TXD16

IFPC_TXD16

IFPD_TXD16

IFPE_TXD16

IFPF_TXD16

IFPA_TXD17

IFPB_TXD17

IFPC_TXD17

IFPD_TXD17

IFPE_TXD17

IFPF_TXD17

IFPA_TXD18

IFPB_TXD18

IFPC_TXD18

IFPD_TXD18

IFPE_TXD18

IFPF_TXD18

IFPA_TXD19

IFPB_TXD19

IFPC_TXD19

IFPD_TXD19

IFPE_TXD19

IFPF_TXD19

IFPA_TXD20

IFPB_TXD20

IFPC_TXD20

IFPD_TXD20

IFPE_TXD20

IFPF_TXD20

IFPA_TXD21

IFPB_TXD21

IFPC_TXD21

IFPD_TXD21

IFPE_TXD21

IFPF_TXD21

IFPA_TXD22

IFPB_TXD22

IFPC_TXD22

IFPD_TXD22

IFPE_TXD22

IFPF_TXD22

IFPA_TXD23

IFPB_TXD23

IFPC_TXD23

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IFPE_TXD24

IFPF_TXD24

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IFPB_TXD25

IFPC_TXD25

IFPD_TXD25

IFPE_TXD25

IFPF_TXD25

IFPA_TXD26

IFPB_TXD26

IFPC_TXD26

IFPD_TXD26

IFPE_TXD26

IFPF_TXD26

IFPA_TXD27

IFPB_TXD27

IFPC_TXD27

IFPD_TXD27

IFPE_TXD27

IFPF_TXD27

IFPA_TXD28

IFPB_TXD28

IFPC_TXD28

IFPD_TXD28

IFPE_TXD28

IFPF_TXD28

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IFPC_TXD29

IFPD_TXD29

IFPE_TXD29

IFPF_TXD29

IFPA_TXD30

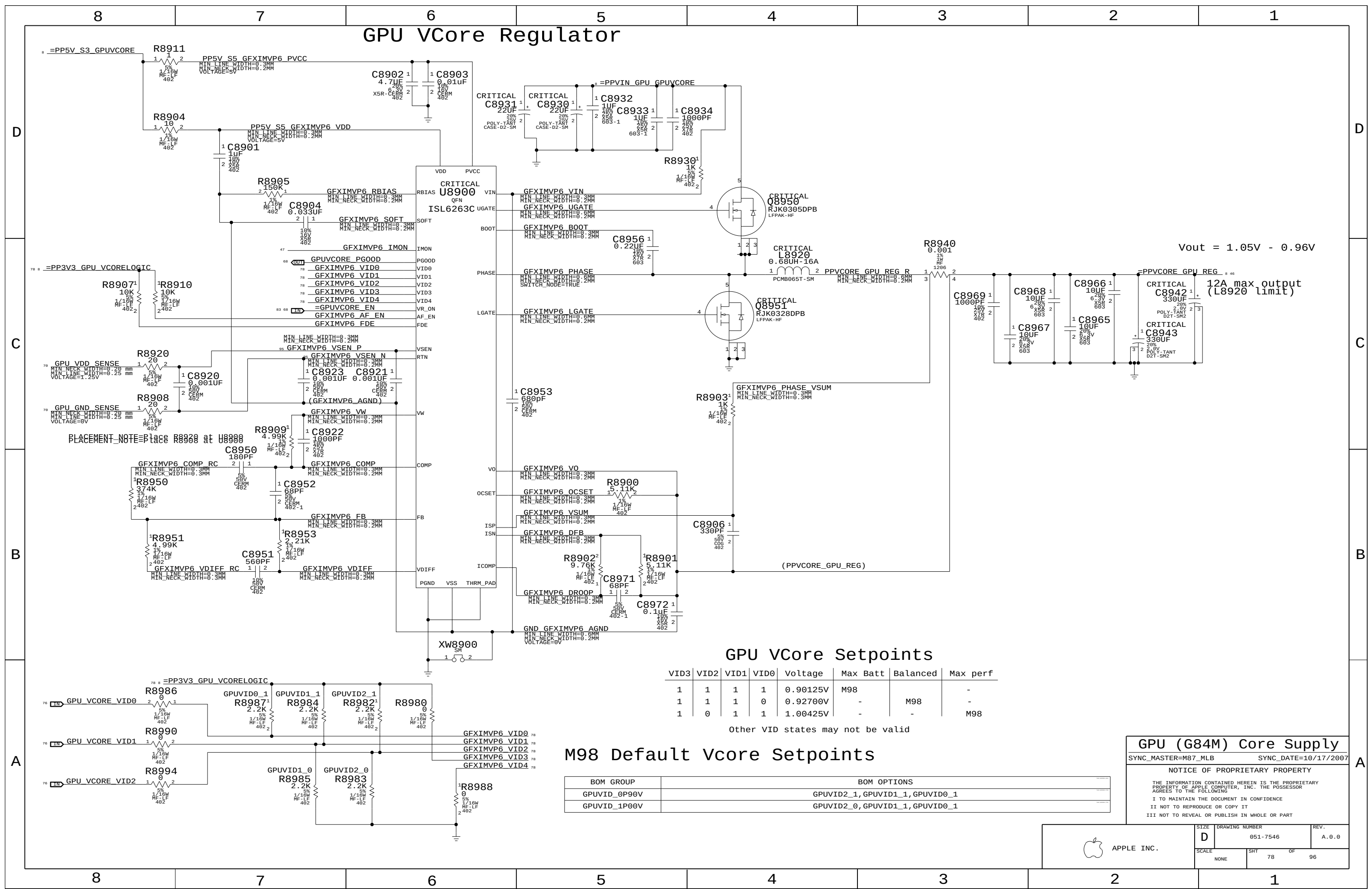
IFPB_TXD30

IFPC_TXD30

IFPD_TXD30

IFPE_TXD30

IFPF_TXD30



M98 Default Vcore Setpoints

GPU (G84M) Core Suppl'y

SYNC_MASTER=M87_MLB	SYNC_DATE=10/17/2007
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SIZE	DRAWING NUMBER	REV.
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D	051-7546	A.0.6
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[illegible]

SCALE	SHT	OF
	78	96

NONE	75	90
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[illegible]

1

[illegible]

[illegible]

518S0651

LVDS Display Connector	
SYNC_MASTER=MUXGX	SYNC_DATE=02/25/2008
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 APPLE INC.	SIZE D	DRAWING NUMBER 051-7546	REV. A.0.0
	SCALE NONE	SHT 79	OF 96

D

C

B

A

5



3



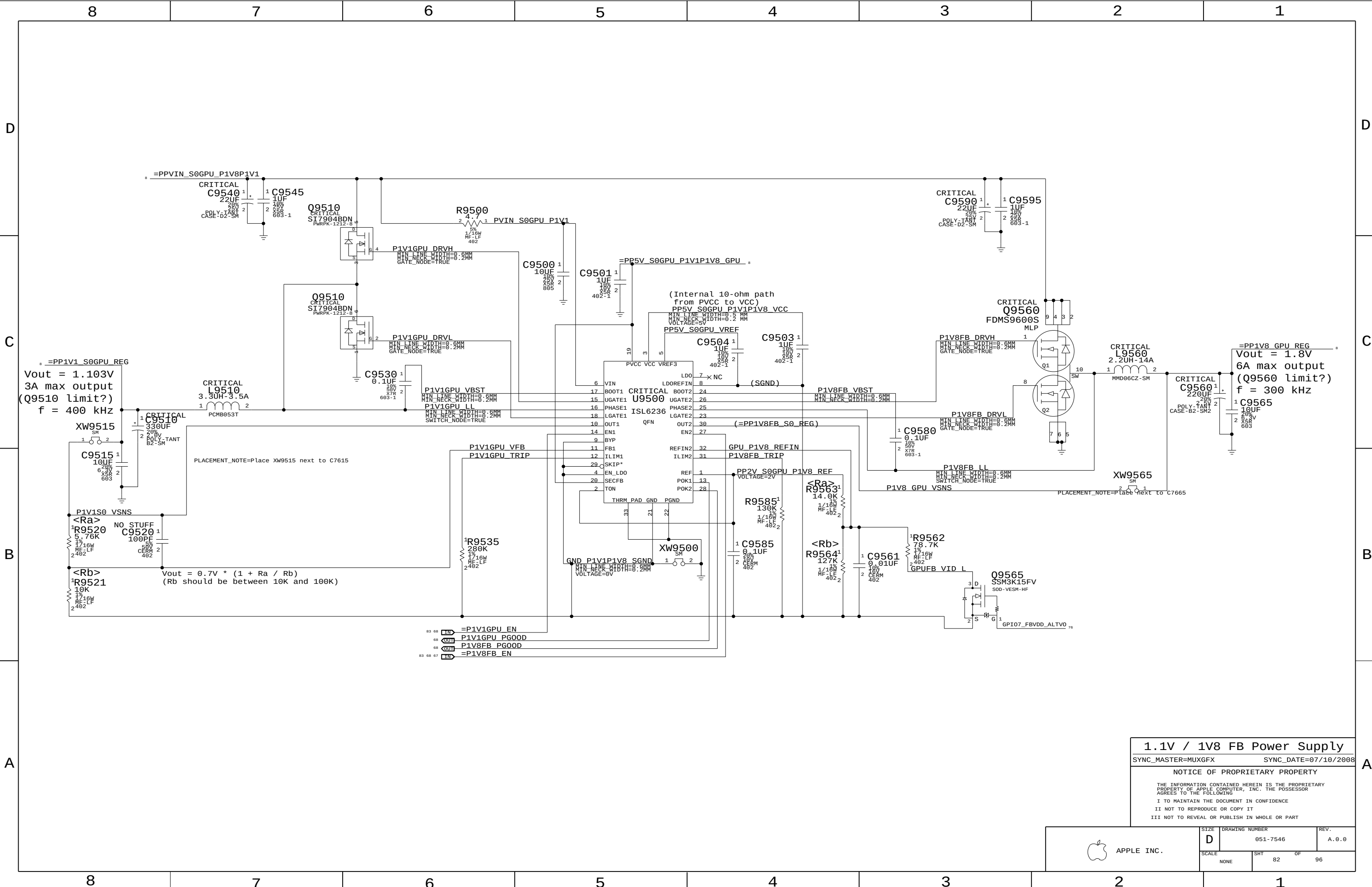
SIZE

REV.
A.0.0

SHT

80

6



1.1V / 1V8 FB Power Supply

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

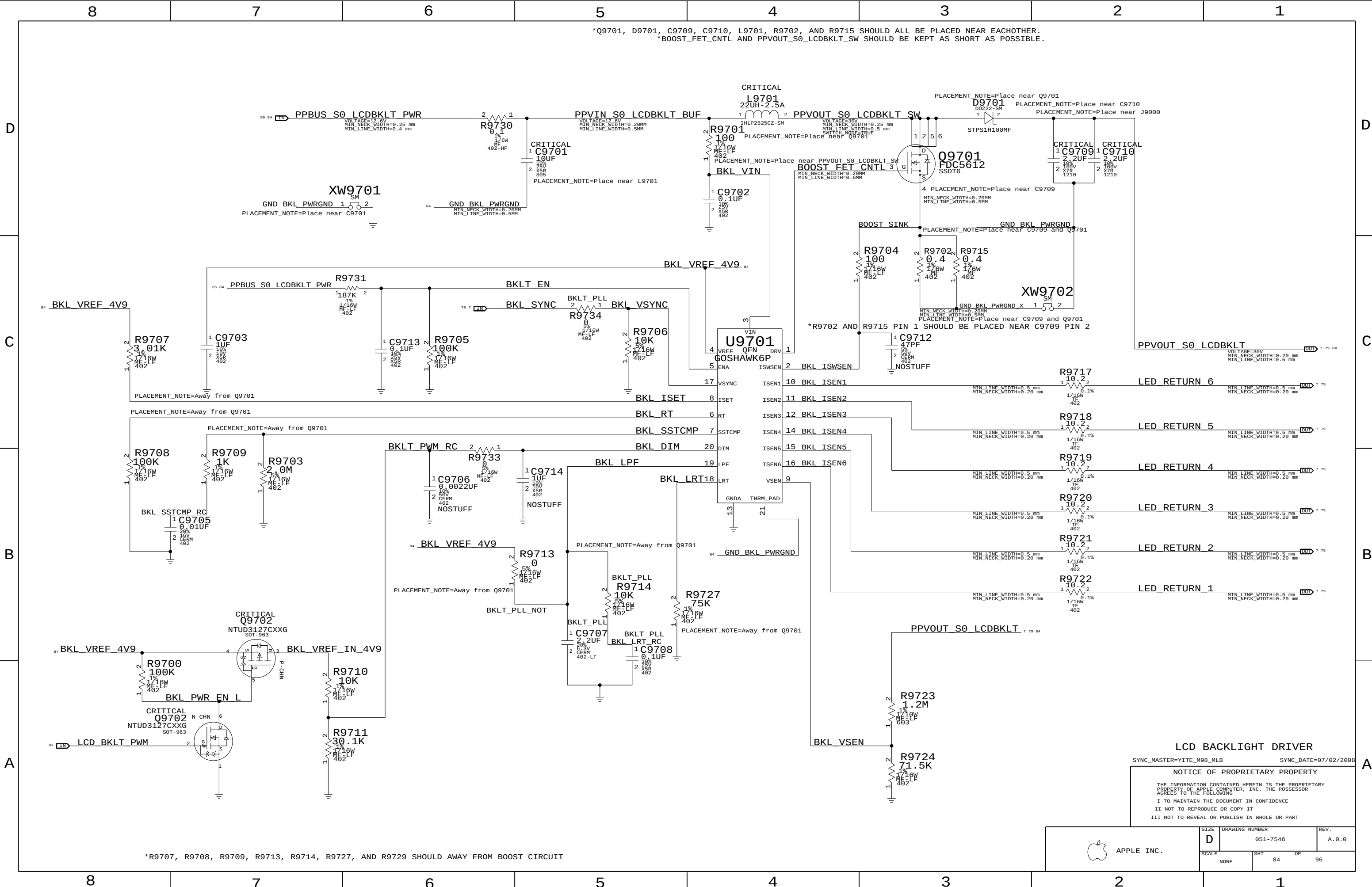
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SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	82	96



*Q9701, D9701, C9709, C9710, L9701, R9702, AND R9715 SHOULD ALL BE PLACED NEAR EACHOTHER.
*BOOST_FET_CNTL AND PPVOUT_S0_LCDBKLT_SW SHOULD BE KEPT AS SHORT AS POSSIBLE.

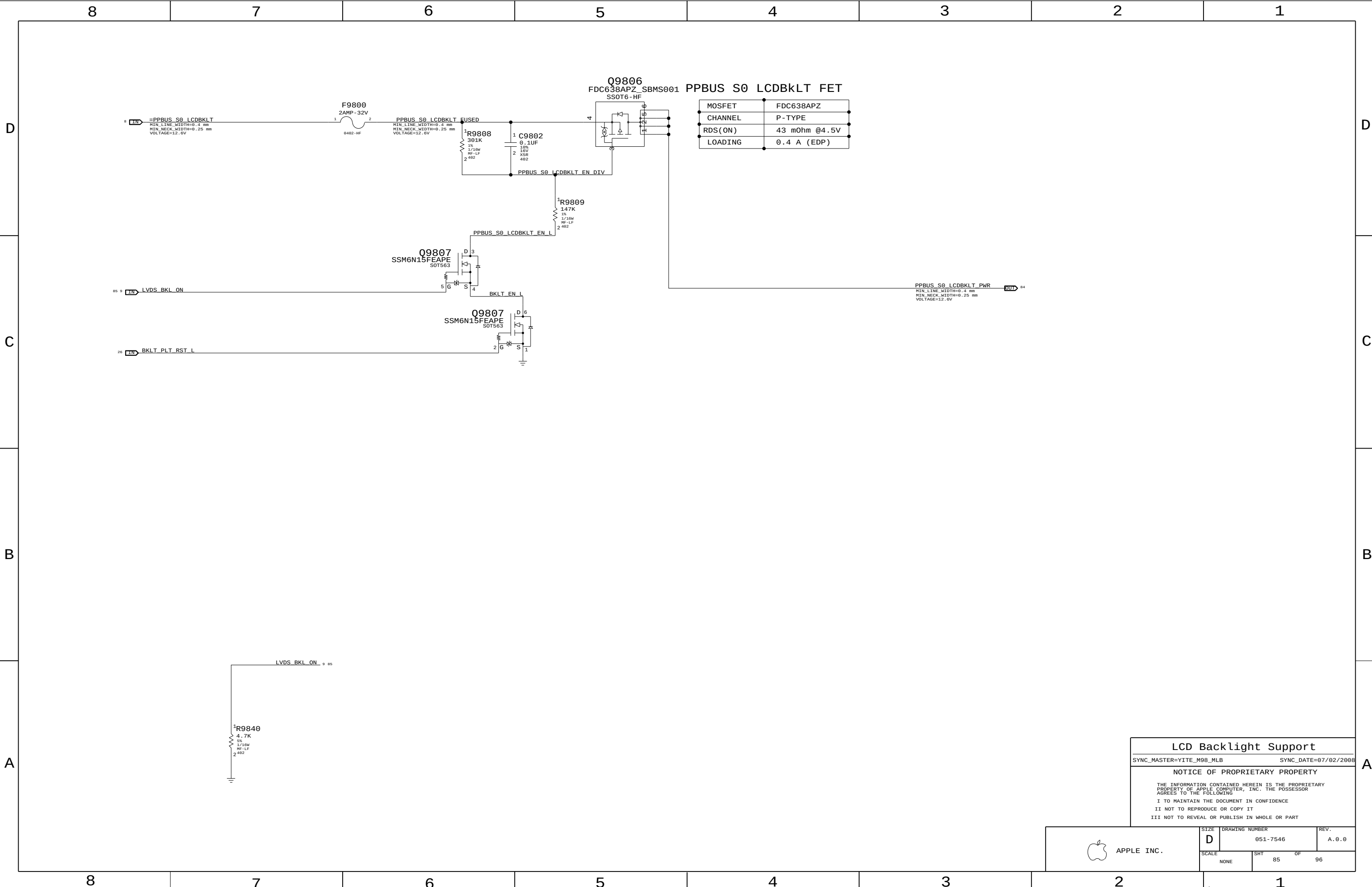
LCD BACKLIGHT DRIVER
SYNC_MASTER=YITE_M98_MLB
SYNC_DATE=07/02/2008
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SIZE D	DRAWING NUMBER 051-7546		REV. A.0.0
	SCALE NONE	SHT 84	OF 96



APPLE INC.

*R9707, R9708, R9709, R9713, R9714, R9727, AND R9729 SHOULD AWAY FROM BOOST CIRCUIT



LCD Backlight Support

SYNC_MASTER=YITE_M98_MLB

SYNC_DATE=07/02/2008

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APPLE INC.	SIZE	DRAWING NUMBER	REV.
	D	051-7546	A.0.0
SCALE		SHT	OF
NONE		85	96

2.5V/1.2V S3 Switcher

The schematic diagram illustrates a 2.5V/1.2V S3 Switcher circuit. The input is connected to a 2.2uF capacitor (C9900) and a 2.2uH inductor (L9980). The input is connected to the VIN pin of the LTC3535-1 buck-boost converter (U9900). The output is connected to a 4.7uF capacitor (C9905) and a 2.2uH inductor (L9980). The output is connected to the SW pin of the LTC3535-1. The feedback network is connected to the FB pin of the LTC3535-1 via a 10k resistor (R9982) and a 150k resistor (R9901). The feedback capacitor (C9985) is connected to the FB pin of the LTC3535-1 via a 10pF capacitor (C9982). The feedback network is connected to the output via a 10k resistor (R9983). The feedback capacitor (C9985) is connected to the feedback network via a 10pF capacitor (C9982). The feedback network is connected to the output via a 10k resistor (R9983). The feedback capacitor (C9985) is connected to the feedback network via a 10pF capacitor (C9982).

Output voltage regulation is achieved using the feedback network, with the output voltage determined by the ratio of the feedback resistors:

$$V_{out} = 0.6V * (1 + R_a/R_b)$$

Key components and their values:

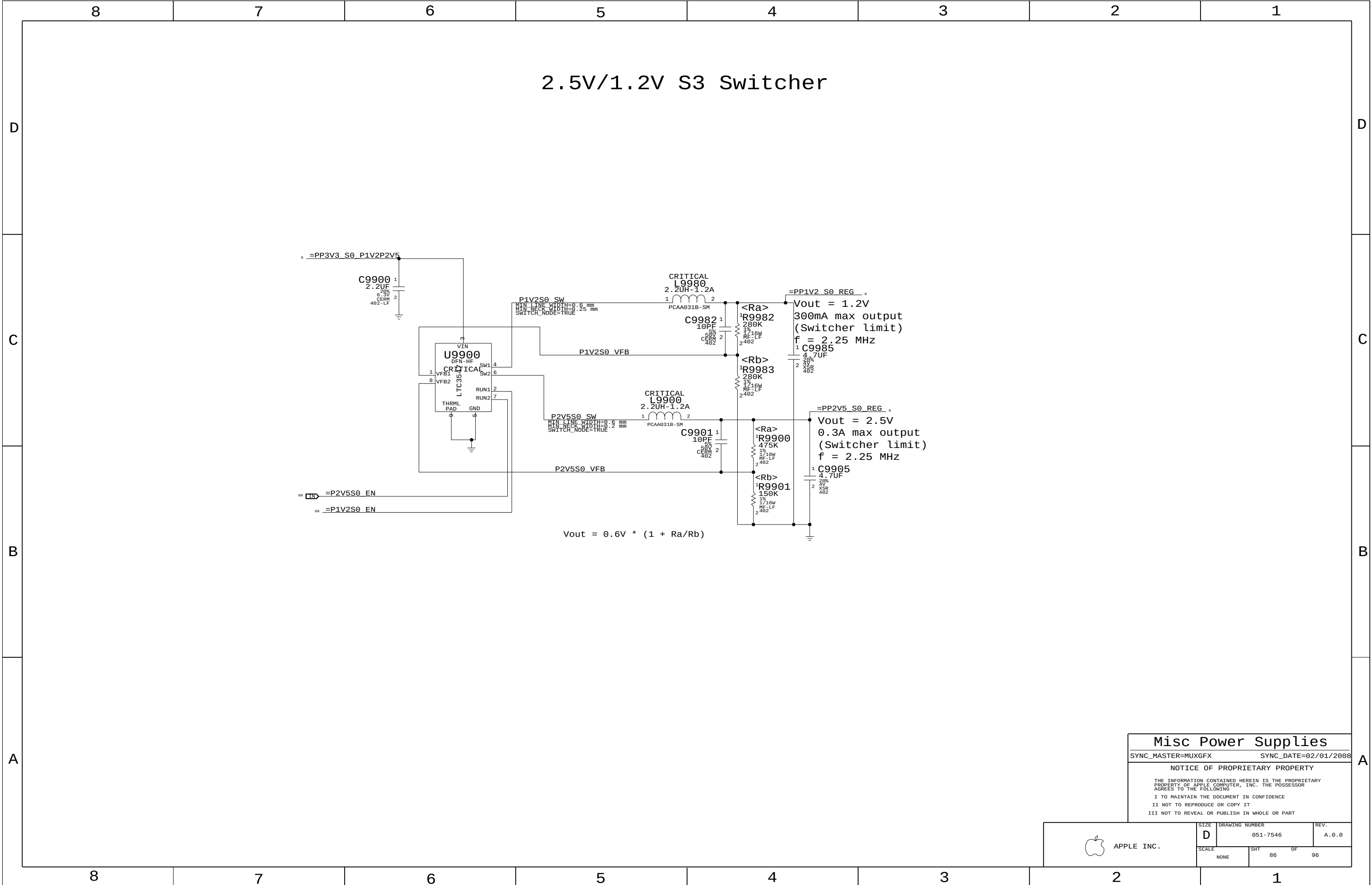
- Input capacitor: C9900 (2.2uF)
- Input inductor: L9980 (2.2uH)
- Output capacitor: C9905 (4.7uF)
- Output inductor: L9980 (2.2uH)
- Feedback resistors: R9982 (10k), R9983 (10k), R9900 (10k), R9901 (150k)
- Feedback capacitor: C9985 (10pF)
- Switcher: LTC3535-1 (U9900)

Output voltage regulation is achieved using the feedback network, with the output voltage determined by the ratio of the feedback resistors:

$$V_{out} = 0.6V * (1 + R_a/R_b)$$

Key components and their values:

- Input capacitor: C9900 (2.2uF)
- Input inductor: L9980 (2.2uH)
- Output capacitor: C9905 (4.7uF)
- Output inductor: L9980 (2.2uH)
- Feedback resistors: R9982 (10k), R9983 (10k), R9900 (10k), R9901 (150k)
- Feedback capacitor: C9985 (10pF)
- Switcher: LTC3535-1 (U9900)



2.5V/1.2V S3 Switcher

The schematic diagram illustrates a 2.5V/1.2V S3 Switcher circuit. The input is connected to a 2.2uF capacitor (C9900) and a 2.2uH inductor (L9980). The input is connected to the VIN pin of the LTC3535-1 buck-boost converter (U9900). The output is connected to a 4.7uF capacitor (C9905) and a 2.2uH inductor (L9980). The output is connected to the VOUT pin of the LTC3535-1. The feedback network is connected to the output via a resistor divider (R9900, R9901). The feedback capacitor (C9905) is connected to the output via a resistor divider (R9900, R9901). The feedback network is connected to the output via a resistor divider (R9900, R9901). The feedback capacitor (C9905) is connected to the output via a resistor divider (R9900, R9901).

Key components and values:

- Input: 2.2uF capacitor (C9900), 2.2uH inductor (L9980)
- Output: 4.7uF capacitor (C9905), 2.2uH inductor (L9980)
- Feedback network: R9982 (280K), R9983 (280K), C9985 (4.7uF)
- Feedback capacitor: C9905 (4.7uF)
- Resistor divider: R9900 (475K), R9901 (150K)

Output voltage regulation:

- Vout = 1.2V, 300mA max output (Switcher limit), f = 2.25 MHz
- Vout = 2.5V, 0.3A max output (Switcher limit), f = 2.25 MHz

Formula for output voltage:

$$V_{out} = 0.6V * (1 + R_a/R_b)$$

Legend:

- CRITICAL: 2.2uH-1.2A
- CRITICAL: 2.2uH-1.2A
- CRITICAL: 2.2uH-1.2A

Notes:

- MIN LINE WIDTH=0.5 mm
- MIN SPACE WIDTH=0.25 mm
- SWITCH_NODE=TRUE

Apple Inc. logo and text: APPLE INC.

Document information:

- SIZE: D
- DRAWING NUMBER: 051-7546
- REV.: A.0.0
- SCALE: NONE
- SHT: 86 OF 96

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2.5V/1.2V S3 Switcher

The schematic diagram illustrates a 2.5V/1.2V S3 Switcher circuit. The input is connected to a 2.2uF capacitor (C9900) and the output is connected to a 2.2uF capacitor (C9901). The output voltage is regulated to 1.2V and 2.5V. The output current is limited to 300mA and 0.3A respectively. The output voltage is also regulated to 0.6V * (1 + Ra/Rb).

Output Specifications:

- Vout = 1.2V, 300mA max output (Switcher limit), f = 2.25 MHz
- Vout = 2.5V, 0.3A max output (Switcher limit), f = 2.25 MHz
- Vout = 0.6V * (1 + Ra/Rb)

Component Values:

- C9900: 2.2uF, 20%, 0.3V, 402-LF
- C9901: 2.2uF, 20%, 0.3V, 402-LF
- C9982: 10pF, 5%, 10V, 402
- C9983: 10pF, 5%, 10V, 402
- C9905: 4.7uF, 20%, 2.5V, X5R, 402
- R9982: 280K, 1%, 1/10W, 402
- R9983: 280K, 1%, 1/10W, 402
- R9900: 475K, 1%, 1/10W, 402
- R9901: 150K, 1%, 1/10W, 402

Notes:

- CRITICAL: 2.2uH-1.2A
- CRITICAL: 2.2uH-1.2A
- CRITICAL: 2.2uH-1.2A

Table:

Misc Power Supplies		
SYNC_MASTER=MUXGFX	SYNC_DATE=02/01/2008	
NOTICE OF PROPRIETARY PROPERTY		
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III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART		
SIZE	DRAWING NUMBER	REV.
D	051-7546	A.0.0
SCALE	SHT	OF
NONE	86	96

APPLE INC.

87654321

FSB (Front-Side Bus) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FSB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
FSB_DSTB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	*	=2x_DIELECTRIC	?
FSB_DSTB	*	=3x_DIELECTRIC	?
FSB_ADDR	*	=STANDARD	?
FSB_ADSTB	*	=2x_DIELECTRIC	?
FSB_1X	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_DSTB	TOP,BOTTOM	=5x_DIELECTRIC	?
FSB_ADDR	TOP,BOTTOM	=3x_DIELECTRIC	?
FSB_ADSTB	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_1X	TOP,BOTTOM	=3x_DIELECTRIC	?

All 4x/2x/1x FSB signals with impedance requirements are 50-ohm single-ended.

FSB 4X signals / groups shown in signal table on right.
Signals within each 4x group should be matched within 5 ps of strobe.
DSTB# complementary pairs should be matched within 1 ps of each other, all DSTB#s matched to +/- 300 ps.
Spacing is 2x dielectric between DATA#, DINV# signals, with 3x dielectric spacing to the DSTB#s.
DSTB# complementary pairs are spaced normally and are NOT routed as differential pairs.

FSB 2X signals / groups shown in signal table on right.
Signals within each 2x group should be matched within 20 ps. ADTSB#s should be matched +/- 300 ps.
Spacing is 1x dielectric between ADDR#, REQ# signals, with 2x dielectric spacing to ADSTB#.

FSB 1X signals shown in signal table on right.
Signals within each 1x group should be matched to CPU clock, +0/-1000 mils.

Design Guide recommends each strobe/signal group is routed on the same layer.
Intel Design Guide recommends FSB signals be routed only on internal layers.

NOTE: Intel Design Guide allows closer spacing if signal lengths can be shortened.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2
SOURCE: Santa Rosa Platform DG, Rev 1.5 (#22294), Sections 4.2 & 4.3

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	25 MIL	?
CPU_GTLREF	*	25 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

SR DG recommends at least 25 mils, >50 mils preferred

Most CPU signals with impedance requirements are 55-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2
SOURCE: Santa Rosa Platform DG, Rev 0.9 (#20517), Sections 4.4 & 5.8.2.4

MCP FSB COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_FSB_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.4

FSB Clock Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_FSB_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	*	=3x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.5

CPU / FSB Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
FSB_DATA_GROUP0	FSB_50S	FSB_DATA	FSB D L<15..0>	7 10 14
FSB_DATA_GROUP0	FSB_50S	FSB_DATA	FSB DINV L<0>	7 10 14
FSB_DSTB0	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L P<0>	7 10 14
FSB_DSTB0	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L N<0>	7 10 14
FSB_DATA_GROUP1	FSB_50S	FSB_DATA	FSB D L<31..16>	7 10 14
FSB_DATA_GROUP1	FSB_50S	FSB_DATA	FSB DINV L<1>	7 10 14
FSB_DSTB1	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L P<1>	7 10 14
FSB_DSTB1	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L N<1>	7 10 14
FSB_DATA_GROUP2	FSB_50S	FSB_DATA	FSB D L<47..32>	7 10 14
FSB_DATA_GROUP2	FSB_50S	FSB_DATA	FSB DINV L<2>	7 10 14
FSB_DSTB2	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L P<2>	7 10 14
FSB_DSTB2	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L N<2>	7 10 14
FSB_DATA_GROUP3	FSB_50S	FSB_DATA	FSB D L<63..48>	7 10 14
FSB_DATA_GROUP3	FSB_50S	FSB_DATA	FSB DINV L<3>	7 10 14
FSB_DSTB3	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L P<3>	7 10 14
FSB_DSTB3	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L N<3>	7 10 14
FSB_ADDR_GROUP0	FSB_50S	FSB_ADDR	FSB A L<16..3>	7 10 14
FSB_ADDR_GROUP0	FSB_50S	FSB_ADDR	FSB REQ L<4..0>	7 10 14
FSB_ADSTB0	FSB_50S	FSB_ADSTB	FSB ADSTB L<0>	7 10 14
FSB_ADDR_GROUP1	FSB_50S	FSB_ADDR	FSB A L<35..17>	7 10 14
FSB_ADSTB1	FSB_50S	FSB_ADSTB	FSB ADSTB L<1>	7 10 14
FSB_1X	FSB_50S	FSB_1X	FSB ADS L	7 10 14
FSB_BREQ0_L	FSB_50S	FSB_1X	FSB_BREQ0_L	9 10 14
FSB_BREQ1_L	FSB_50S	FSB_1X	FSB_BREQ1_L	14
FSB_1X	FSB_50S	FSB_1X	FSB BNR L	10 14
FSB_1X	FSB_50S	FSB_1X	FSB BPRT L	10 14
FSB_1X	FSB_50S	FSB_1X	FSB DBSY L	10 14
FSB_1X	FSB_50S	FSB_1X	FSB DEFER L	10 14
FSB_1X	FSB_50S	FSB_1X	FSB DRDY L	10 14
FSB_1X	FSB_50S	FSB_1X	FSB HIT L	7 10 14
FSB_1X	FSB_50S	FSB_1X	FSB HITM L	7 10 14
FSB_1X	FSB_50S	FSB_1X	FSB LOCK L	7 10 14
FSB_CPURST_L	FSB_50S	FSB_1X	FSB_CPURST_L	9 10 13 14
FSB_1X	FSB_50S	FSB_1X	FSB RS L<2..0>	10 14
FSB_1X	FSB_50S	FSB_1X	FSB TRDY L	10 14
CPU_ASYNC	CPU_50S	CPU_AGTL	CPU A20M L	10 14
CPU_BSEL	CPU_50S	CPU_AGTL	CPU_BSEL<2..0>	9 10
CPU_FERR_L	CPU_50S	CPU_8MTL	CPU_FERR L	10 14
CPU_ASYNC	CPU_50S	CPU_AGTL	CPU IGNNE L	10 14
CPU_INIT_L	CPU_50S	CPU_AGTL	CPU_INIT L	10 14
CPU_ASYNC_R	CPU_50S	CPU_AGTL	CPU_INTR	9 10 14
CPU_ASYNC_R	CPU_50S	CPU_AGTL	CPU NMI	9 10 14
CPU_PROCHOT_L	CPU_50S	CPU_AGTL	CPU_PROCHOT_L	10 14 43 62
CPU_PWRGD	CPU_50S	CPU_AGTL	CPU_PWRGD	10 13 14
CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_SMI L	10 14
CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_STPCLK_L	10 14
PM_THRMTRIP_L	CPU_50S	CPU_8MTL	PM_THRMTRIP_L	10 14 43
FSB_CPUSLP_L	CPU_50S	CPU_AGTL	FSB_CPUSLP_L	10 14
CPU_FROM_SB	CPU_50S	CPU_AGTL	CPU_DPSLP_L	10 14
CPU_DPRSTP_L	CPU_50S	CPU_AGTL	CPU_DPRSTP_L	9 10 14 62
CPU_ASYNC	CPU_50S	CPU_AGTL	FSB DPWR L	10 14
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_BCLK_VML_COMP_VDD	14
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_BCLK_VML_COMP_GND	14
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_CPU_COMP_VCC	14
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_CPU_COMP_GND	14
FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB_CLK_CPU_P	10 14
FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB_CLK_CPU_N	10 14
FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB_CLK_ITP_P	13 14
FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB_CLK_ITP_N	13 14
FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB_CLK_MCP_P	14
FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB_CLK_MCP_N	14
CPU_TFERR_L	CPU_50S		CPU_IERR L	10
PM DPRSLPVR	CPU_50S	CPU_AGTL	PM DPRSLPVR	21 62
(See above)	CPU_50S	CPU_AGTL	IMVP DPRSLPVR	62
CPU_GTLREF	CPU_50S	CPU_GTLREF	CPU_GTLREF	10 27
CPU_COMP	CPU_50S	CPU_COMP	CPU_COMP<3>	10
CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP<2>	10
CPU_COMP	CPU_50S	CPU_COMP	CPU_COMP<1>	10
CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP<0>	10
XDP_TDI	CPU_50S	CPU_ITP	XDP_TDI	6 10 13
XDP_TDO	CPU_50S	CPU_ITP	XDP_TDO	6 10
XDP_TMS	CPU_50S	CPU_ITP	XDP_TMS	6 10 13
XDP_TCK	CPU_50S	CPU_ITP	XDP_TCK	6 10 13
XDP_TRST_L	CPU_50S	CPU_ITP	XDP_TRST_L	6 10 13
XDP_BPM_L	CPU_50S	CPU_ITP	XDP_BPM L<4..0>	10 13
XDP_BPM_L5	CPU_50S	CPU_ITP	XDP_BPM L<5>	10 13
(FSB_CPURST_L)	CPU_50S	CPU_ITP	XDP_CPURST_L	13
	CPU_50S	CPU_8MTL	CPU_VID<6..0>	9 11
	CPU_50S	CPU_8MTL	IMVP6_VID<6..0>	9 62
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	11 62
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	11 62
(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6_VSEN_P	62
(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6_VSEN_N	62

87654321

CPU/FSB Constraints

SYNC_MASTER=MUXGFXSYNC_DATE=02/18/2008

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PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	13.1 MM	=90_OHM_DIFF	=90_OHM_DIFF
CLK_PCIE_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?
MCP_PEX_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.4

Analog Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CRT_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CRT	*	=4:1_SPACING	?
CRT_2CRT	*	=STANDARD	?
CRT_2CLK	*	50 MIL	?
CRT_2SWITCHER	*	250 MIL	?
CRT_SYNC	*	16 MIL	?
MCP_DAC_COMP	*	=2:1_SPACING	?

CRT signal single-ended impedance varies by location:
- 37.5-ohm from MCP to first termination resistor.
- 50-ohm from MCP to second termination resistor.
- 75-ohm from output of three-pole filter to connector (if possible).
R/G/B signals should be matched as close as possible and < 10 inches.
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Sections 2.5.1 & 2.5.2.

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
LVDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
MCP_DV_COMP	*	Y	20 MIL	20 MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3X_DIELECTRIC	?
LVDS	*	=3X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4X_DIELECTRIC	?
LVDS	TOP,BOTTOM	=4X_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length. DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps. DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals. Max length of LVDS/DisplayPort/TMDS traces: 12 inches.
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Sections 2.5.3 & 2.5.4.

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA	*	=4X_DIELECTRIC	?
SATA_TERM	*	8 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA	TOP,BOTTOM	=3X_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.7.1.

ELECTRICAL_CONSTRAINT_SET

NET_TYPE

PHYSICAL

SPACING

PEG_R2D

PCIE_90D

PCIE

PEG_R2D_P<15..0>

70

PEG_R2D

PCIE_90D

PCIE

PEG_R2D_N<15..0>

70

PEG_D2R

PCIE_90D

PCIE

PEG_R2D_C_P<15..0>

9 70

PEG_D2R

PCIE_90D

PCIE

PEG_R2D_C_N<15..0>

9 70

PEG_D2R

PCIE_90D

PCIE

PEG_D2R_P<15..0>

9 70

PEG_D2R

PCIE_90D

PCIE

PEG_D2R_N<15..0>

9 70

PEG_D2R

PCIE_90D

PCIE

PEG_D2R_C_P<15..0>

70

PEG_D2R

PCIE_90D

PCIE

PEG_D2R_C_N<15..0>

70

PCIE_MINI_R2D_P

PCIE_90D

PCIE

PCIE_MINI_R2D_P

31 95

PCIE_MINI_R2D_N

PCIE_90D

PCIE

PCIE_MINI_R2D_N

31 95

PCIE_MINI_R2D_C_P

PCIE_90D

PCIE

PCIE_MINI_R2D_C_P

17 31

PCIE_MINI_R2D_C_N

PCIE_90D

PCIE

PCIE_MINI_R2D_C_N

17 31

PCIE_MINI_D2R_P

PCIE_90D

PCIE

PCIE_MINI_D2R_P

17 31

PCIE_MINI_D2R_N

PCIE_90D

PCIE

PCIE_MINI_D2R_N

17 31

PCIE_FW_R2D_P

PCIE_90D

PCIE

PCIE_FW_R2D_P

36

PCIE_FW_R2D_N

PCIE_90D

PCIE

PCIE_FW_R2D_N

36

PCIE_FW_R2D_C_P

PCIE_90D

PCIE

PCIE_FW_R2D_C_P

17 36

PCIE_FW_R2D_C_N

PCIE_90D

PCIE

PCIE_FW_R2D_C_N

17 36

PCIE_FW_D2R_P

PCIE_90D

PCIE

PCIE_FW_D2R_P

17 36

PCIE_FW_D2R_N

PCIE_90D

PCIE

PCIE_FW_D2R_N

17 36

PCIE_FW_D2R_C_P

PCIE_90D

PCIE

PCIE_FW_D2R_C_P

36

PCIE_FW_D2R_C_N

PCIE_90D

PCIE

PCIE_FW_D2R_C_N

36

PCIE_EXCARD_R2D_P

PCIE_90D

PCIE

PCIE_EXCARD_R2D_P

7 32 95

PCIE_EXCARD_R2D_N

PCIE_90D

PCIE

PCIE_EXCARD_R2D_N

7 32 95

PCIE_EXCARD_R2D_C_P

PCIE_90D

PCIE

PCIE_EXCARD_R2D_C_P

17 32

PCIE_EXCARD_R2D_C_N

PCIE_90D

PCIE

PCIE_EXCARD_R2D_C_N

17 32

PCIE_EXCARD_D2R_P

PCIE_90D

PCIE

PCIE_EXCARD_D2R_P

7 17 32

PCIE_EXCARD_D2R_N

PCIE_90D

PCIE

PCIE_EXCARD_D2R_N

7 17 32

MCP_PEG0_REFCLK

CLK_PCIE_100D

CLK_PCIE

PEG_CLK100M_P

17 70

MCP_PEG0_REFCLK

CLK_PCIE_100D

CLK_PCIE

PEG_CLK100M_N

17 70

MCP_PEF1_REFCLK

CLK_PCIE_100D

CLK_PCIE

PCIE_CLK100M_MINI_P

17 31

MCP_PEF1_REFCLK

CLK_PCIE_100D

CLK_PCIE

PCIE_CLK100M_MINI_N

17 31

MCP_PEF2_REFCLK

CLK_PCIE_100D

CLK_PCIE

PCIE_CLK100M_FW_P

17 36

MCP_PEF2_REFCLK

CLK_PCIE_100D

CLK_PCIE

PCIE_CLK100M_FW_N

17 36

MCP_PEF3_REFCLK

CLK_PCIE_100D

CLK_PCIE

PCIE_CLK100M_EXCARD_P

17 32

MCP_PEF3_REFCLK

CLK_PCIE_100D

CLK_PCIE

PCIE_CLK100M_EXCARD_N

17 32

MCP_PEX_CLK_COMP

MCP_PEX_COMP

MCP_PEX_CLK_COMP

17

CRT_RED

CRT_50S

CRT

CRT_IG_R_C_PR

18 25

CRT_GREEN

CRT_50S

CRT

CRT_IG_G_Y_Y

18 25

CRT_BLUE

CRT_50S

CRT

CRT_IG_B_COMP_PB

18 25

CRT_SYNC

CRT_50S

CRT_SYNC

CRT_IG_VSYNC

18 25

CRT_SYNC

CRT_50S

CRT_SYNC

CRT_IG_VSYNC

18 25

MCP_DAC_RSET

MCP_DAC_COMP

MCP_TV_DAC_RSET

18 25

MCP_DAC_VREF

MCP_DAC_COMP

MCP_TV_DAC_VREF

18 25

TMDS_IG_TXC

DP_100D

DISPLAYPORT

TMDS_IG_TXC_P

TMDS_IG_TXC

DP_100D

DISPLAYPORT

TMDS_IG_TXC_N

TMDS_IG_TXD

DP_100D

DISPLAYPORT

TMDS_IG_TXD_P<2..0>

TMDS_IG_TXD

DP_100D

DISPLAYPORT

TMDS_IG_TXD_N<2..0>

DP_ML

DP_100D

DISPLAYPORT

DP_IG_ML_P<3..0>

9 80

DP_ML

DP_100D

DISPLAYPORT

DP_IG_ML_N<3..0>

9 80

DP_AUX_CH

DP_100D

DISPLAYPORT

DP_IG_AUX_CH_P

18 80

DP_AUX_CH

DP_100D

DISPLAYPORT

DP_IG_AUX_CH_N

18 80

MCP_HDMI_RSET

MCP_DV_COMP

MCP_HDMI_RSET

18 25

MCP_HDMI_VPROBE

MCP_DV_COMP

MCP_HDMI_VPROBE

18 25

LVDS_IG_A_CLK

LVDS_100D

LVDS

LVDS_IG_A_CLK_P

18 83

LVDS_IG_A_CLK

LVDS_100D

LVDS

LVDS_IG_A_CLK_N

18 83

LVDS_IG_A_DATA

LVDS_100D

LVDS

LVDS_IG_A_DATA_P<2..0>

18 83

LVDS_IG_A_DATA

LVDS_100D

LVDS

LVDS_IG_A_DATA_N<2..0>

18 83

LVDS_IG_A_DATA3

LVDS_100D

LVDS

LVDS_IG_A_DATA_P<3>

9 18

LVDS_IG_A_DATA3

LVDS_100D

LVDS

LVDS_IG_A_DATA_N<3>

9 18

LVDS_IG_B_CLK

LVDS_100D

LVDS

LVDS_IG_B_CLK_P

9 18

LVDS_IG_B_CLK

LVDS_100D

LVDS

LVDS_IG_B_CLK_N

9 18

LVDS_IG_B_DATA

LVDS_100D

LVDS

LVDS_IG_B_DATA_P<2..0>

18 83

LVDS_IG_B_DATA

LVDS_100D

LVDS

LVDS_IG_B_DATA_N<2..0>

18 83

LVDS_IG_B_DATA3

LVDS_100D

LVDS

LVDS_IG_B_DATA_P<3>

9 18

LVDS_IG_B_DATA3

LVDS_100D

LVDS

LVDS_IG_B_DATA_N<3>

9 18

MCP_IFPAB_RSET

MCP_DV_COMP

MCP_IFPAB_RSET

18 25

MCP_IFPAB_VPROBE

MCP_IFPAB_VPROBE

18 25

SATA_HDD_R2D

SATA_100D

SATA

SATA_HDD_R2D_C_P

20 39

SATA_HDD_R2D

SATA_100D

SATA

SATA_HDD_R2D_C_N

20 39

SATA_HDD_R2D

SATA_100D

SATA

SATA_HDD_R2D_P

39

SATA_HDD_R2D

SATA_100D

SATA

SATA_HDD_R2D_N

39

SATA_HDD_D2R

SATA_100D

SATA

SATA_HDD_D2R_P

20 39

SATA_HDD_D2R

SATA_100D

SATA

SATA_HDD_D2R_N

20 39

SATA_HDD_D2R

SATA_100D

SATA

SATA_HDD_D2R_C_P

39

SATA_HDD_D2R

SATA_100D

SATA

SATA_HDD_D2R_C_N

39

SATA_ODD_R2D

SATA_100D

SATA

SATA_ODD_R2D_C_P

20 39

SATA_ODD_R2D

SATA_100D

SATA

SATA_ODD_R2D_C_N

20 39

SATA_ODD_R2D

SATA_100D

SATA

SATA_ODD_R2D_P

7 39

SATA_ODD_R2D

SATA_100D

SATA

SATA_ODD_R2D_N

7 39

SATA_ODD_D2R

SATA_100D

SATA

SATA_ODD_D2R_P

20 39

SATA_ODD_D2R

SATA_100D

SATA

SATA_ODD_D2R_N

20 39

SATA_ODD_D2R

SATA_100D

SATA

SATA_ODD_D2R_C_P

7 39

SATA_ODD_D2R

SATA_100D

SATA

SATA_ODD_D2R_C_N

7 39

MCP_SATA_TERM

SATA_TERM

MCP_SATA_TERM

20

8

7

6

5

4

3

2

1

MCP Constraints 1

SYNC_MASTER=MUXGFX

SYNC_DATE=02/18/2008

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SIZE

D

DRAWING NUMBER

051-7546

REV.

A.0.0

SCALE

NONE

SHT

89

OF

96

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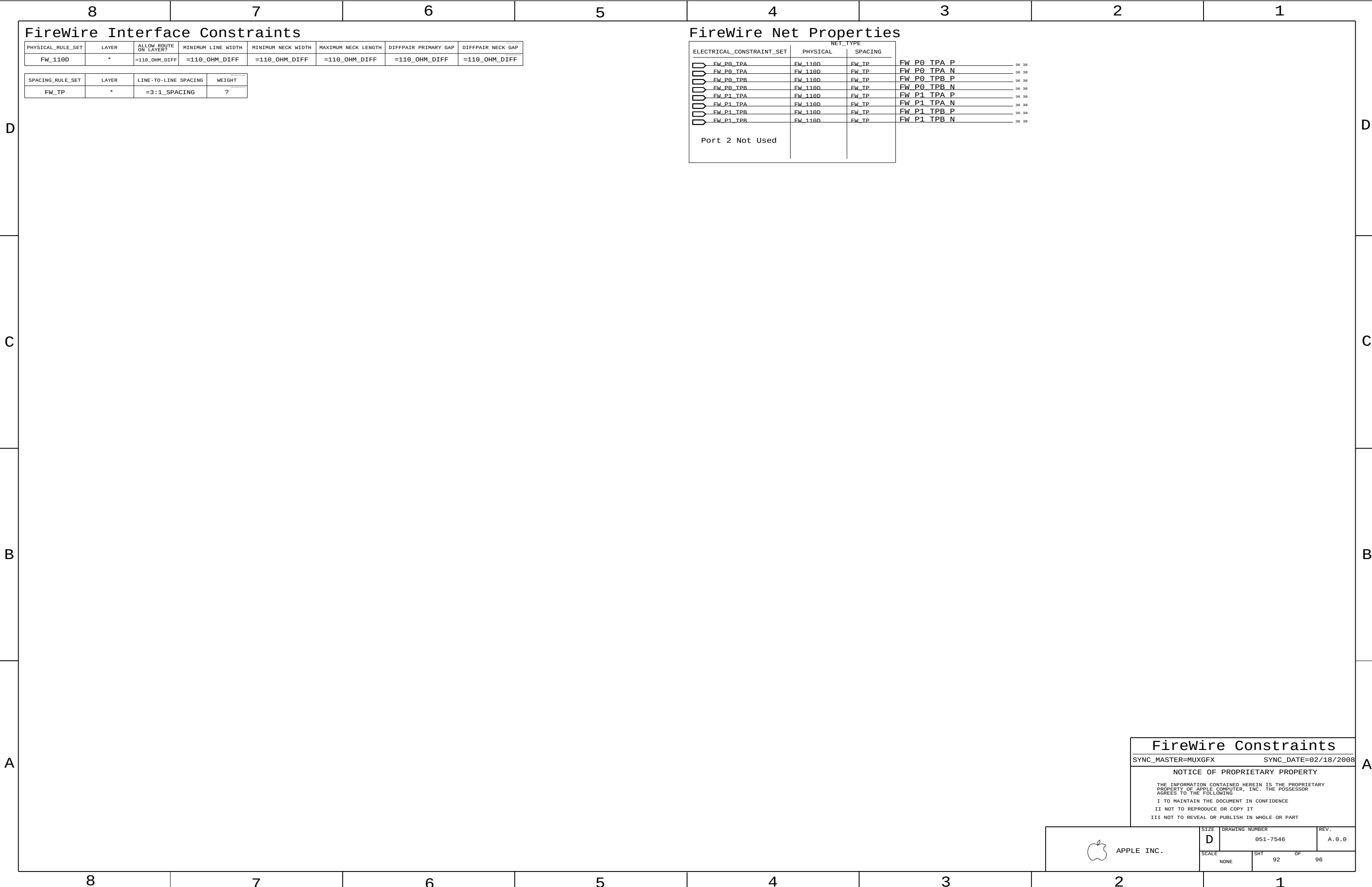
5

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D

C

From T18 MXM: Digital Video Signal Constraints

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.
DisplayPort/TMD5 intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMD5 traces: 12 inches.
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Sections 2.5.3 & 2.5.4.

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A

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A

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_I101_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_I101_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
PP1V8_MEM	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	"	GND_P2MM
MEM_CMD	GND	"	GND_P2MM
MEM_CTRL	GND	"	GND_P2MM
MEM_DATA	GND	"	GND_P2MM
MEM_DQS	GND	"	GND_P2MM

				NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	CLK_FSB	GND	*	GND_P2MM
CLK_PCIE	GND	*	GND_P2MM	CPU_COMP	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM	CPU_GTLREF	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM	CPU_VCCSENSE	GND	*	GND_P2MM
				FSB_DSTB	FSB_DSTB	*	GND_P2MM

USB	GND		GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

Memory Constraint Relaxations

Allow 0.127 mm necks for >0.127 mm lines for GMCH fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_78D	BOTTOM			0.127 MM	6.35 MM		

Graphics , SATA Constraint Relaxations

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
LVDS_100D	BGA	100_DIFF_BGA
DP_100D	BGA	100_DIFF_BGA
SATA_100D	BGA	100_DIFF_BGA

M99 Specific Net Properties

ELECTRICAL CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
		ENET_MDT_100D	ENETCONN	ENETCONN_P<3..0>
		ENET_MDT_100D	ENETCONN	ENETCONN_N<3..0>
U640		SATA_100D	SATA	SATA_ODD_R2D_UF_P
U640		SATA_100D	SATA	SATA_ODD_R2D_UF_N
U640		SATA_100D	SATA	SATA_ODD_D2R_UF_P
U640		SATA_100D	SATA	SATA_ODD_D2R_UF_N
U640		SATA_100D	SATA	SATA_HDD_D2R_UF_P
U640		SATA_100D	SATA	SATA_HDD_D2R_UF_N
U640		SATA_100D	SATA	SATA_HDD_R2D_UF_P
U640		SATA_100D	SATA	SATA_HDD_R2D_UF_N
	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	GFXXIMVP6_VSEN_P
		SENSE_1T01_55S	SENSE	GFXXIMVP6_VSEN_N
	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	MPCOREISNS_P
		SENSE_1T01_55S	SENSE	MPCOREISNS_N
U324	CPUTHMSNS_D2_DP	THERM_1T01_55S	THERM	CPUTHMSNS_D2_P
U325		THERM_1T01_55S	THERM	CPUTHMSNS_D2_N
U327	CPU_THERMO_DP	THERM_1T01_55S	THERM	CPU_THERMO_P
U327		THERM_1T01_55S	THERM	CPU_THERMO_N
U328	GPUTHMSNS_D_DP	THERM_1T01_55S	THERM	GPUTHMSNS_D_P
U328		THERM_1T01_55S	THERM	GPUTHMSNS_D_N
U329	GPU_THERMO_DP	THERM_1T01_55S	THERM	GPU_TDIODE_P
U329		THERM_1T01_55S	THERM	GPU_TDIODE_N
U330	MCPTHMSNS_D_DP	THERM_1T01_55S	THERM	MCPTHMSNS_D_P
U330		THERM_1T01_55S	THERM	MCPTHMSNS_D_N
U331	MCP_THERMO_DP	THERM_1T01_55S	THERM	MCP_THMDIODE_P
U331		THERM_1T01_55S	THERM	MCP_THMDIODE_N
U332	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	1V05CPUISNS_R_P
U332		SENSE_1T01_55S	SENSE	1V05CPUISNS_R_N
U333	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	DDRISNS_R_P
U333		SENSE_1T01_55S	SENSE	DDRISNS_R_N
U334	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	GPUISENS_P
U334		SENSE_1T01_55S	SENSE	GPUISENS_N
U335	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	1V05CPU_P
U335		SENSE_1T01_55S	SENSE	1V05CPU_N
U336	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	DDRISNS_P
U336		SENSE_1T01_55S	SENSE	DDRISNS_N
U340	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	P1V8GPU_P
U340		SENSE_1T01_55S	SENSE	P1V8GPU_N
U341	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	ISNS_CPU_P
U341		SENSE_1T01_55S	SENSE	ISNS_CPU_N
			GND	GND
			SB_POWER	PP3V3_S5
			SB_POWER	PP3V3_S0
			SB_POWER	PP1V5_S0
U337	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	P1V8GPUISNS_P
U337		SENSE_1T01_55S	SENSE	P1V8GPUISNS_N
U338	SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	P1V8GPUISNS_R_P
U338		SENSE_1T01_55S	SENSE	P1V8GPUISNS_R_N

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	5.8 MM OVERRIDE	OVERRIDE	OVERRIDE
MEM_40S_VDD OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	5.8 MM OVERRIDE	OVERRIDE	OVERRIDE
MEM_70D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	5.8 MM OVERRIDE	OVERRIDE	OVERRIDE
MEM_70D_VDD OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_90D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
USB_90D OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MEM_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MII_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_USB_RBIA5 OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	*	OVERRIDE	OVERRIDE	0.25 MM OVERRIDE	250 MIL OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S OVERRIDE	BOTTOM OVERRIDE	OVERRIDE	OVERRIDE	0.23 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S_OVERRIDE	ISL4, ISL9	VERRIDE	VERRIDE	VERRIDE	VERRIDE	VERRIDE	VERRIDE
MEM_40S_VDD_OVERRIDE	ISL3, ISL10	N_OVERRIDE	VERRIDE	VERRIDE	VERRIDE	VERRIDE	VERRIDE
MEM_70D_OVERRIDE	ISL4, ISL9	VERRIDE	VERRIDE	VERRIDE	VERRIDE	VERRIDE	VERRIDE
MEM_70D_VDD_OVERRIDE	ISL3, ISL10	N_OVERRIDE	VERRIDE	VERRIDE	VERRIDE	VERRIDE	VERRIDE

Ground-referenced memory signals (DQ,DQM,DQS) MAY route on ISL9 (VDD-referenced plane)but not next to VDD island. Forces power-referenced memory signals (CLK,ADDR,CTRL) to not route on ISL3, ISL4 & ISL10(GND-referenced planes).

M99 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
H100	(PCIE_EXCARD)	PCIE_900	PCIE	PCIE_EXCARD_R2D_P 7 32 89
H100	(PCIE_EXCARD)	PCIE_900	PCIE	PCIE_EXCARD_R2D_N 7 32 89
H100	(PCIE_MINI)	PCIE_900	PCIE	PCIE_MINI_R2D_P 31 89
H100	(PCIE_MINI)	PCIE_900	PCIE	PCIE_MINI_R2D_N 31 89
H100		CLK_PCIE_1000	CLK_PCIE	PCIE_CLK100M_MINI_CONN_P 31
H100		CLK_PCIE_1000	CLK_PCIE	PCIE_CLK100M_MINI_CONN_N 31
H100		1T01_DIEFPAIR		CHGR_CSI_R_P 61
H100		1T01_DIEFPAIR		CHGR_CSI_R_N 61
H100		1T01_DIEFPAIR		CHGR_CSO_R_P 46 61
H100		1T01_DIEFPAIR		CHGR_CSO_R_N 46 61
H100	(USB_EXT_A)	USB_900	USB	USB2_EXT_A_MUXED_P 40
H100	(USB_EXT_A)	USB_900	USB	USB2_EXT_A_MUXED_N 40
H100	(USB_EXT_A)	USB_900	USB	USB2_LT1_P 40
H100	(USB_EXT_A)	USB_900	USB	USB2_LT1_N 40
H100	(USB_EXT_D)	USB_900	USB	CONN_TPAD_USB_P
H100	(USB_EXT_D)	USB_900	USB	CONN_TPAD_USB_N
H100	(USB_CAMERA)	USB_900	USB	USB_CAMERA_CONN_P 31
H100	(USB_CAMERA)	USB_900	USB	USB_CAMERA_CONN_N 31
H100		USB_900	USB	CONN_USB2_BT_P 31
H100		USB_900	USB	CONN_USB2_BT_N 31
H100		USB_900	USB	USB_LT2_P 40
H100		USB_900	USB	USB_LT2_N 40
H100		USB_900	USB	USB2_EXCARD_CONN_P 7 32
H100		USB_900	USB	USB2_EXCARD_CONN_N 7 32
H100		DP_1000	DISPLAYPORT	DP_IG_AUX_CH_C_P 80
H100		DP_1000	DISPLAYPORT	DP_IG_AUX_CH_C_N 80
H100	MCP_PE4_REFCLK	CLK_PCIE_1000	CLK_PCIE	PCIE_CLK100M_FC_P 9 32
H100		CLK_PCIE_1000	CLK_PCIE	PCIE_CLK100M_FC_N 9 32
H100	PCIE_FC_R2D	PCIE_900	PCIE	PCIE_FC_R2D_C_P 9 32
H100		PCIE_900	PCIE	PCIE_FC_R2D_C_N 9 32
H100	PCIE_FC_D2R	PCIE_900	PCIE	PCIE_FC_D2R_P 9 32
H100		PCIE_900	PCIE	PCIE_FC_D2R_N 9 32
H100		PCIE_900	PCIE	PCIE_FC_R2D_P 32
H100		PCIE_900	PCIE	PCIE_FC_R2D_N 32
H100		CLK_PCIE_1000	CLK_PCIE	PCIE_CLK100M_EXCARD_CONN_N 7 32
H100		CLK_PCIE_1000	CLK_PCIE	PCIE_CLK100M_EXCARD_CONN_P 7 32
H100	SPK_OUT	DIEFPAIR	AUDIO	SPKRCONN_L_P_OUT 7 57
H100		DIEFPAIR	AUDIO	SPKRCONN_L_N_OUT 7 57
H100	SPK_OUT	DIEFPAIR	AUDIO	SPKRCONN_S_P_OUT 7 57
H100		DIEFPAIR	AUDIO	SPKRCONN_S_N_OUT 7 57
H100	SPK_OUT	DIEFPAIR	AUDIO	SPKRCONN_R_P_OUT 7 57
H100		DIEFPAIR	AUDIO	SPKRCONN_R_N_OUT 7 57
H100		DIEFPAIR	AUDIO	SPKRAMP_L_P_OUT 57
H100		DIEFPAIR	AUDIO	SPKRAMP_L_N_OUT 57
H100		DIEFPAIR	AUDIO	SPKRAMP_R_P_OUT 57
H100		DIEFPAIR	AUDIO	SPKRAMP_R_N_OUT 57
H100		DIEFPAIR	AUDIO	SPKRAMP_S_P_OUT 57
H100		DIEFPAIR	AUDIO	SPKRAMP_S_N_OUT 57

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